VNIVERSIDAD D SALAMANCA

Creating Electronic Books with Code and Artificial Intelligence Assistants

Course for the Faculties of Sciences and Chemical Sciences

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CONTENTS

I Tutorial 3

1 1. What is a TeachBook? 5
1.1 Attribution and origin . . . 6
1.2 What goes in and what comes out . . . 6
1.3 Why this is useful for teaching . . . 6
1.4 What this template adds on top of the original base . . . 7
1.5 What a TeachBook is NOT . . . 8
1.6 Practical advice to start . . . 8
1.7 Bibliography for this page . . . 8

2 2. Workflow 9
2.1 Book Development Cycle . . . 9
2.2 File Structure . . . 9
2.3 Directory Tree . . . 11

3 3. Editing with AI 13
3.1 How to Ask the AI for Help . . . 13

4 4. My first book from scratch 17
4.1 Step 0. Prepare the basic requirements . . . 17
4.2 Step 1. Create a repository from the template . . . 18
4.3 Step 2. Enable GitHub Actions and GitHub Pages . . . 19
4.4 Step 3. Clone the repository in Antigravity . . . 20
4.5 Step 4. Ask the agent to configure the environment . . . 21
4.6 Step 5. Make a first edit with the agent . . . 22
4.7 Step 6. View the local preview . . . 23
4.8 Step 7. Generate the PDFs . . . 24
4.9 Step 8. Publish the changes . . . 25

5 5. The “X as code” Philosophy 27
5.1 What is “X as code”? . . . 27
5.2 Why is this philosophy interesting? . . . 28
5.3 Code as a bridge to the final result . . . 28
5.4 Why this fits so well with AI . . . 29
5.5 Reproducibility, not just automation . . . 29
5.6 The mindset shift . . . 30

II Examples by Degree 31

6 Degree in Physics 33

6.1 Physics Example: Harmonic Oscillator . . . 33
6.2 Drawing Circuits with SchemDraw . . . 34
6.3 Advanced CircuitikZ for Precise Circuits . . . 35
6.4 Technical Diagrams with Kroki for Physics . . . 37
6.5 Ohm's Law: Linear Fit of V-I Data . . . 39
6.6 Lissajous figures: sound, oscillations, and visualization . . . 41

7 Degree in Mathematics **47**
7.1 Mathematics Example: Symbolic Calculus . . . 47
7.2 Quadratic Equation . . . 47
7.3 Newton's Method for Finding Roots . . . 48
7.4 2D Linear Transformations . . . 50
7.5 Planar Point Sets with Many Unit Distances . . . 52

8 Degree in Statistics **57**
8.1 Statistics Example: Data Generation . . . 57
8.2 Linear Regression with Synthetic Data . . . 57
8.3 Estimating π with Monte Carlo . . . 59
8.4 Practice Questions: Statistics . . . 60

9 Degree in Computer Science **63**
9.1 Entity-Relationship Diagram with Kroki (Mermaid) . . . 63
9.2 UML Class Diagram with Kroki (Mermaid) . . . 64
9.3 Diagrams with Kroki for Computer Science . . . 67
9.4 Algorithmic complexity: asymptotic growth and Big-O notation . . . 70

10 Degree in Geology **73**
10.1 Topographic Profile . . . 73
10.2 Stratigraphic Column . . . 74
10.3 The Rock Cycle with Kroki (Mermaid) . . . 75

11 Chemical Engineering Degree **77**
11.1 Process Flow Diagram . . . 78
11.2 McCabe-Thiele Diagram . . . 79
11.3 Recycle Solved with `fsolve` . . . 82
11.4 Concentration Gradient in a Catalyst Particle . . . 84
11.5 Fast Reading of the Moody Chart . . . 86

12 Chemistry Degree **89**
12.1 pH Curve in a Buffer Solution . . . 90
12.2 Michaelis-Menten Kinetics . . . 91
12.3 Interactive Periodic Table . . . 93
12.4 Interactive IR Spectrum . . . 94
12.5 Nernst Equation . . . 94

III Cross-disciplinary Examples **97**

13 Cross-disciplinary Examples **99**
13.1 Example families . . . 99
13.2 The common idea . . . 100
13.3 Growth and Saturation Models . . . 100
13.4 Questionnaire Analysis . . . 101
13.5 Text Corpus Frequencies . . . 102
13.6 Rubrics and Learning Sequences . . . 103

13.7 Simulated Data and Variable Relationships . . . 105
13.8 Balances and Conservation . . . 106

IV Examples from Other Books **109**

14 Examples from Other Books **111**

V Basic Content **115**

15 Basic Content **117**
15.1 What you will find here . . . 117
15.2 How to use this section . . . 118

16 MyST Markdown: The Basics **119**
16.1 Essential Syntax . . . 119
16.2 Directives . . . 120
16.3 Math . . . 120
16.4 Tables . . . 121
16.5 Code Blocks . . . 121

17 Figures and Images **123**
17.1 Centered image with alt text . . . 123
17.2 Numbered figure with caption . . . 124
17.3 Cross-reference to a figure . . . 124
17.4 Margin figure . . . 124
17.5 Useful parameters . . . 125

18 Equations and Mathematics **127**
18.1 Inline math . . . 127
18.2 Display equation (unnumbered) . . . 127
18.3 Numbered equation with label . . . 127
18.4 Cross-reference to an equation . . . 128
18.5 Common math symbols . . . 128

19 Citations and Bibliography **129**
19.1 Bibliography file . . . 129
19.2 Textual citation: `{cite:t}` . . . 129
19.3 Parenthetical citation: `{cite:p}` . . . 130
19.4 Multiple citations . . . 130
19.5 Printing the bibliography . . . 130
19.6 Workflow . . . 131

20 Cross-References **133**
20.1 Label a section . . . 133
20.2 Reference a section . . . 133
20.3 Label and reference figures . . . 133
20.4 Label and reference equations . . . 134
20.5 Label and reference tables . . . 134
20.6 Roles summary . . . 135

21 Tables **137**
21.1 Simple Markdown table . . . 137
21.2 Numbered table with caption . . . 137

21.3 List-table directive . . . 138
21.4 Tips for readable tables . . . 139

22 Admonitions and Dropdowns **141**
22.1 Note ({note}) . . . 141
22.2 Warning ({warning}) . . . 141
22.3 Tip ({tip}) . . . 142
22.4 Important ({important}) . . . 142
22.5 Caution ({caution}) . . . 142
22.6 Custom admonition . . . 142
22.7 Dropdown (collapsible content) . . . 143
22.8 Teaching use cases . . . 143

23 Tabs and Cards **145**
23.1 Simple tabs ({tab-set} / {tab-item}) . . . 145
23.2 Grouped tabs ({tab-set} / {tab-item}) . . . 146
23.3 Tabs for procedure steps . . . 146
23.4 Cards ({card}) . . . 147

24 Code Blocks **149**
24.1 Syntax highlighting . . . 149
24.2 Line numbers . . . 149
24.3 Highlight specific lines . . . 150
24.4 Other languages . . . 150
24.5 Common languages in science . . . 150

25 Videos **153**
25.1 YouTube video (HTML + PDF) . . . 153
25.2 How to get a YouTube video ID . . . 153
25.3 Local video (HTML5) . . . 154
25.4 Externally generated videos . . . 154
25.5 Video with description . . . 154
25.6 Recommended complete pattern . . . 154

26 Diagrams with Kroki (Mermaid) **157**
26.1 Flowchart (flowchart) . . . 157

27 HTML and PDF Compatibility **161**
27.1 Compatibility table . . . 161
27.2 Universal pattern: HTML + LaTeX fallback . . . 161
27.3 Example: video with fallback . . . 162
27.4 Example: diagram with Kroki . . . 162
27.5 Practical rules . . . 162

VI Interactivity and Assessment **163**

28 Interactivity and Assessment **165**
28.1 What will you find here? . . . 165
28.2 Thebe + Binder: Live Code . . . 166
28.3 Interactive Notebooks . . . 168
28.4 Interactive Widgets (ipywidgets) . . . 172
28.5 Quizzes with jupyterquiz . . . 173
28.6 Parametric Questions . . . 174
28.7 Exercises with Solutions . . . 175

28.8 Interactive HTML (no frameworks) . . . 177
28.9 H5P (Optional) . . . 179

VII Web Publication **181**

29 Web Publication **183**
29.1 GitHub Pages . . . 183
29.2 Own Server . . . 184

VIII Information **187**

30 About **189**
30.1 How this book is made . . . 189
30.2 About the authors . . . 189

31 Credits and licenses **191**
31.1 Project credits . . . 191
31.2 Acknowledgements . . . 191
31.3 Book license . . . 191
31.4 External resources . . . 191
31.5 Code and tools . . . 192

32 How to Cite **193**
32.1 Text Citation . . . 193
32.2 BibTeX . . . 193
32.3 DOI . . . 193

33 Bibliography **195**

Welcome to **Creating Electronic Books with Code and Artificial Intelligence Assistants**.

What is this?

This is the course material and a template designed for the teaching staff of the **Faculties of Sciences and Chemical Sciences at USAL** to create interactive teaching books easily. The main idea of this template is for it to be broad enough to cover many use cases and for each student to adapt it to their own course, while also being sufficiently equipped to be used easily with the help of AI assistants (such as GitHub Copilot, Gemini, Claude, Codex, etc.) and a code editor such as VS Code.

Content

In this book you will find:

- *Tutorials* to learn how to use the template
- *Examples by Degree* to see real cases
- Information on *how to cite* and *licenses*

PDF version

You can also download the printable version of the book:

- Download PDF in English
- Descargar PDF en español

Note

This project is designed to be used with **VS Code** and **AI** assistants.

Part I

Tutorial

1. WHAT IS A TEACHBOOK?

A **TeachBook** is a digital teaching book, built from text files, code, and multimedia resources, which are then compiled into a **navigable website** and, in the case of this course, also a **PDF version**.

In simple terms:

- you edit the **source content**
- the system **builds** it
- you get an **attractive, functional, publishable** book

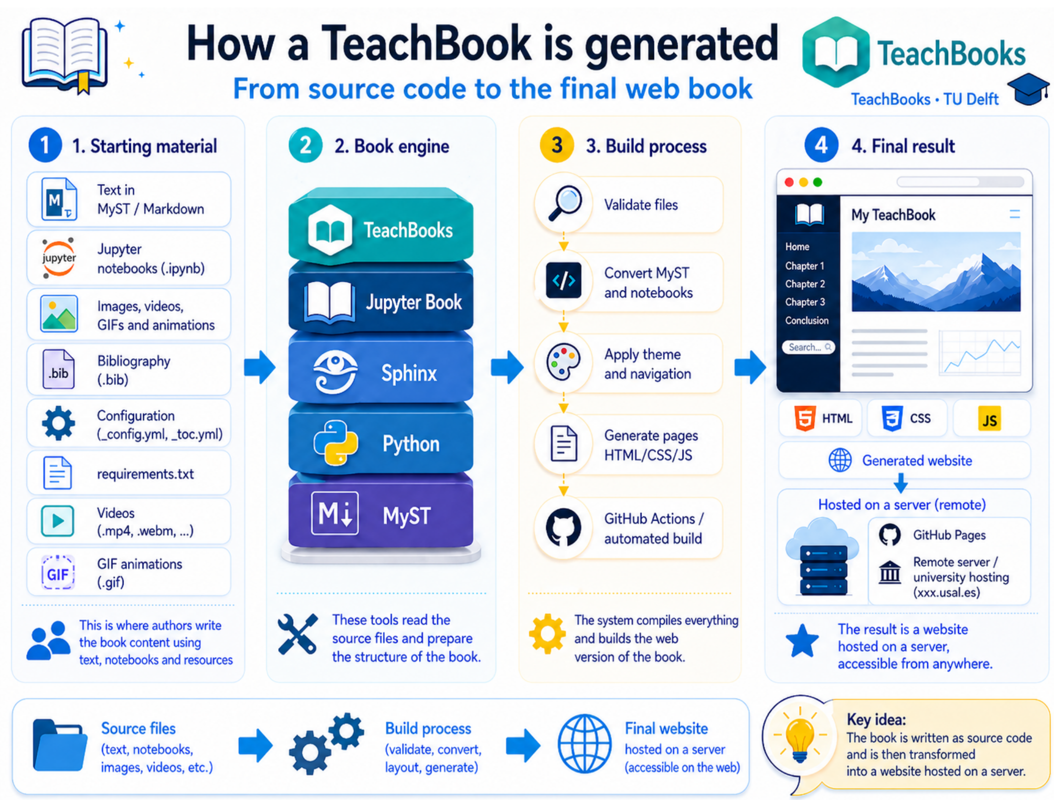


Figure1.1: A TeachBook starts from editable content and produces a publishable teaching website.

1.1 Attribution and origin

This template is built on the original ecosystem of:

- TeachBooks
- Jupyter Book

To situate the template, it is useful to cite both the TeachBooks project documentation [TeachBooksDTeam25] and the reference work for Jupyter Book [ProjectJupyterBC+25]: the former provides the teaching-oriented approach, while the latter provides the underlying system for building publishable computational books.

This specific adaptation has been prepared for the course entitled "Elaboración de libros electrónicos mediante código y asistentes de Inteligencia Artificial", taught at the **Faculties of Sciences and Chemical Sciences at the University of Salamanca (USAL)**.

This matters: **we are not starting from scratch**. We are starting from existing tools and previous projects, and building on top of them a template adapted to a specific real teaching use (adapted to the needs of each course, but with reuse potential).

1.2 What goes in and what comes out

The core idea is this:

The following table summarizes the main elements of this section.

Table. What goes in and what comes out.

You start with...	And you get...
`.md` and `.ipynb` files	book web pages
images, bibliography, and static assets	enriched content
book configuration (`_config_*.yml`)	navigation, theme, language, extensions
table of contents (`_toc_*.yml`)	visible book structure

1.3 Why this is useful for teaching

1. **Accessibility**: the book can be read on the web from any device.
2. **Reproducibility**: the content comes from source files that can be versioned.
3. **Maintainability**: updating a page is much easier (thanks to AI agents) than rebuilding a whole set of notes.
4. **Scalability**: you can start with one page or chapter and grow gradually.
5. **Compatibility**: the same project can serve as both a website and a classic PDF book.

1.4 What this template adds on top of the original base

On top of TeachBooks and Jupyter Book, this project adds several practical improvements for teachers and students:

1.4.1 1. Multilingual structure

The book is prepared to work in:

- Spanish
- English

with synchronized indexes and content. More languages could be added easily.

1.4.2 2. PDF export

Besides the website, this template prepares a PDF output designed for printing or offline distribution. Not all multimedia and interactive content can be exported, but the rest of the valuable content can, making it possible to have a traditional version of the book.

1.4.3 3. Simple automation scripts

You do not need to learn a complicated workflow on day one. The project includes SKILLS to:

- prepare the environment (install dependencies, configure variables, etc.)
- build the website (generate the navigable book locally so you can review it)
- open live preview
- export PDF
- convert PDF to Markdown (to reuse content you already have in PDF from other subjects or courses)
- etc.

1.4.4 4. AI agent integration

The template is prepared so an agent can help you:

- create content
- reorganize chapters
- add multimedia
- maintain the structure of the book
- etc.

1.5 What a TeachBook is NOT

It is not:

- a PowerPoint presentation
- a React-style web app with heavy JS frameworks
- a custom-designed website (although the theme can be customized, that is not the main goal)

The main idea is that the content remains **editable, structured, and reusable**.

1.6 Practical advice to start

Do not try to build a huge book on the first day.

Start small:

- a cover page
- one short chapter
- one image
- one equation
- one section with two or three pages

Once that works well and you have been able to publish it, expand.

1.7 Bibliography for this page

2. WORKFLOW

This is the recommended workflow for your TeachBook:

1. **Write content** in Markdown files (`.md`) or Notebooks (`.ipynb`). In the case of Notebooks, it is important to execute them beforehand so that the book keeps the cell outputs. The AI agent can help you write the content and execute the Notebooks.
2. **Preview** with `python scripts/launch_preview.py` to see the same build that will be published. In this course, you can ask the AI agent to do it for you, because it has the information it needs.
3. **Save versions** with Git (`commit`). The AI agent will help you save changes in an orderly way.
4. **Publish** to GitHub (`push`) so the website updates automatically through GitHub Pages.

2.1 Book Development Cycle

1. Modify or add content in Markdown files or Notebooks.
2. Build the book and start a local server to preview the changes.
3. Generate the PDF versions of the book when needed. That process first creates LaTeX and then compiles the PDFs linked from the website.
4. If everything is correct, commit the changes with Git.
5. Publish the changes to GitHub to deploy the website.
6. If you find errors or want to improve something, go back to step 1 and repeat the cycle.

Fig. 2.1 summarizes the full process: write content, preview it, review the result, version the changes, and publish the website.

2.2 File Structure

To work with the book, we need to understand the file and folder structure. The content is organized by language, and each language has its own table of contents and configuration. This makes it possible to maintain versions in several languages with the same content or with content adapted to each language. We follow the same philosophy as the original TeachBooks project, with a few adaptations to make it easier to use with AI assistants.

The Spanish content is organized in `book/es/`. The English content is organized in `book/en/`. Each language has its own table of contents (`_toc_*.yml`) and configuration (`_config_*.yml`), but the structure must remain coherent across languages.

Main files and folders:

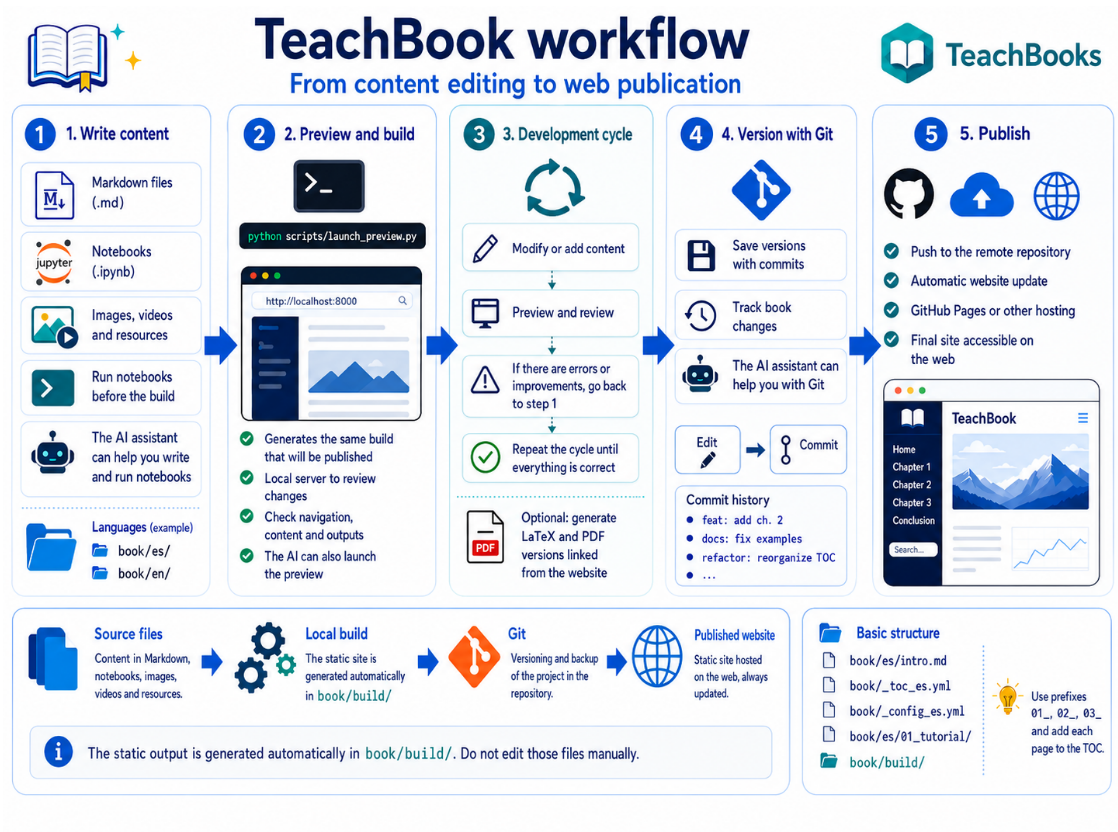


Figure2.1: TeachBook workflow from content editing to web publication.

- `book/es/intro.md`: main page of the Spanish version.
- `book/en/intro.md`: main page of the English version.
- `book/_toc_es.yml` and `book/_toc_en.yml`: book tables of contents; they define the order of chapters and sections.
- `book/_config_es.yml` and `book/_config_en.yml`: language-specific configuration files.
- `book/_static/`: shared images, videos, styles, scripts, PDFs, and references.
- `book/es/01_tutorial/` and `book/en/01_tutorial/`: folders for the initial tutorial.

Good practices for keeping things organized:

- Use numerical prefixes in folders and files (`01_`, `02_`, `03_`) to reflect the sequence.
- Add each new file to the TOC of the corresponding language; if it is not in the TOC, it will not appear in the navigation.
- Keep names clear and consistent, for example `03_experimental_activity.md`.
- When you add or change content in Spanish, also review the equivalent English page.

The build results are generated in `book/_build/html/`, which is the static version of the book published on GitHub Pages. You do not need to edit anything in that folder, because it is generated automatically from the source files.

The `book/_static/` folder is intended for static files such as images, videos, CSS styles, JavaScript scripts, PDFs, and references. You can place your images there and reference them from Markdown files using relative paths. If you do not know how to do that, the AI agent can help you place the images and reference them correctly.

2.3 Directory Tree

```
teachbook_usal_template/
├── book/
│   ├── _config_es.yml
│   ├── _config_en.yml
│   ├── _toc_es.yml
│   ├── _toc_en.yml
│   ├── _static/
│   ├── _build/
│   ├── es/
│   │   ├── intro.md
│   │   └── 01_tutorial/
│   │       ├── 01_que_es_un_teachbook.md
│   │       └── 02_flujo_trabajo.md
│   └── en/
│       ├── intro.md
│       └── 01_tutorial/
│           ├── 01_what_is_a_teachbook.md
│           └── 02_workflow.md
└── scripts/
    └── launch_preview.py
```

3. EDITING WITH AI

This course is aimed at people with any level of programming experience. The possibilities opened up by AI assistants are enormous, because they allow people with no programming experience to create content more easily and people with programming experience to work faster and more efficiently.

You can edit this book in two ways:

- **Manually**: by editing Markdown files (`.md`) or Notebooks (`.ipynb`) directly with a text editor or an IDE such as VS Code. This also involves running scripts to build and preview the book, committing changes with Git, and pushing them to GitHub to publish updates.
- **With AI**: by using an AI assistant to help you write content, generate code, review spelling, create graphs, and so on. You can ask the AI to perform specific tasks or help you maintain the structure of the book while you write. We are no longer talking about code autocompletion, but about an assistant that understands the context of the book and helps you create quality content without needing to know how to program.

For this to work well, the project must be well structured and the AI must have all the information it needs to help you. That is why this template is designed to be easy to use with AI assistants, with a clear structure and simple scripts for building and publishing the book.

Vibe Coding: a development methodology that emphasizes a fluid programming experience through integrated AI assistants. It consists of writing code intuitively, letting AI agents edit the code and propose solutions while you keep control of the creative flow. The main problem with this methodology is that it can lead to excessive dependence on AI, which could affect code quality and the programmer's ability to solve problems independently. However, when used in a balanced way, it can significantly increase productivity and creativity.

As shown in Fig. 3.1, the term was coined by Andrej Karpathy in a post that framed this way of working as useful for quick, exploratory projects.

In addition, figures such as Linus Torvalds, closely associated with well-crafted code, have shown interest in this new way of programming, while also raising concerns about the quality of AI-generated code and the need to keep human control over the process. Fig. 3.2 illustrates that combination of practical interest and technical judgement.

3.1 How to Ask the AI for Help

This template is designed for use with AI assistants such as **GitHub Copilot** or **Antigravity**.

You can ask it things like:

- "Explain this Python code to me"
- "Generate a graph of a sine function"
- "Review the spelling of this paragraph"
- "Add a mathematical equation in LaTeX format"

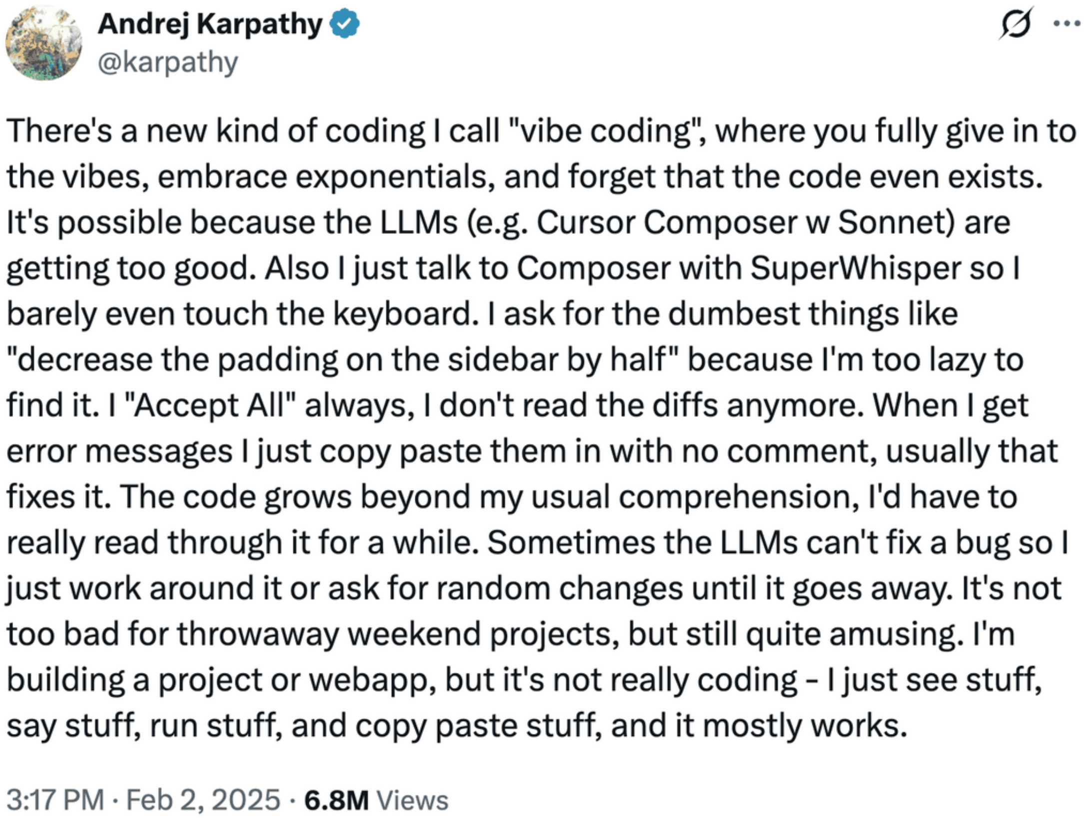


Figure3.1: Andrej Karpathy's post introducing the term *vibe coding*.

Figure3.2: Linus Torvalds' comment on an integration done with help from Google Antigravity.

Or even more complex tasks, such as:

- Starting from a folder with your resources in the repository (PDFs, images, articles, videos), helping you write a complete book chapter that follows the structure and style you have defined.
- Translating the new chapter you have just written into another language while preserving the book's format and structure.
- Running all the commands needed to build the project, deploy it locally for preview, commit the changes with Git, and push them to GitHub to publish the updates.
- Regenerating the book PDF in several languages.
- Publishing the book.

As mentioned earlier, this project contains several SKILLS. You can think of them as cheat sheets or mini-tutorials for specific tasks that AI agents have available. They are located in the `.github/skills/` folder and are kept synchronized in other folders so that the project is compatible with other AI agents such as Claude, Codex, and others.

Taking this idea one step further, you can create your own SKILLS **by asking the assistant to generate them itself following the project's approach**. For example, if you want a SKILL for generating graphs with a library specific to your discipline, or for reviewing the spelling of a paragraph according to a particular style guide, you can ask the AI to generate that SKILL using the format of the existing project SKILLS. This way, you can build a set of custom SKILLS adapted to your needs and to the needs of your project.

4. MY FIRST BOOK FROM SCRATCH

This page shows a first complete workflow for creating a new book from this template: create the repository, open it on your computer, ask the agent for help, preview locally, generate PDFs, and publish the website.

The goal is not to learn Git, Python, LaTeX, or GitHub in depth before starting. The goal is to understand what is happening on each screen, what you can ask the agent to do, and where to look when you need to check that everything worked.

> **Note**
>
> The GIFs on this page are sped up so the process can be shown compactly. The initial environment setup and PDF generation can take much longer, especially the first time the LaTeX engine is installed.

4.1 Step 0. Prepare the basic requirements

Before working with the template, check that the computer has the basic tools ready: a GitHub account, Python installed, Git available, and an editor with an AI agent, such as Antigravity or VS Code. As shown in Fig. 4.1, this step sets up the starting point before creating the repository.

GitHub will be the place where the book is stored on the internet. Python will run the scripts included in the template. Git is the tool that records the history of changes and sends them to GitHub. Antigravity or VS Code will be the workspace: you will open the book folder there, ask the agent to perform tasks, and review the files.

It is also important to configure Git for its first use. If you have never done it on that computer, open a terminal and run:

```
git config --global user.name "Your name"
git config --global user.email "your.email@example.com"
```

These details are attached to commits. A commit is a snapshot of the project at a specific moment. You do not write a password or token here: only your name and email address. The email should be associated with your GitHub account if you want GitHub to recognize those changes as yours.

If you do not know whether Git is already configured, you can check it with:

```
git config --global user.name
git config --global user.email
```

If those commands return your name and email, you can continue. If they return nothing, configure Git with the previous commands. This step usually only needs to be done once per computer.

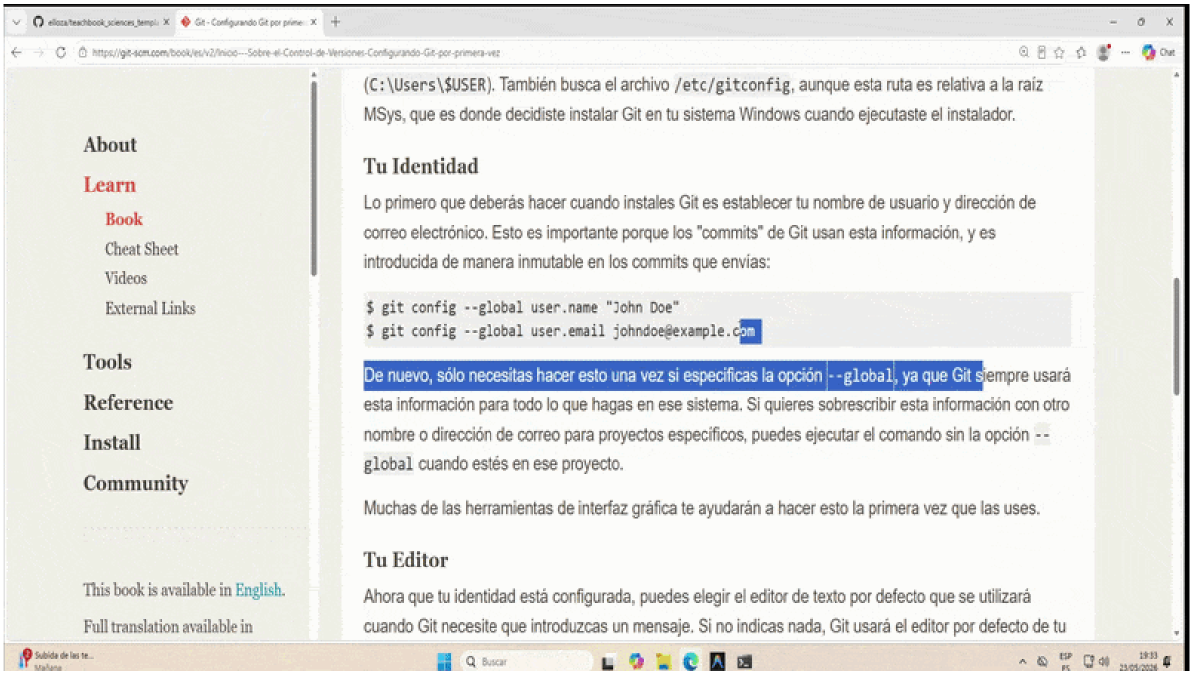


Figure4.1: Basic requirements: GitHub, Python, configured Git, and an editor with an AI agent.

4.2 Step 1. Create a repository from the template

The next step is to open the template URL on GitHub and use **Use this template** to create your own repository. In this first walkthrough we will make it **public**, as shown in Fig. 4.2, because that simplifies the initial publication with GitHub Pages. Private repositories can be considered later.

A repository is the project folder inside GitHub. When you use the template button, GitHub does not modify the original repository: it creates an independent copy in your account. That copy will be your book. You can change the title, remove examples, add chapters, and publish the website without affecting anyone else.

When GitHub asks for the repository details, pay special attention to three fields:

- **Owner**: the account or organization where the book will be created.
- **Repository name**: the technical name of the repository. Avoid spaces and unusual characters; for example, `my-first-book`.
- **Public**: for this first exercise, public is recommended because GitHub Pages is easier to activate.

After creating the repository, GitHub will take you to a page whose address looks similar to:

```
https://github.com/your-user/my-first-book
```

Keep that page in mind: it is the main GitHub page for your project. From there you can copy the clone URL, open **Settings**, inspect **Actions**, and check whether the website has been published correctly.

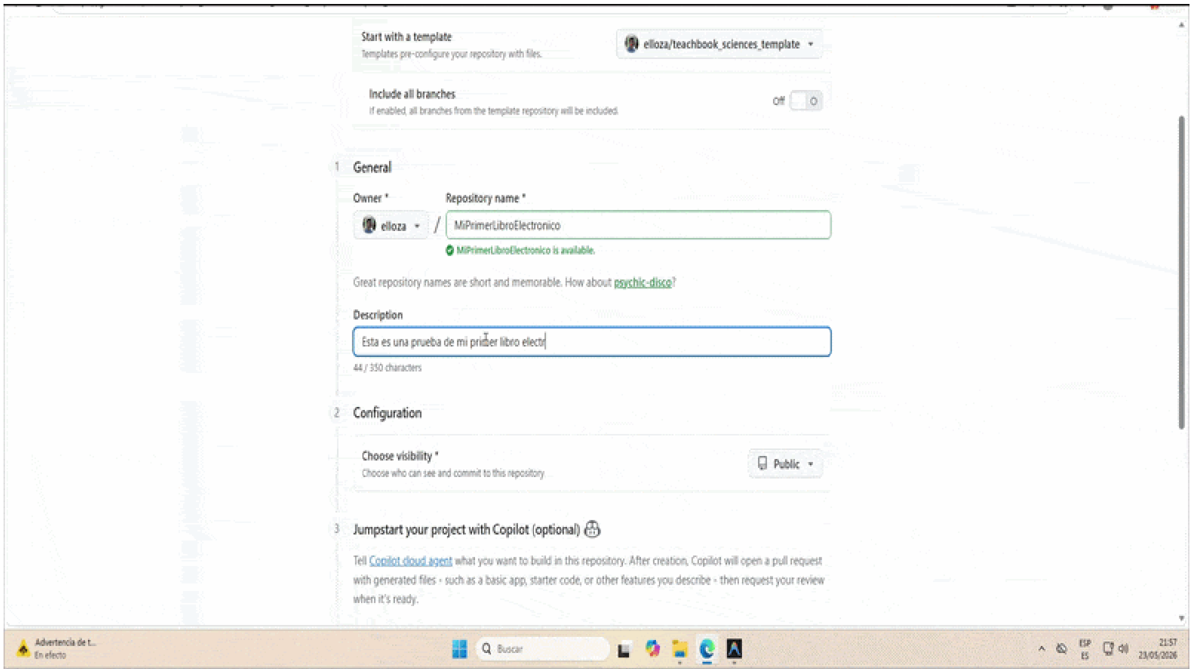


Figure4.2: Creation of a new public repository using GitHub's template button.

4.3 Step 2. Enable GitHub Actions and GitHub Pages

After creating the repository, allow GitHub to run the project workflows and publish the website. Fig. 4.3 shows the **GitHub Pages** configuration using **GitHub Actions** as the source.

GitHub Actions is GitHub's automation system. In this template it does in the cloud what you could do on your computer: install the environment, check the project, generate the PDFs, build the website, and publish it. GitHub Pages is the service that takes that built website and makes it available at a public URL.

To configure it, open your GitHub repository and go to:

```
Settings -> Pages -> Build and deployment -> Source -> GitHub Actions
```

If this is the first use of Actions in that account or repository, GitHub may ask for an additional confirmation to enable workflows. That detail may not appear in the GIF, but it must be accepted so the book can be built and published automatically.

The public website URL does not always appear immediately. It usually appears after the first successful deployment. There are two places where you can find it:

- In **Settings -> Pages**, GitHub usually shows a link to the published website once a successful deployment exists.
- In the **Actions** tab, open the deployment workflow run and look for the deployment summary. When everything is green, there is often a link to the published site.

The typical GitHub Pages URL for a project repository has this form:

```
https://your-user.github.io/my-first-book/
```

If the repository belongs to an organization, the first part changes to the organization name. If you later use a custom domain, such as `libro.usal.es`, that URL can be different.

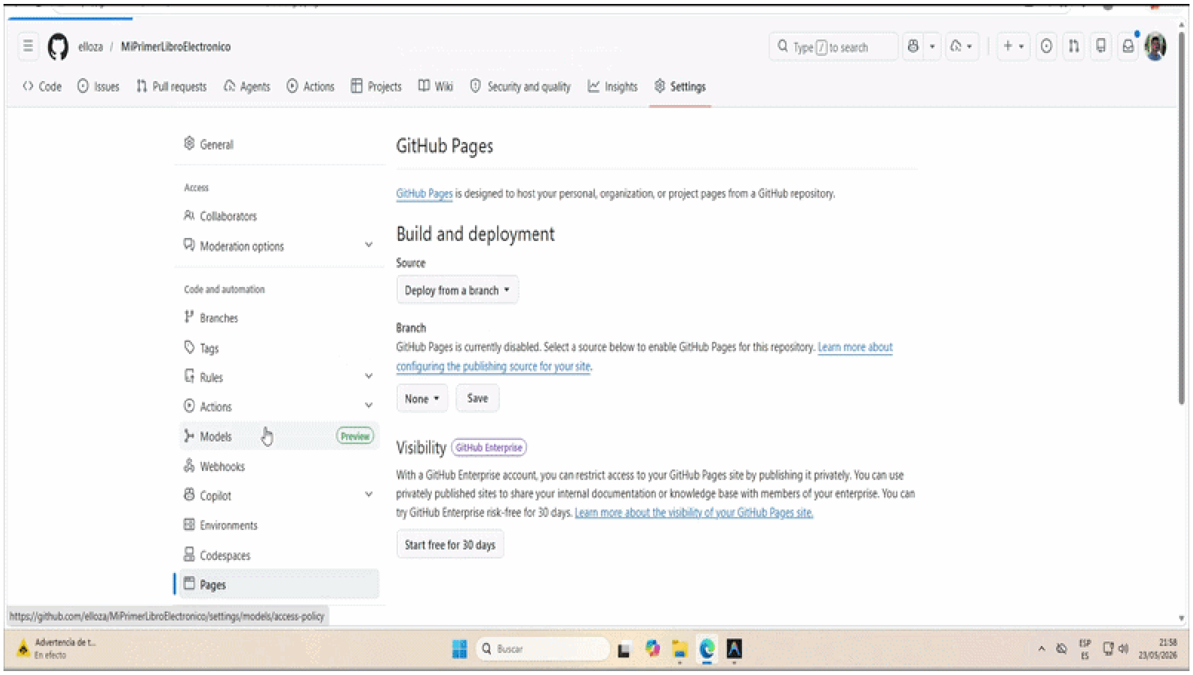


Figure4.3: Activation of GitHub Pages with GitHub Actions as the publication source.

4.4 Step 3. Clone the repository in Antigravity

With the repository created on GitHub, copy its URL and clone it from Antigravity. Fig. 4.4 shows the general flow: open the editor, choose the clone option, and paste the repository URL.

Cloning means downloading a complete copy of the repository to your computer. It is not just downloading a ZIP file: after cloning, the local folder remains connected to GitHub. That lets you work locally and, when you are satisfied, upload the changes with `commit` and `push`.

To find the URL you need, go back to the main page of the repository on GitHub and click the green **Code** button. In the **HTTPS** tab you will see an address similar to:

```
https://github.com/your-user/my-first-book.git
```

Copy that URL and paste it into Antigravity's clone option. The editor will ask where on your computer you want to store the project. Choose a folder that is easy to find, such as `Documents`, `Desktop`, or a folder dedicated to your courses.

At this point GitHub may ask for authentication. That is expected: the editor needs permission to download the repository and, later, to push changes back. If a browser window or GitHub sign-in screen appears, follow the authentication flow. Do not paste passwords into project files.

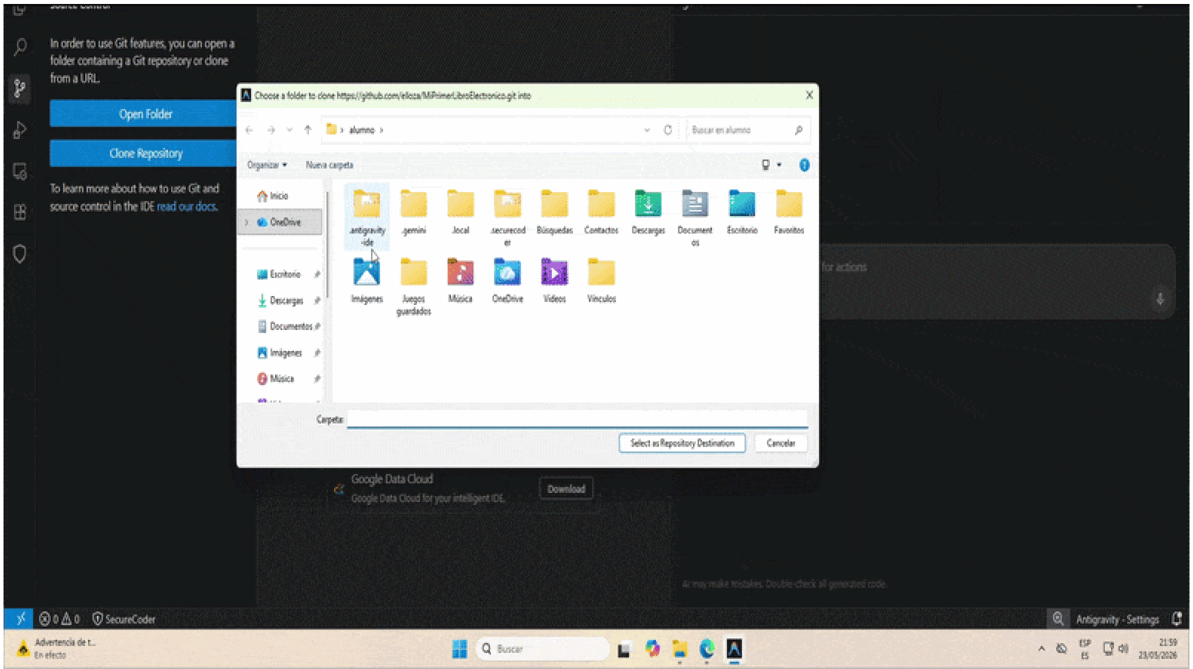


Figure4.4: Repository cloned from GitHub in Antigravity.

4.5 Step 4. Ask the agent to configure the environment

When the project is opened for the first time, ask the agent to prepare the environment. It is sensible to use a low-cost model, for example **Gemini 3.1 Pro (Low)** if available, because this task is fairly mechanical and may require several interactions.

A useful prompt would be:

```
Configure this project for the first time following the repository instructions.
Use the .venv environment and continue with the required installations.
```

The `.venv` environment is a local folder where the Python tools required by this book are installed. It prevents the project dependencies from being mixed with other programs on the computer. For a non-technical user, the key point is this: if the agent uses `.venv`, the project stays organized and the installation is easier to repeat on another machine.

As shown in Fig. 4.5, the agent will run commands and may ask several times whether it can continue. During this first setup, the usual answer is yes. It may ask for permission to install `uv`, create `.venv`, install dependencies, prepare LaTeX, or synchronize skills. Those questions are expected.

This step takes longer than the GIF. It depends on the connection, the computer, and whether Python, Git, or `uv` were already installed. During installation, avoid closing the editor or shutting down the computer. If it looks like nothing is happening for a few minutes, check whether the agent is waiting for confirmation in the terminal.

When it finishes, the project should normally have a `.venv` folder and the agent should be able to run the verification scripts. If something fails, do not try to fix it manually by copying random commands: ask the agent to read the error, review `AGENTS.md`, and run the setup again cleanly.

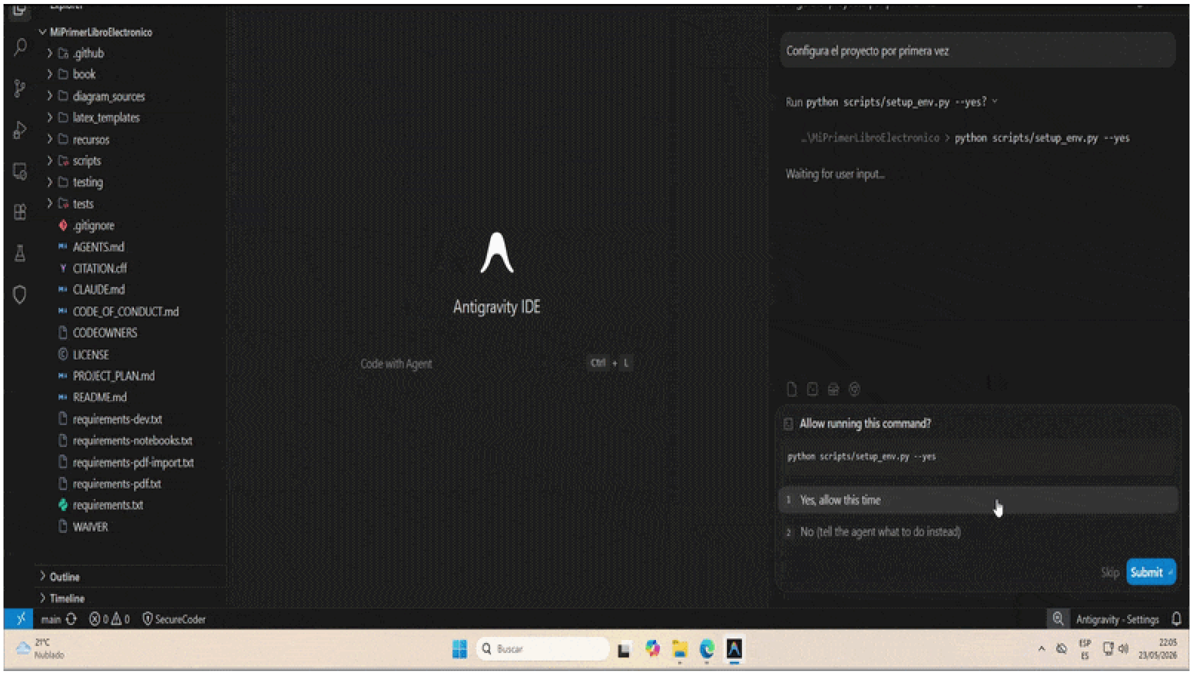


Figure4.5: Initial project environment setup through the agent.

4.6 Step 5. Make a first edit with the agent

When the environment is ready, ask for a first real change. In the example shown in Fig. 4.6, the agent is asked to hide the current template chapters and leave only a sample chapter about the **Central Limit Theorem**, written with LaTeX and MyST.

This first change helps you learn the workflow. You do not need to start by writing the whole book. It is better to request a small modification, check that it appears in the local website, and then publish it. That way you understand the full cycle before investing many hours in content.

The request should clearly state that the change must be made in both languages:

```
Temporarily hide the current template chapters and leave only one sample chapter
explaining the Central Limit Theorem with LaTeX and MyST.
Do it in Spanish and English and keep the TOCs synchronized.
```

There are two important ideas in that prompt. The first is **hide**, not delete. The agent can remove sections from the menu by commenting or adjusting the TOCs, but it is useful to keep the example files because they may serve as references later. The second is **in Spanish and English**. This template is designed for multiple languages, so every visible change should have its equivalent in the configured languages.

It is also useful to ask the agent to validate the result:

```
When you finish, check encoding, multi-language structure, and build the website.
```

If the agent modifies a Spanish page but forgets the English one, the book can become inconsistent. That is why the automatic checks are part of the normal workflow, not an extra for advanced users.

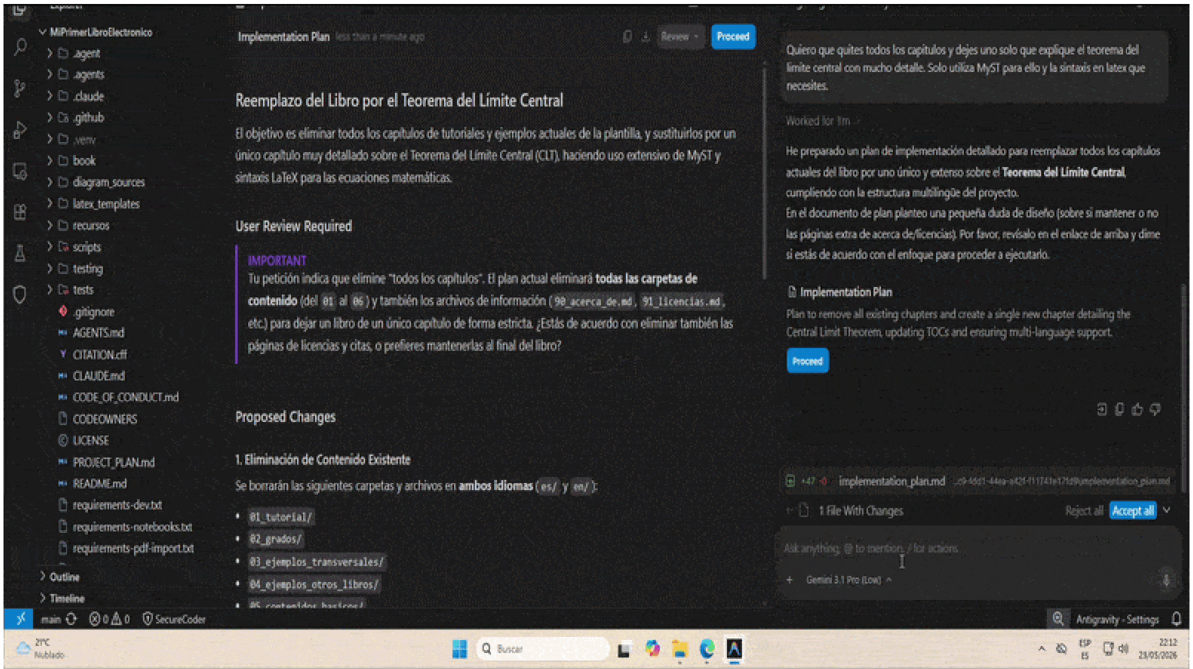


Figure4.6: First assisted edit: hide example content and leave a minimal chapter in both languages.

4.7 Step 6. View the local preview

After editing, ask the agent to launch a local preview. Fig. 4.7 shows how the agent starts a server on the computer and then reports the URL where the book can be opened.

You can ask for it simply:

```
I want to see a local preview of the book. Start the server and tell me the exact URL.
```

The URL is usually local, for example `http://127.0.0.1:8000` or `http://localhost:8000`. That means the website is not published on the internet yet: it is being served from your own PC so you can review changes before uploading them. That address only works on your computer while the local server is running.

It is important to distinguish these two URLs:

- `http://localhost:8000` or `http://127.0.0.1:8000`: local preview on your PC.
- `https://your-user.github.io/my-first-book/`: website published on GitHub Pages.

If port `8000` is already in use, the agent may choose another one, such as `8001` or `8080`. That is fine: open exactly the URL the agent gives you. If you see an old version, ask the agent to rebuild the book and restart the preview. In this template, the preview should be built with the same workflow as production, so you do not review an outdated website.

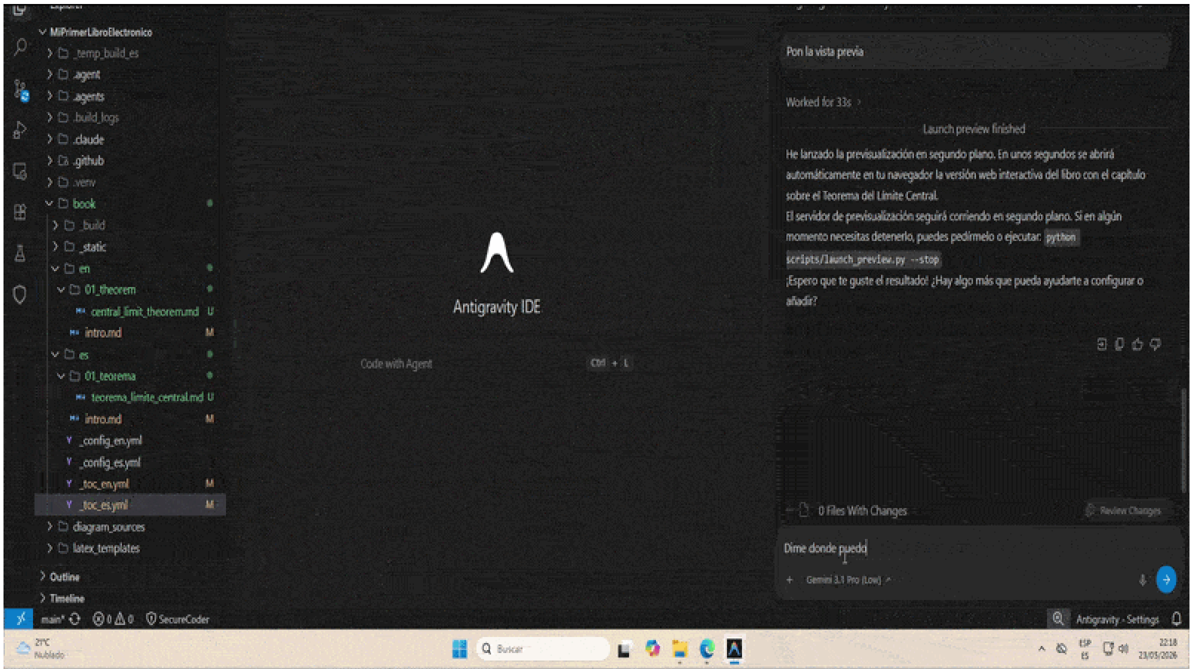


Figure4.7: Local book preview through a server running on the user's own computer.

4.8 Step 7. Generate the PDFs

When the local website looks right, ask the agent to generate the book PDFs. Fig. 4.8 shows the flow in compressed form, but this process may take longer than the GIF, especially the first time.

You can ask:

```
Generate the book PDFs in all languages and check that they were created correctly.
```

The template normally generates one PDF per language and places them in `book/_static/`, for example:

```
book/_static/teachbook_es.pdf
book/_static/teachbook_en.pdf
```

The reason for the initial wait is that the project needs a LaTeX engine to convert the book to PDF. LaTeX is the system that produces academic documents with good typography, equations, indexes, and references. You do not need to learn LaTeX to use the template, but the engine that compiles the document must be installed.

During the first generation it may look like many things are being installed. That is normal. Later runs are usually faster because the installation has already been prepared. If it fails, the important thing is to keep the error message and ask the agent to review it. It is not advisable to delete folders or reinstall tools manually without a diagnosis.

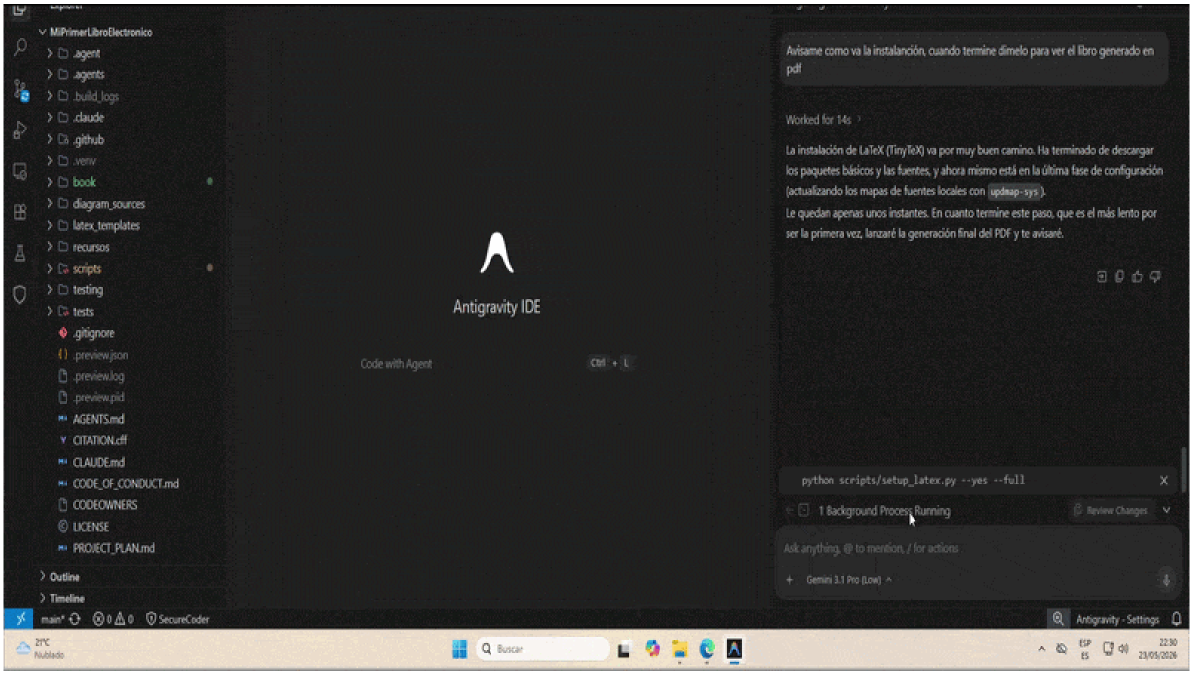


Figure4.8: Book PDF generation, including the initial LaTeX installation when required.

4.9 Step 8. Publish the changes

Finally, ask the agent to publish the changes. As shown in Fig. 4.9, publishing in this template means a complete Git cycle: add files, create a commit, and push to GitHub.

A good request would be:

```
Save the changes, commit, push, and check that the GitHub Pages deployment works.
```

The commit saves a specific version of the project with a descriptive message. The push uploads that version to GitHub. When it reaches GitHub, the deployment workflow starts. That workflow builds the book in the cloud, regenerates the PDFs, and publishes the website.

After the push, open your repository on GitHub and go to the **Actions** tab. There you will see a new workflow run. While it is running, it may appear in yellow or with a progress icon. When it finishes correctly, it appears in green. If it appears in red, the deployment failed and you need to open that run to read the error.

To find the published website URL, check these places:

- **Settings -> Pages**: after a successful deployment, GitHub usually shows the public site URL here.
- **Actions -> deployment run**: the run that published the website may include a deployment link.
- **Usual URL pattern**: if you do not use a custom domain, it will look similar to `https://your-user.github.io/my-first-book/`.

Keep in mind that the published website does not always update in the browser in the same second. If the workflow is green but you still see old content, wait a little and reload the page. You can also open a private browser window to avoid cache.

Because this template generates the PDFs first and then builds the website, the download buttons or links on the site should point to the updated PDFs. That is why manually uploading only HTML is not recommended: the full deployment keeps the website and printable documents synchronized.

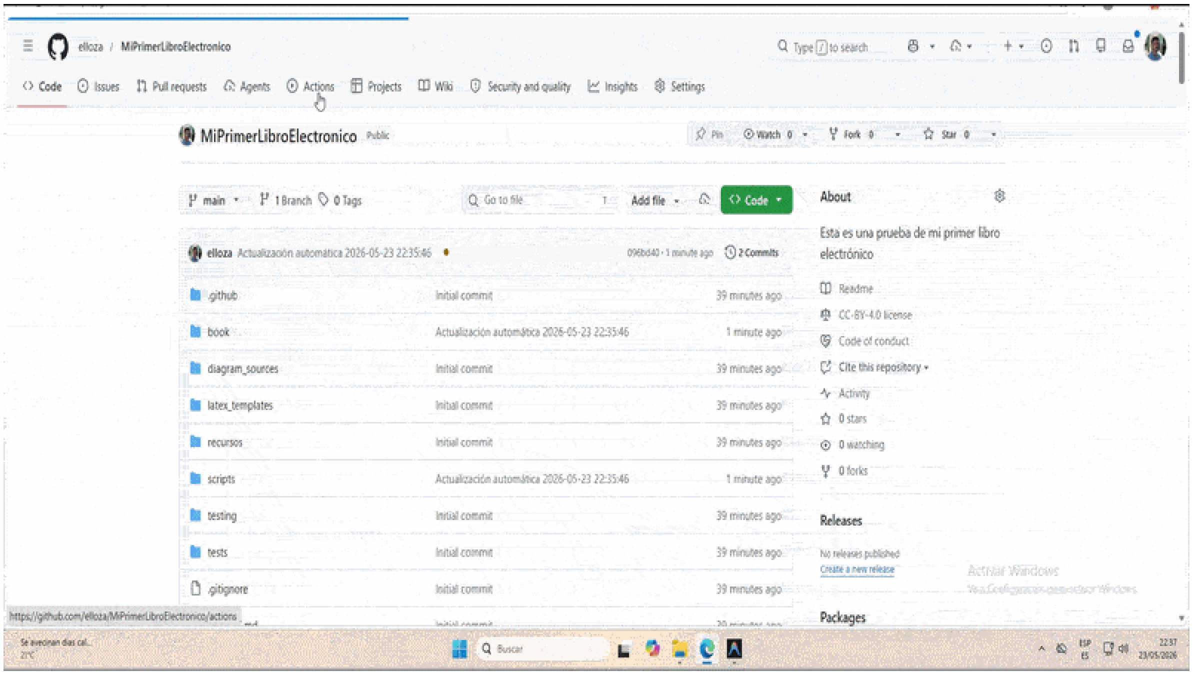


Figure4.9: Book publication through commit, push, and automatic deployment with GitHub Actions.

At the end of this walkthrough you have the basic full cycle: create the repository, edit with the agent, review locally, generate PDFs, and publish the website. From that point on, daily work consists of repeating the same pattern with smaller changes: write, preview, validate, and publish.

5. THE “X AS CODE” PHILOSOPHY

One of the important ideas we want you to take from this course is the possibility of editing and generating teaching artifacts through programming languages and markup languages. This is what we call the “X as code” philosophy. It means using code as a medium to describe and generate educational resources, so that we can benefit from the fact that LLMs can understand and manipulate that code to help us create quality materials without manually editing every artifact.

5.1 What is “X as code”?

When we talk about “X as code”, we mean the idea that an educational resource (X) can be described and generated through a programming language or a markup language. Instead of creating a diagram, plot, animation, or quiz directly in a visual tool, we describe it with code. Then that code is compiled or executed to generate the final artifact.

The idea is to use programming languages and markup languages as **intermediate languages** for generating teaching artifacts: images, diagrams, animations, videos, simulations, quizzes, tables, notebooks, and PDFs.

In other words: working with **X as code**.

- *Diagrams as code*: diagrams written as text.
- *Figures as code*: plots and figures generated from data or formulas.
- *Animations as code*: educational videos rendered from a programmed scene.
- *Quizzes as code*: questions, solutions, and exercise banks defined as structured text.
- *Books as code*: a complete book compiled from Markdown files, notebooks, and configuration.

Key idea

A teaching artifact generated from code is not edited “by hand” at the end of the process. It is described with an intermediate language, where AI assistants can understand the structure and content, and then the final result is generated through a compilation or execution process.

5.2 Why is this philosophy interesting?

One of the problems with generating content using AI tools is that the final result is often an image, PDF, or video that cannot be edited easily. If you want to change something, you have to regenerate the artifact from scratch or edit it manually, which can be complicated and time-consuming.

In the end, the advantage AI offers for generating content quickly is lost if the result is not editable. In contrast, if the artifact is generated from code, you can ask the AI to edit that code to make changes, generate new versions, or adapt the content to different contexts.

Having this intermediate language is sometimes key for AI to help you generate quality content. For example, if you want to generate a flowchart, you can describe it with Mermaid or Kroki code. The AI can understand that description and help you modify, translate, or adapt it without needing to edit the final image.

5.3 Code as a bridge to the final result

When you draw a diagram directly in a visual tool, the result is locked inside that interface. If you want to change ten arrows, translate labels, or adapt the style, you have to edit it manually again.

When you describe that same diagram with code, an intermediate layer appears:

As shown in Fig. 5.1, the diagram is versioned as a static image.

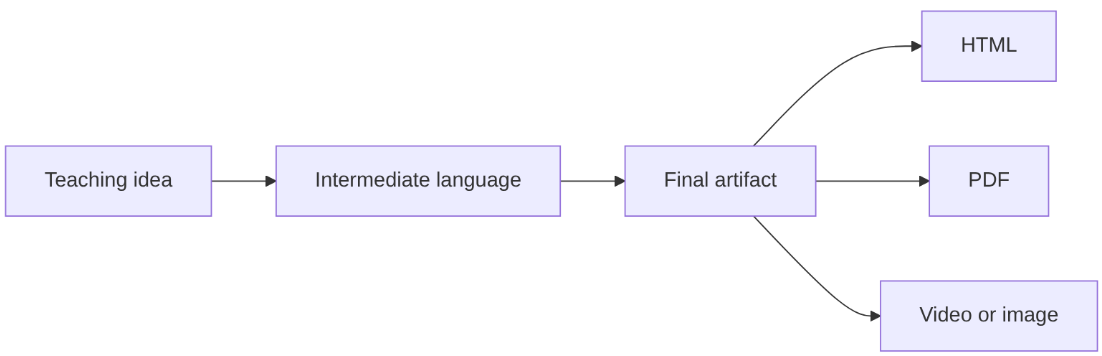


Figure5.1: Diagram: Code as a bridge to the final result.

The following table summarizes the main elements of this section.

Table. What artifacts can be generated from code?.

Final artifact	Intermediate language	Teaching use case
Diagram	Mermaid/Kroki, GraphViz, PlantUML	Concept map, experiment flow, process schema
Scientific plot	Python + matplotlib	Represent real or simulated data
Electrical circuit	Schemdraw or CircuitikZ	Generate consistent schematics for Physics problems
Animation or video	External tool + MP4	Explain a transformation, a wave, or a proof step by step without loading the base installation
Table or dataset	Python + pandas	Create clean tables from raw data
Quiz	MyST, JupyterQuiz, structured Markdown	Review questions with collapsible solutions
Executable notebook	Jupyter Notebook	Simulation, reproducible calculation, guided practice
Web book and PDF	Jupyter Book/TeachBooks	Publish the same material as a website, download, and printable version

5.4 Why this fits so well with AI

AI does not "understand" a final image the way it understands a code-based description. If you give it a screenshot of a diagram, it can suggest changes; if you give it the Mermaid or Kroki code that generates it, it can **make** those changes.

The practical advantage

We do not ask AI to edit pixels. We ask it to edit instructions. Then the system recompiles those instructions and produces the final artifact.

This makes it possible to ask for things like:

- "Turn this flowchart into a state diagram".
- "Generate an English version while keeping the same structure".
- "Adapt the example so it uses Biology data instead of Physics data".
- "Prepare the visual script for a 20-second animation showing this function growing".
- "Create three versions of the quiz with different difficulty levels".

The key is that the final result is not handcrafted every time. It is produced from an editable recipe.

5.5 Reproducibility, not just automation

Working "as code" does not mean making things more complicated. It means every artifact has a clear recipe:

1. **Input**: data, formulas, text, or a teaching idea.
2. **Intermediate language**: Markdown, Python, Mermaid, LaTeX, etc.
3. **Compilation**: the system transforms that description.
4. **Output**: image, video, website, PDF, notebook, or quiz.

If you change the recipe, you regenerate the artifact. If something fails, you fix the recipe. If another teacher wants to adapt it, they do not start from scratch: they modify the recipe.

5.6 The mindset shift

The question stops being: “Which program should I use to draw this?”

The question becomes: “Which intermediate language best describes this artifact?”

Not every material needs code. But when a resource will be revised, translated, versioned, reused, or generated with AI support, it is worth asking whether it can be expressed as **X as code**.

That is the teaching-as-code philosophy: using code not as an end in itself, but as a **generative medium** between the teaching intention and the final artifact.

Part II

Examples by Degree

DEGREE IN PHYSICS

In this section, you will find examples adapted to Physics Degree courses. The goal is to show how to integrate:

- Complex mathematical formulas
- Plots of physical simulations
- Experimental data analysis code

6.1 Physics Example: Harmonic Oscillator

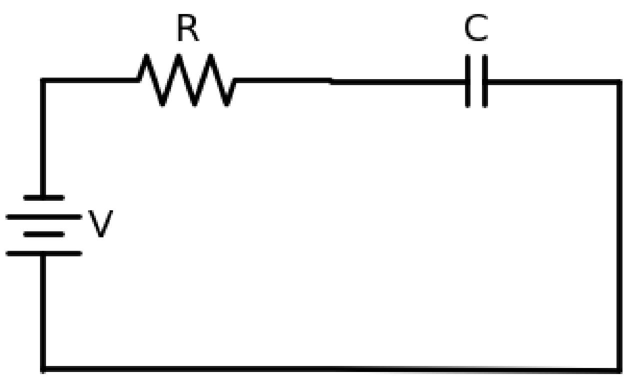
This notebook simulates a simple harmonic oscillator.

```
import numpy as np
import matplotlib.pyplot as plt

# Parameters
k = 1.0  # Spring constant
m = 1.0  # Mass
omega = np.sqrt(k / m)
t = np.linspace(0, 20, 100)

# Position
x = np.cos(omega * t)

# Plot
plt.figure(figsize=(8, 4))
plt.plot(t, x)
plt.title('Simple Harmonic Oscillator')
plt.xlabel('Time (s)')
plt.ylabel('Position (m)')
plt.grid(True)
plt.show()
```

6.2 Drawing Circuits with SchemDraw

In this example we draw a series RC circuit (battery + resistor + capacitor) using the `schemdraw` library. This type of circuit is fundamental in electronics and appears in low-pass filters, timers, and coupling circuits.

```
import schemdraw
import schemdraw.elements as elm

with schemdraw.Drawing() as d:
    battery = d.add(elm.Battery().up().label('V', loc='bot'))
    d.add(elm.Resistor().right().label('R'))
    d.add(elm.Capacitor().right().label('C'))
    d.add(elm.Line().down())
    d.add(elm.Line().tox(battery.start).left())
```



6.2.1 Circuit Components

- **Battery (V):** provides the electromotive force that drives the current.
- **Resistor (R):** limits current flow and dissipates energy as heat.
- **Capacitor (C):** stores energy as an electric field between its plates.

The time constant of the circuit is $\tau = RC$, which determines the charging and discharging speed of the capacitor.

```
R = 1000     # 1 kOhm
C = 1e-6     # 1 uF
tau = R * C
print(f'Time constant = RC = {tau*1e3:.2f} ms')
print(f'Time for full charge (~5 tau) = {5*tau*1e3:.2f} ms')
```

```
Time constant = RC = 1.00 ms
Time for full charge (~5 tau) = 5.00 ms
```

6.2.2 Conclusion

With `schemdraw` we can generate clean, professional circuit diagrams. The time constant $\tau = RC$ is the key parameter: after a time 5τ the capacitor is considered fully charged (≈99.3% of the source voltage).

6.3 Advanced CircuitikZ for Precise Circuits

CircuitikZ is the advanced option for drawing circuits with LaTeX-quality output. If you want highly precise schematics, similar to technical notes or scientific papers, this is the right tool.

> **Practical recommendation**
>
> Use **SchemDraw** for most teaching examples. Use **CircuitikZ** when you need a more formal and precise finish.

6.3.1 Recommended workflow in TeachBook

In this project, CircuitikZ is not inserted directly as `raw latex` in the page. Instead:

1. You write the circuit in a `.tex` file
2. You convert it to an image with a script
3. You insert the image with `{figure}`

This makes it work in both HTML and PDF.

6.3.2 Example source file

File `rc_circuit.tex`:

```
\begin{circuitikz}
\draw
  (0,0) to[battery1,l=$V$] (0,3)
  to[R,l=$R$] (3,3)
  to[C,l=$C$] (3,0)
  -- (0,0);
\end{circuitikz}
```

6.3.3 Command to render it

```
python scripts/render_circuitikz.py rc_circuit.tex book/_static/generated/rc_circuit.
↪png
```

6.3.4 Insert it in a MyST page

````
```{figure} ../../../_static/generated/rc_circuit.png
:alt: RC circuit generated with CircuitikZ
:width: 70%
:align: center
:name: fig-advanced-circuitikz-1

RC circuit generated with CircuitikZ.
```
````

6.3.5 Rendered result

The Fig. 6.1 visually summarizes this part of the explanation.

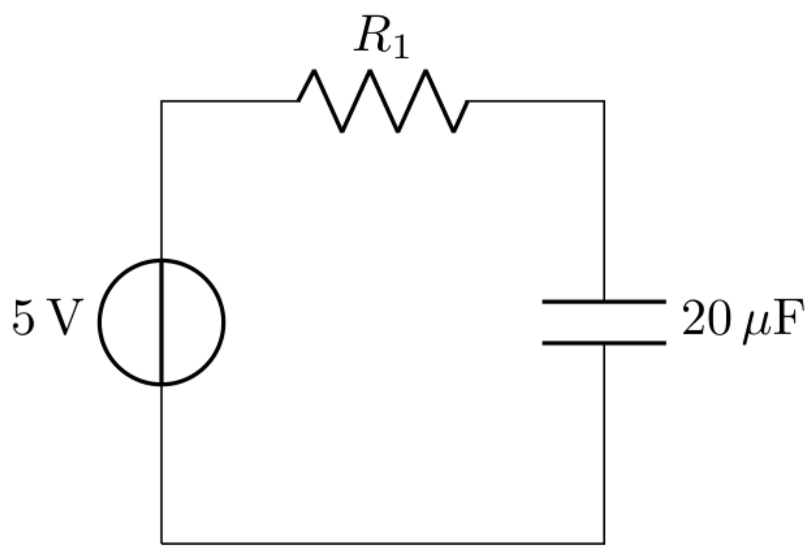


Figure6.1: RC circuit generated from a `.tex` file with CircuitikZ and converted to a PNG image.

The advantage of this workflow is that the final circuit behaves like a normal image: it appears on the website, is included in the PDF, and does not require the reader to have LaTeX installed.

6.3.6 When to use CircuitikZ

- formal electrical schematics
- advanced electronics notes
- material with a technical-paper style finish

6.3.7 When to use SchemDraw instead

- quick classroom examples
- introductory material
- situations where you want to edit the circuit with less friction

6.3.8 Prerequisite

If you do not have Tectonic installed yet:

```
python scripts/setup_latex.py --yes
```

6.4 Technical Diagrams with Kroki for Physics

Kroki is useful for **block diagrams, signal flow, timing diagrams, and conceptual schematics**. For **electrical circuits with standard symbols**, the recommended tool is **CircuitikZ rendered as an image**.

6.4.1 When to use each tool

- **Use Kroki** for conceptual explanations, functional blocks, timing, instrumentation, and signal flow.
- **Use WaveDrom through Kroki** for digital signals, buses, clocks, and timing protocols.
- **Use CircuitikZ** for resistors, capacitors, sources, switches, and formal electrical schematics.
- **Use SchemDraw** when you want to build simple circuits from Python inside a notebook.

6.4.2 1. Mermaid: instrumentation chain

As shown in Fig. 6.2, the diagram is versioned as a static image.

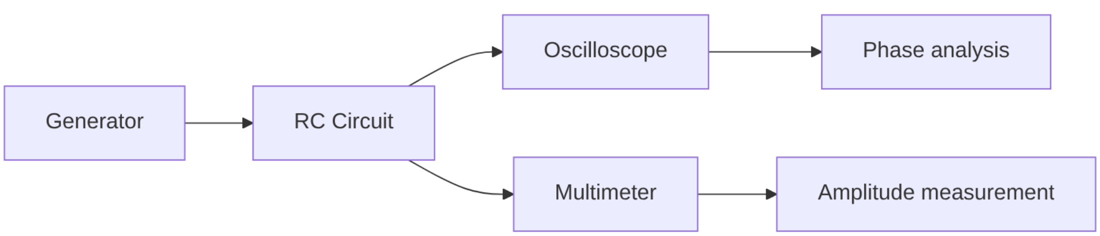


Figure6.2: Diagram: Mermaid: instrumentation chain.

As shown in Fig. 6.3, the diagram is versioned as a static image.

The Fig. 6.4 visually summarizes this part of the explanation.

```
\begin{circuitikz}
\draw
  (0,0) to[V, l=$5\,\mathrm{V}$] (0,3)
  to[R, l=$R_1$] (3,3)
  to[C, l=$20\,\mu\mathrm{F}$] (3,0)
```

(continues on next page)

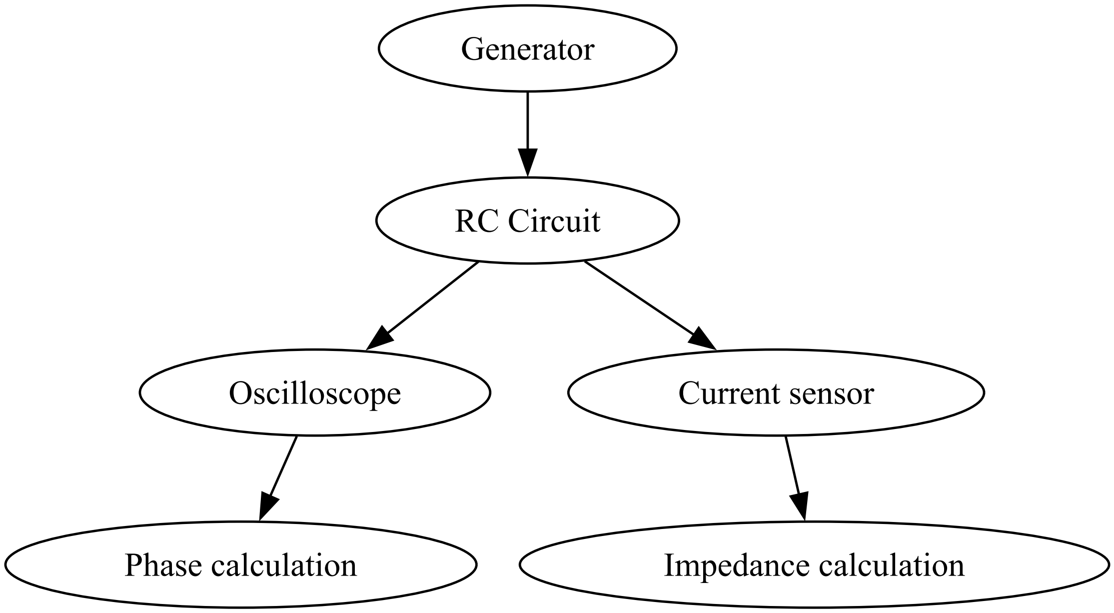


Figure6.3: Diagram: GraphViz: signal flow in an experimental setup.

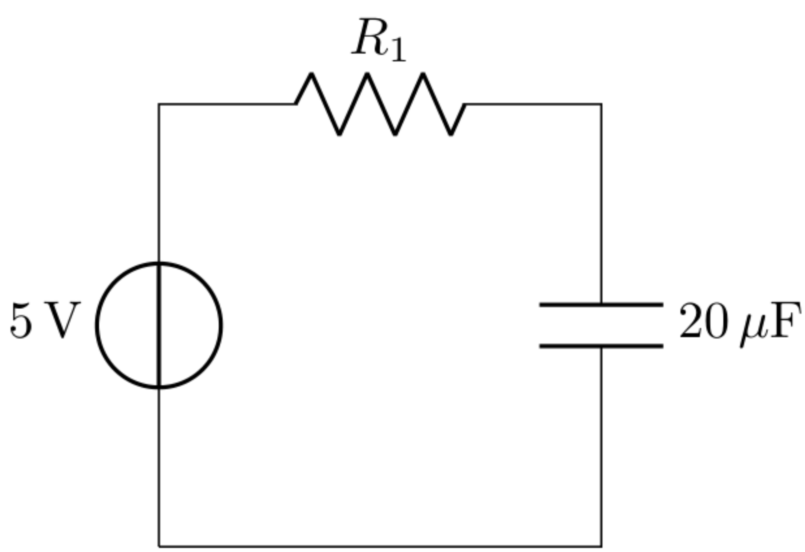


Figure6.4: RC circuit generated from CircuitikZ code.

(continued from previous page)

```
  -- (0,0);
\end{circuitikz}
```

Important

Kroki supports TikZ, but the public Kroki service does not guarantee that the `circuitikz` package is available. For robust teaching circuits, this project renders CircuitikZ locally as an image.

6.4.3 4. Wavedrom: input and output signals

As shown in Fig. 6.5, the diagram is versioned as a static image.

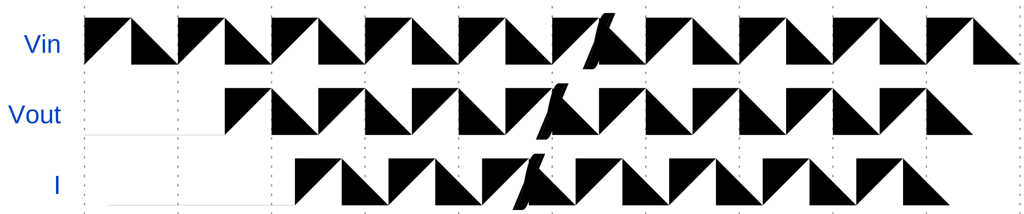


Figure6.5: Diagram: Wavedrom: input and output signals.

As shown in Fig. 6.6, the diagram is versioned as a static image.

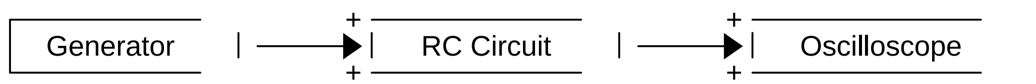


Figure6.6: Diagram: Ditaa: quick experimental bench sketch.

- **Kroki** = system-level visual explanation, blocks, flows, and timing diagrams.
- **CircuitikZ** = formal circuit with standard symbols and LaTeX-quality output.
- **SchemDraw** = programmable circuit from Python, ideal for teaching notebooks.

6.5 Ohm's Law: Linear Fit of V-I Data

We generate synthetic voltage-current data with noise and fit a linear regression $V = IR$ to verify Ohm's Law. We calculate the R^2 coefficient to evaluate the quality of the fit.

```
import numpy as np
import matplotlib.pyplot as plt

R_true = 220.0  # True resistance in
np.random.seed(42)

I = np.linspace(0.001, 0.05, 30)
```

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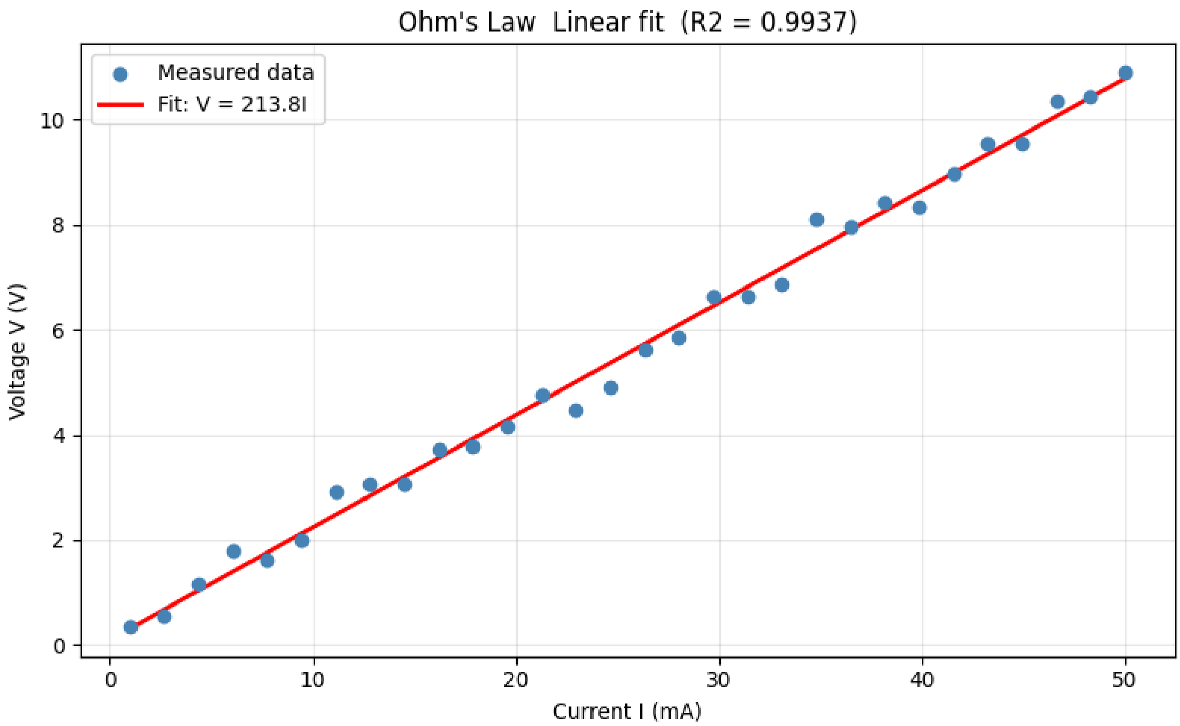
```
V_true = R_true * I
noise = np.random.normal(0, 0.3, size=V_true.shape)
V_measured = V_true + noise

coef = np.polyfit(I, V_measured, 1)
R_fitted = coef[0]
V_fit = np.polyval(coef, I)

SS_res = np.sum((V_measured - V_fit) ** 2)
SS_tot = np.sum((V_measured - np.mean(V_measured)) ** 2)
R2 = 1 - SS_res / SS_tot

print(f'True resistance:     {R_true:.1f} ')
print(f'Fitted resistance:   {R_fitted:.1f} ')
print(f'R2 = {R2:.6f}')
```

```
True resistance:     220.0
Fitted resistance:   213.8
R2 = 0.993681
```

6.5.1 Visualization

The plot shows experimental data (dots) and the fitted line. An R^2 close to 1 indicates that the linear model explains the variability in the data.

```
plt.figure(figsize=(8, 5))
plt.scatter(I * 1000, V_measured, color='steelblue', label='Measured data', zorder=3)
plt.plot(I * 1000, V_fit, 'r-', linewidth=2, label=f'Fit: V = {R_fitted:.1f}I')
plt.xlabel('Current I (mA)')
plt.ylabel('Voltage V (V)')
plt.title(f"Ohm's Law  Linear fit  (R2 = {R2:.4f})")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



6.5.2 Conclusion

The linear fit recovers the resistance value with great accuracy. The $R^2 \approx 1$ coefficient confirms that the $V = IR$ relationship is linear, validating Ohm's Law for this synthetic dataset.

6.6 Lissajous figures: sound, oscillations, and visualization

Goal. Connect simple harmonic motion with a two-dimensional visualization: Lissajous figures. The central idea is that two sinusoidal oscillations, one along the horizontal axis and one along the vertical axis, can produce closed curves whose shape reveals frequency and phase relationships.

This example replaces the second harmonic-oscillator example to avoid repeating the same one-dimensional model. The basic oscillator remains in the first Physics page; here we extend it to two perpendicular directions.

6.6.1 Historical context: seeing sound

Julio Mulero's article on the music of Lissajous figures develops a useful teaching idea: sound can be understood as vibration and therefore as a periodic signal that can be represented mathematically [Mul18]. A tuning fork produces an almost pure tone; to first approximation, its motion is sinusoidal.

Jules Antoine Lissajous, a nineteenth-century French physicist, looked for a way to visualize such vibrations. The historical setup attached small mirrors to tuning forks, reflected a light beam from them, and projected the result onto a screen. When the beam combined two perpendicular vibrations, the screen no longer showed a simple time wave, but a plane curve. That curve encoded the relation between the two oscillations.

If both frequencies were equal, segments, ellipses, or circles could appear depending on phase difference. If the frequencies had simple ratios, richer figures appeared. These curves therefore connect physics, trigonometry, acoustics, instrumentation, and visual aesthetics.

6.6.2 Mathematical model

A Lissajous figure can be described by the parametric equations

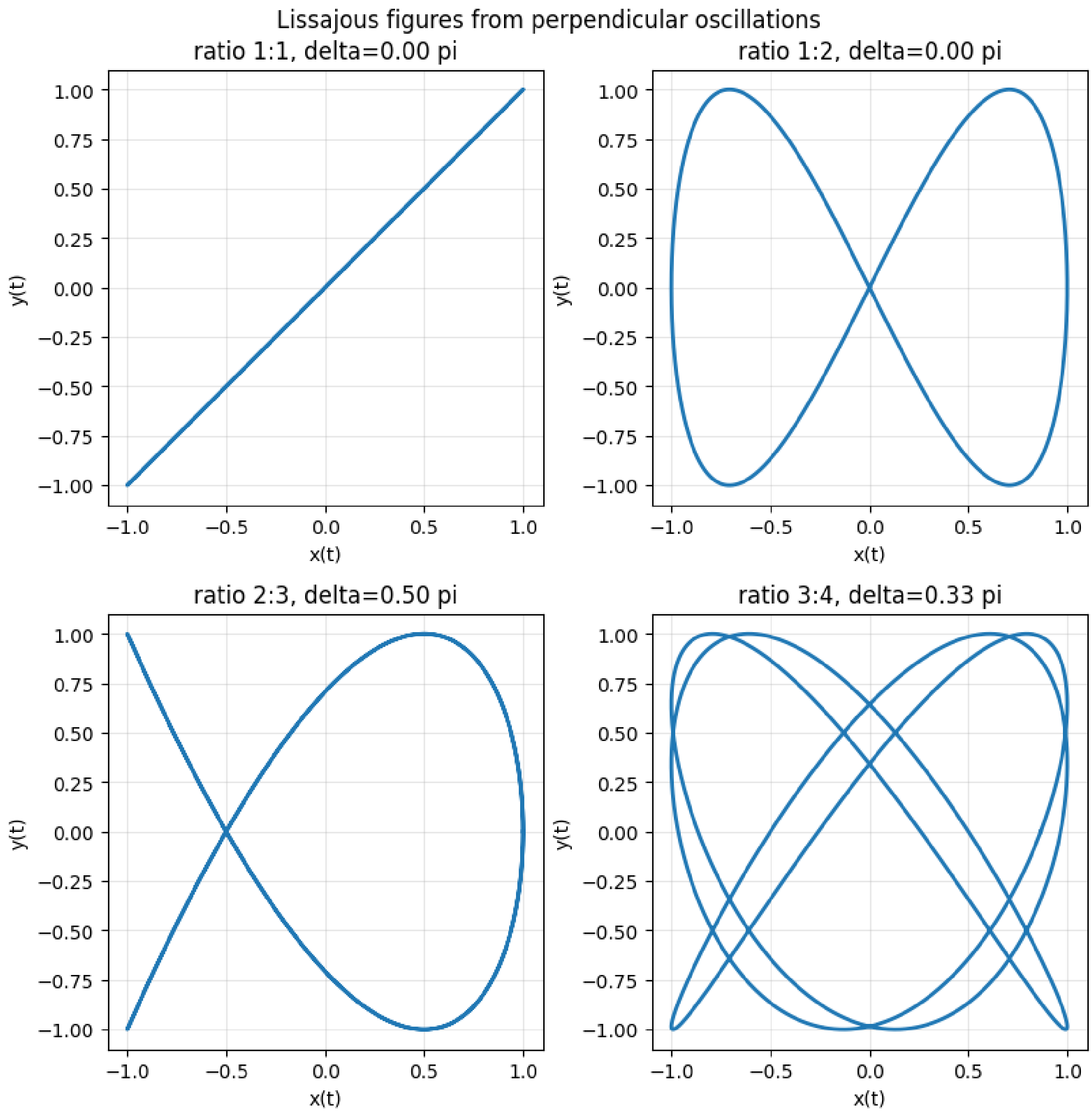
$$x(t) = A_x \sin(at + \delta), \qquad y(t) = A_y \sin(bt),$$

where A_x and A_y are amplitudes, a and b control the frequencies in each direction, and δ is the phase difference. The shape of the curve mainly depends on two elements:

- the frequency ratio $a : b$;
- the phase shift δ between the oscillations.

When a/b is rational, the trajectory closes after a finite number of oscillations. When the relation is not commensurable, the trajectory may not close over the observed interval.

```
import numpy as np
import matplotlib.pyplot as plt

# Generate several Lissajous curves from two sine motions.
t = np.linspace(0, 2 * np.pi, 2000)
examples = [(1, 1, 0.0), (1, 2, 0.0), (2, 3, np.pi / 2), (3, 4, np.pi / 3)]

fig, axes = plt.subplots(2, 2, figsize=(8, 8), constrained_layout=True)
for ax, (a, b, delta) in zip(axes.ravel(), examples):
    x = np.sin(a * t + delta)
    y = np.sin(b * t)
    ax.plot(x, y, linewidth=2)
    ax.set_aspect("equal")
    ax.set_title(f"ratio {a}:{b}, delta={delta / np.pi:.2f} pi")
    ax.set_xlabel("x(t)")
    ax.set_ylabel("y(t)")
    ax.grid(True, alpha=0.3)

plt.suptitle("Lissajous figures from perpendicular oscillations", y=1.02)
plt.show()
```



6.6.3 Two video demonstrations

Video 1. Lissajous's figures, Wheatstone apparatus for compounding two orthogonal oscillations

The first video shows a classical Wheatstone apparatus for compounding two orthogonal oscillations, an experimental setup close to the historical spirit of Lissajous. The source is the video published by the `fstfirenze` YouTube channel on 17 August 2015 [fstfirenze15].

Video 1: Lissajous's figures, Wheatstone apparatus for compounding two orthogonal oscillations
Channel: fstfirenze. Date: 2015-08-17. Source: https://www.youtube.com/watch?v=xPjNPb8h8Ok. [**?**]

Video 2. Figuras de Lissajous / Lissajous figures

The second video gives a direct visualization of Lissajous figures and helps identify how the shapes change when the signals vary. The source is the video published by the `PIE "oscar" UCM` YouTube channel on 18 June 2012 [PIEoscarUCM12].

Video 2: Figuras de Lissajous / Lissajous figures

Channel: PIE "oscar" UCM. Date: 2012-06-18. Source: https://www.youtube.com/watch?v=mLhXPBvllWw. [**?**]

6.6.4 Experiment with the parameters

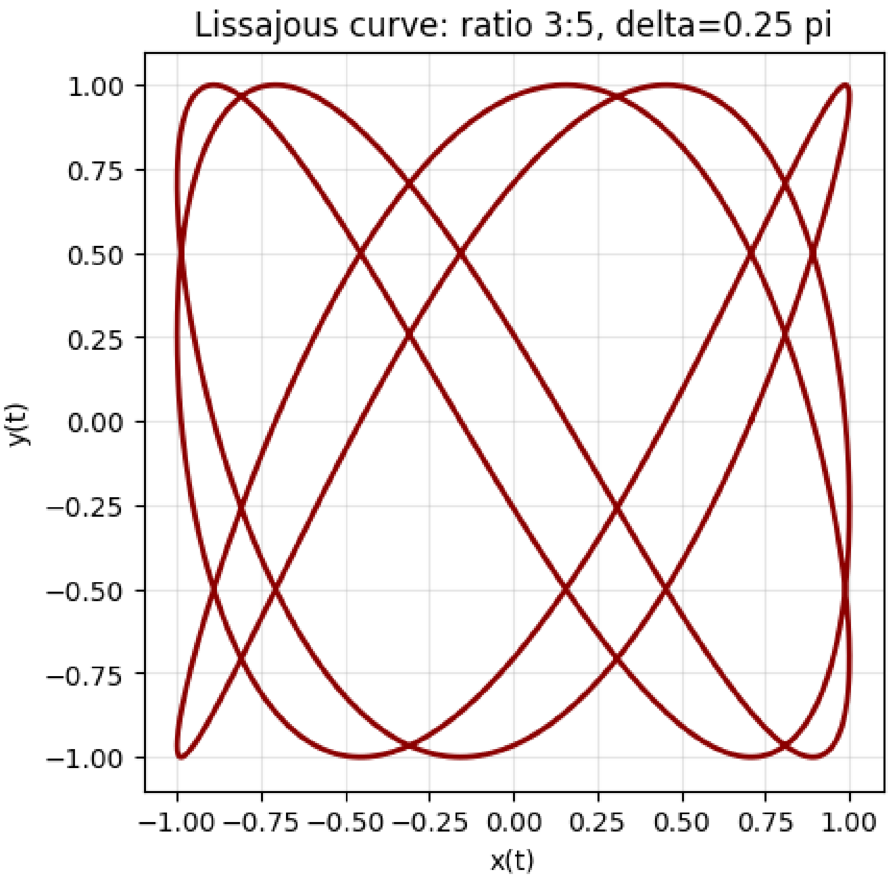
Change `a`, `b`, and `delta` in the next cell. Start with small integer values to observe closed curves, and then try different phase shifts such as `0`, `np.pi / 4`, `np.pi / 2`, and `np.pi`.

```python
import numpy as np
import matplotlib.pyplot as plt

# Change these three values and rerun the cell.
a = 3
b = 5
delta = np.pi / 4

t = np.linspace(0, 2 * np.pi, 3000)
x = np.sin(a * t + delta)
y = np.sin(b * t)

plt.figure(figsize=(5, 5))
plt.plot(x, y, color="darkred", linewidth=2)
plt.gca().set_aspect("equal")
plt.title(f"Lissajous curve: ratio {a}:{b}, delta={delta / np.pi:.2f} pi")
plt.xlabel("x(t)")
plt.ylabel("y(t)")
plt.grid(True, alpha=0.3)
plt.show()
```



6.6.5 Interpretation

Lissajous figures are not arbitrary drawings: they are geometric traces of two simple harmonic motions. In an oscilloscope in XY mode, for example, one signal controls the horizontal axis and another controls the vertical axis. If both signals are sinusoidal, the screen turns temporal relationships into geometry.

This representation has physical value. It can help estimate frequency ratios, detect phase shifts, and recognize whether two signals are synchronized. It also has cultural value: the same mathematical structure can appear as sound, a light curve, a laboratory experiment, or an artistic image.

6.6.6 Exercises

1. Change the ratio `a:b` to `1:3`, `2:5`, and `4:5`. Describe how the number of lobes changes.
2. Keep `a = b = 1` and try several values of `delta`. When do you get a line? When do you get a circle or an ellipse?
3. Explain why a simple frequency ratio produces a closed and stable figure.

DEGREE IN MATHEMATICS

Examples for mathematics courses. Priorities here include:

- Rigor in mathematical notation (LaTeX)
- Visual proofs
- Symbolic algorithms (SymPy)

7.1 Mathematics Example: Symbolic Calculus

We will use SymPy to differentiate and integrate functions.

```
from sympy import symbols, diff, integrate, sin, exp

x = symbols('x')
f = exp(-x) * sin(x)

# Derivative
df = diff(f, x)
print(f"Derivative: {df}")

# Indefinite Integral
int_f = integrate(f, x)
print(f"Integral: {int_f}")
```

7.2 Quadratic Equation

The quadratic equation is a useful example for combining mathematical explanation, formulas, and tables in a page that works in both HTML and PDF.

We start from the general form:

$$ax^2 + bx + c = 0, \qquad a \neq 0$$

The behavior of its solutions depends on the discriminant:

$$\Delta = b^2 - 4ac$$

When $\Delta \geq 0$, the real solutions are computed with:

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}$$

7.2.1 Interpreting the discriminant

Value of Δ	Type of solutions	Geometric interpretation
$\Delta > 0$	Two distinct real solutions	The parabola intersects the x-axis at two points
$\Delta = 0$	One repeated real solution	The parabola is tangent to the x-axis
$\Delta < 0$	Two complex conjugate solutions	The parabola does not intersect the x-axis

7.2.2 Example

For the equation:

$$x^2 - 5x + 6 = 0$$

we have $a = 1$, $b = -5$, and $c = 6$. Therefore:

$$\Delta = (-5)^2 - 4 \cdot 1 \cdot 6 = 1$$

Since $\Delta > 0$, there are two real solutions:

$$x_1 = \frac{5+1}{2} = 3, \qquad x_2 = \frac{5-1}{2} = 2$$

Tip

This static example is intentionally lightweight: it requires no video dependencies and exports reliably to both the website and the PDF.

7.3 Newton's Method for Finding Roots

The Newton-Raphson method is an iterative algorithm for finding roots of $f(x) = 0$:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

We apply it to the equation $x^3 - 2x - 5 = 0$, which has a real root near $x = 2$.

```
import numpy as np
import matplotlib.pyplot as plt

def f(x):
    return x**3 - 2*x - 5

def df(x):
    return 3*x**2 - 2

x0 = 2.0
steps = [x0]
for _ in range(5):
    x_new = steps[-1] - f(steps[-1]) / df(steps[-1])
    steps.append(x_new)
```

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```
for i, x in enumerate(steps):
    print(f'x_{i} = {x:.10f}   f(x) = {f(x):.2e}')
```

```
x_0 = 2.0000000000   f(x) = -1.00e+00
x_1 = 2.1000000000   f(x) = 6.10e-02
x_2 = 2.0945681211   f(x) = 1.86e-04
x_3 = 2.0945514817   f(x) = 1.74e-09
x_4 = 2.0945514815   f(x) = -8.88e-16
x_5 = 2.0945514815   f(x) = -8.88e-16
```

7.3.1 Convergence Visualization

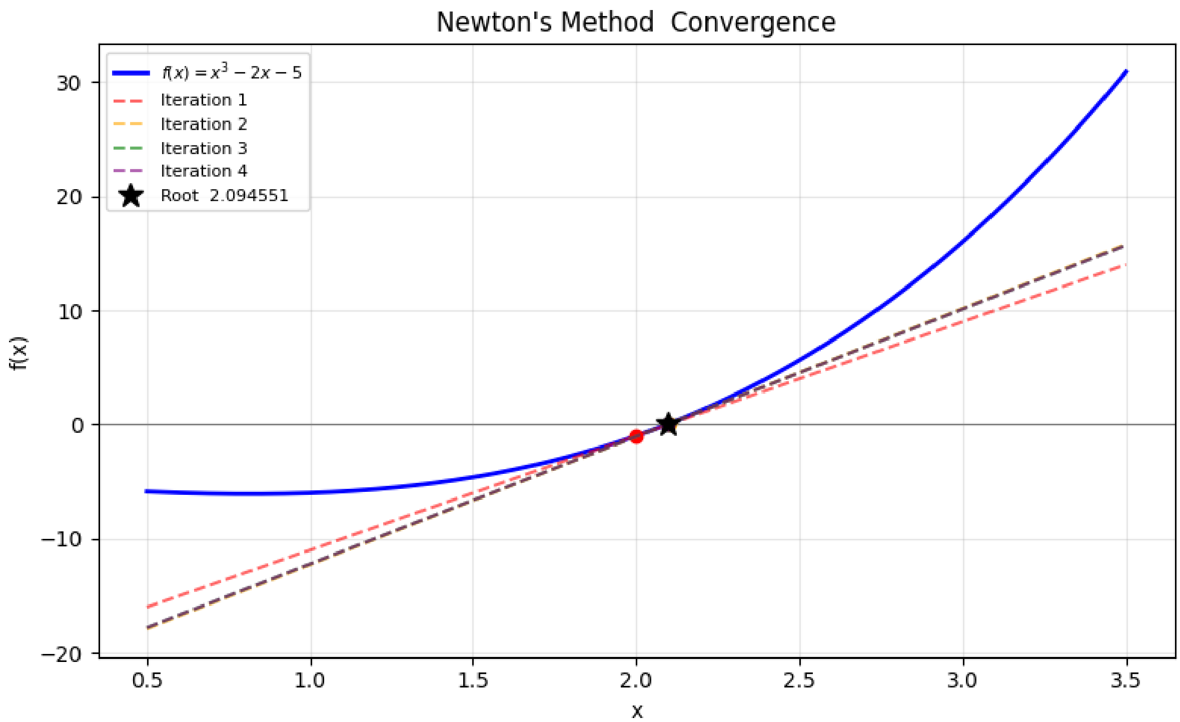
The plot shows the function $f(x)$ and the tangent lines used at each iteration. Each tangent intersects the x-axis at the next approximation of the root.

```
x_plot = np.linspace(0.5, 3.5, 300)

plt.figure(figsize=(8, 5))
plt.plot(x_plot, f(x_plot), 'b-', linewidth=2, label=r'$f(x) = x^3 - 2x - 5$')
plt.axhline(0, color='gray', linewidth=0.8)

colors = ['red', 'orange', 'green', 'purple']
for i in range(min(4, len(steps)-1)):
    xi = steps[i]
    tangent = f(xi) + df(xi) * (x_plot - xi)
    plt.plot(x_plot, tangent, '--', color=colors[i], alpha=0.6,
             label=f'Iteration {i+1}')
    plt.plot(xi, f(xi), 'o', color=colors[i], markersize=6)

plt.plot(steps[-1], 0, 'k*', markersize=12, label=f'Root  {steps[-1]:.6f}')
plt.xlabel('x')
plt.ylabel('f(x)')
plt.title("Newton's Method  Convergence")
plt.legend(fontsize=8)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



7.3.2 Conclusion

Newton's method converges very quickly (quadratic convergence): in just 4-5 iterations we reach a precision of 10^{-10}. A good starting point is required, and $f'(x_n) \neq 0$ must hold at each step.

7.4 2D Linear Transformations

A linear transformation in $\mathbb{R}^2$ is represented by a 2×2 matrix. Applied to the unit square, it lets us visualize how the transformation distorts space. We explore three classic examples: rotation, scaling, and shear.

```
import numpy as np
import matplotlib.pyplot as plt

square = np.array([[0,0],[1,0],[1,1],[0,1],[0,0]]).T

theta = np.pi / 4
R_rot = np.array([[np.cos(theta), -np.sin(theta)],
                   [np.sin(theta),  np.cos(theta)]])
R_scl = np.array([[2, 0], [0, 0.5]])
R_shr = np.array([[1, 0.5], [0, 1]])

transforms = [
    (R_rot, 'Rotation 45'),
    (R_scl, 'Scaling (2, 0.5)'),
    (R_shr, 'Shear'),
]
```

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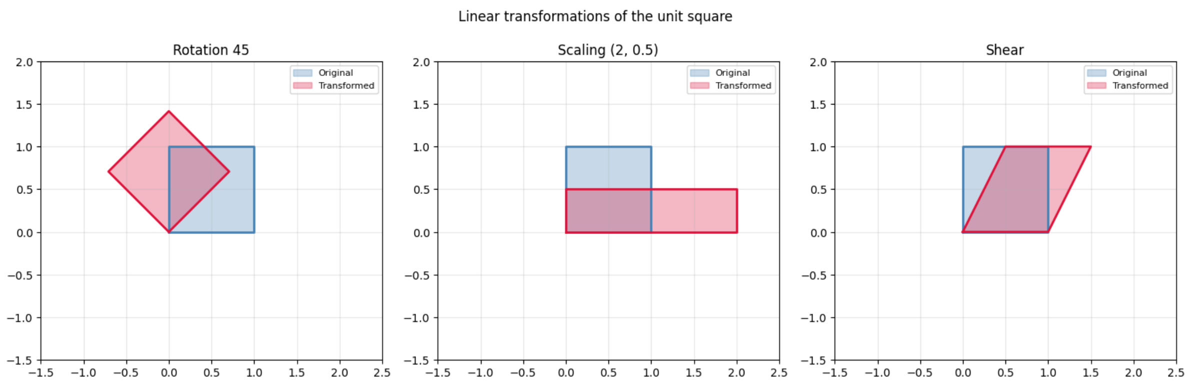
```
fig, axes = plt.subplots(1, 3, figsize=(15, 4.5))
for ax, (M, title) in zip(axes, transforms):
    transformed = M @ square
    ax.fill(square[0], square[1], alpha=0.3, color='steelblue', label='Original')
    ax.plot(square[0], square[1], 'steelblue', linewidth=2)
    ax.fill(transformed[0], transformed[1], alpha=0.3, color='crimson', label=
↪'Transformed')
    ax.plot(transformed[0], transformed[1], 'crimson', linewidth=2)
    ax.set_xlim(-1.5, 2.5)
    ax.set_ylim(-1.5, 2.0)
    ax.set_aspect('equal')
    ax.set_title(title)
    ax.legend(fontsize=8)
    ax.grid(True, alpha=0.3)

plt.suptitle('Linear transformations of the unit square', y=1.02)
plt.tight_layout()
plt.show()
```



7.4.1 Interpretation

- **Rotation:** preserves areas and angles (orthogonal transformation, $\det = 1$).
- **Scaling:** stretches one axis and compresses another ($\det = 2 \times 0.5 = 1$).
- **Shear:** shifts one axis proportionally to the other ($\det = 1$, preserves area).

```
for M, name in transforms:
    print(f'{name}:  det = {np.linalg.det(M):.2f}')
```

```
Rotation 45:  det = 1.00
Scaling (2, 0.5):  det = 1.00
Shear:  det = 1.00
```

7.4.2 Conclusion

The matrix determinant tells us whether the transformation preserves area ($|\det| = 1$), expands it ($|\det| > 1$), or contracts it ($|\det| < 1$). Visualizing the effect on the unit square is the most intuitive way to understand any linear transformation.

7.5 Planar Point Sets with Many Unit Distances

This page explains the manuscript *Planar Point Sets with Many Unit Distances* [OpenAI26] from the perspective of a mathematics course. The goal is not to reproduce every technical detail from algebraic number theory, but to make the architecture of the proof clear: an arithmetic construction in number fields eventually produces finite planar point sets with many pairs at distance exactly one.

7.5.1 The Unit-Distance Problem

Let $P \subset \mathbb{R}^2$ be a finite point set. Define

$$\nu(P) = \#\{\{p, q\} \subset P : \|p - q\|_2 = 1\},$$

the number of unordered pairs of points of P at Euclidean distance one. Also define

$$\nu(n) = \max_{|P|=n} \nu(P),$$

the largest possible number of unit distances among n planar points.

The conjecture discussed in the manuscript goes back to Erdős. In one asymptotic form, it predicted an absolute constant C such that, for all sufficiently large n,

$$\nu(n) \leq n^{1+C/\log\log n}. \tag{7.1}$$

The main result of the manuscript asserts the opposite: there is a fixed constant $\delta > 0$ and infinitely many values of n for which

$$\nu(n) \geq n^{1+\delta}. \tag{7.2}$$

Equation (7.2) is not merely a small improvement over a known estimate. It says that, along an infinite sequence of sizes, the number of unit distances grows with an exponent strictly larger than one.

Conceptual reading

The argument is not an elementary recipe for drawing a small example by hand. It is an existence proof: it builds planar point sets from infinite towers of number fields and then projects them to the plane.

7.5.2 Proof Map

The proof has two halves. The first is geometric: if we have enough algebraic norm-one elements, they can be interpreted as translations of length one. The second is arithmetic: it constructs number fields in which many such elements exist.

Table. Logical map of the argument.

Block	Main object	Contribution
Planar problem	Finite set $P \subset \mathbb{R}^2$	We want many pairs at distance 1
Geometry of numbers	Minkowski lattice in $\mathbb{C}^f$	Counts points and translations inside a bounded window
Arithmetic	Fields L, $K = L(i)$ and split primes	Produces many elements u with $u\,c(u) = 1$
Projection	One complex coordinate $\mathbb{C}^f \to \mathbb{C}$	Turns algebraic translations into planar unit segments

7.5.3 Admissible Data

The geometric part starts with an admissible datum:

- a totally real field L of degree $f = [L : \mathbb{Q}]$;
- the CM field $K = L(i)$, with relative complex conjugation c;
- distinct rational primes $q_1, \dots, q_t$, all congruent to 1 modulo 4 and splitting completely in L.

If $Q = \prod_b q_b$, the complete splitting of these primes produces many conjugate pairs of prime ideals in K:

$$\{P_s, cP_s\}, \qquad s = 1, \dots, m, \qquad m = tf.$$

For each binary vector $\varepsilon \in \{0, 1\}^m$, choose an ideal

$$A_\varepsilon = \prod_{\varepsilon_s = 1} P_s \prod_{\varepsilon_s = 0} cP_s.$$

There are 2^m choices but only $h(K)$ ideal classes. By the pigeonhole principle, many of those choices lie in the same ideal class. Comparing one such choice with another gives elements

$$u_\varepsilon = \frac{\alpha_\varepsilon}{c(\alpha_\varepsilon)}$$

which satisfy

$$u_\varepsilon\, c(u_\varepsilon) = 1. \tag{7.3}$$

Since L is totally real, under any complex embedding $\sigma : K \hookrightarrow \mathbb{C}$ the automorphism c becomes ordinary complex conjugation. Therefore (7.3) implies

$$|\sigma(u_\varepsilon)| = 1.$$

This is the key idea: the algebraic elements constructed in the proof have length one in every relevant complex coordinate.

7.5.4 Minkowski Window and Counting

Choose a Minkowski embedding

$$\Phi : K \longrightarrow \mathbb{C}^f, \qquad \Phi(x) = (\sigma_1(x), \dots, \sigma_f(x)).$$

The fractional ideal $D^{-1}\mathcal{O}_K$, with $D = Q^2$, becomes a lattice $\Lambda \subset \mathbb{C}^f$. The elements $u \in U$ constructed above belong to this lattice and have every coordinate of modulus one.

Now take a window

$$B_R = \{z \in \mathbb{C}^f : \|z\|_\infty \le R\},$$

which is a product of discs of radius R. For a translate $a + \Lambda$, consider the points

$$X_a = (a + \Lambda) \cap B_R.$$

The manuscript averages over the torus $\mathbb{C}^f/\Lambda$. The intuition is simple: if the lattice is shifted at random, the expected number of points inside the window is proportional to the volume of B_R, and the expected number of pairs $(x, x + u)$ with $u \in U$ depends on the overlap of B_R with $B_R - u$.

If $\gamma = t \log 2 - \log H > 0$, this averaging gives a translate with many pairs:

$$E_a \geq e^{\gamma f/2} |X_a|. \tag{7.4}$$

Then $X_a \subset \mathbb{C}^f$ is projected to the first complex coordinate. The projection is injective on the lattice coset being used: if two points have the same first coordinate, their difference comes from an algebraic integer with one zero embedding, and hence it is zero. This gives a planar set

$$P = \pi_1(X_a) \subset \mathbb{C} \simeq \mathbb{R}^2.$$

Every pair counted in (7.4) projects to a segment of length one, because the first coordinate of each $u \in U$ has modulus one.

7.5.5 Controlling the Size of P

To turn the estimate into a statement about $n = |P|$, one must control how many points fit inside the window. The manuscript uses a packing argument: two distinct lattice points cannot be too close in every coordinate at once, because the product of their coordinate differences is controlled by the algebraic norm.

This gives a uniform exponential upper bound

$$|P| \leq e^{Bf}$$

for a constant B fixed by the parameters. Together with the unit-pair estimate, this gives

$$\nu(P) \geq \frac{1}{2} |P| e^{\gamma f/2}.$$

Since $|P| \leq e^{Bf}$, the factor $e^{\gamma f/2}$ can be rewritten as a positive power of $|P|$. Absorbing constants for large degree yields

$$\nu(P) \geq |P|^{1+\delta}$$

for some $\delta > 0$.

7.5.6 Arithmetic Construction of the Fields

The arithmetic step produces a sequence of totally real fields

$$F = F_0 \subset F_1 \subset F_2 \subset \cdots$$

with degrees $f_j = [F_j : \mathbb{Q}]$ tending to infinity. The tower is everywhere unramified and has Galois groups of 3-power order. This matters for two reasons:

1. because the tower is unramified, the root discriminant stays controlled;
2. because the proof works with pro-3 groups, the generator-relation count can keep the tower infinite.

The starting point is a cyclic cubic field F built from cubic characters and cyclotomic fields. Rational primes q_b are then chosen by Chebotarev so that they split completely in every level of the tower. To impose this splitting, the proof kills Frobenius classes lying in the Frattini subgroup. The Golod-Shafarevich inequality ensures that, even after adding those relations, the resulting pro-3 group remains infinite.

Finally, one sets

$$K_j = F_j(i).$$

The root discriminant of K_j is bounded independently of j, and a Minkowski class-number estimate gives

$$h(K_j) \leq H^{f_j}.$$

The parameter t is chosen on the order of ℓ^2, while $\log H$ grows only like $O(\ell \log \ell)$. Therefore, for large ℓ,

$$t \log 2 - \log H > 0.$$

This inequality supplies the positive parameter $\gamma > 0$ used in the geometric part.

Student takeaway

The proof looks abstract because it combines combinatorial geometry, lattices, number fields, and pro-p groups. A useful way to read it is to follow the gains and losses: one gains 2^{tf} choices of ideals, loses a controlled class-number factor, and then turns the remaining exponential gain into many planar unit distances.

7.5.7 Bibliographic Context

The original problem and the classical unit-distance bounds are connected to Erdős, Kővári-Sós-Turán, Spencer-Szemerédi-Trotter, Székely, Ágoston-Pálvölgyi, and Guth-Katz [ErdHos46, GK15, KHovariSosTuran54, SSzemerediT84, Szekely97, AgostonPalvolgyi22]. Variants for norms and related convexity problems appear in work by Brass, Brass-Moser-Pach, Eisenbrand-Pach-Rothvoß-Sopher, Bílka-Buchin-Fulek-Kiyomi-Okamoto-Tanigawa-Tóth, Matoušek, Alon-Bucić-Sauermann, Greilhuber-Schildkraut-Tidor, Valtr, and Erdős-Fishburn [ABucicS25, Bra96, BMP05, BilkaBF+10, EPRothvossS08, ErdHosF97, GST25, Matouvsek11, Val05]. The algebraic part uses standard tools from class field towers, pro-p theory, Chebotarev, cyclotomic fields, discriminants, and class numbers [Dav00, DdSMS99, GS64, GS65, HM01, HMR21, Koc02, Lan94, Neu99, NSW08, RZ10, Sha63, Sha66, Tsc26, Was97].

7.5.8 Page References

DEGREE IN STATISTICS

Examples oriented towards data analysis and probability. Highlights:

- DataFrame handling (Pandas)
- Statistical visualization (Seaborn)
- Probabilistic models

8.1 Statistics Example: Data Generation

We generate a normal distribution and visualize its histogram.

```
import numpy as np
import matplotlib.pyplot as plt

data = np.random.normal(0, 1, 1000)

plt.hist(data, bins=30, alpha=0.7, color='green', density=True)
plt.title('Normal Distribution Histogram')
plt.xlabel('Values')
plt.ylabel('Density')
plt.show()
```

8.2 Linear Regression with Synthetic Data

We generate bivariate data with an underlying linear relationship and Gaussian noise. We fit a linear regression model using least squares and evaluate the fit quality with R^2 and residual analysis.

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(12)
n = 50
x = np.linspace(2, 12, n)
y_true = 3.5 * x + 10
y = y_true + np.random.normal(0, 5, n)

coef = np.polyfit(x, y, 1)
y_fit = np.polyval(coef, x)
residuals = y - y_fit
```

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```
SS_res = np.sum(residuals**2)
SS_tot = np.sum((y - np.mean(y))**2)
R2 = 1 - SS_res / SS_tot

print(f'Fitted equation: y = {coef[0]:.3f}x + {coef[1]:.3f}')
print(f'R2 = {R2:.4f}')
```

```
Fitted equation: y = 3.350x + 9.963
R2 = 0.7609
```

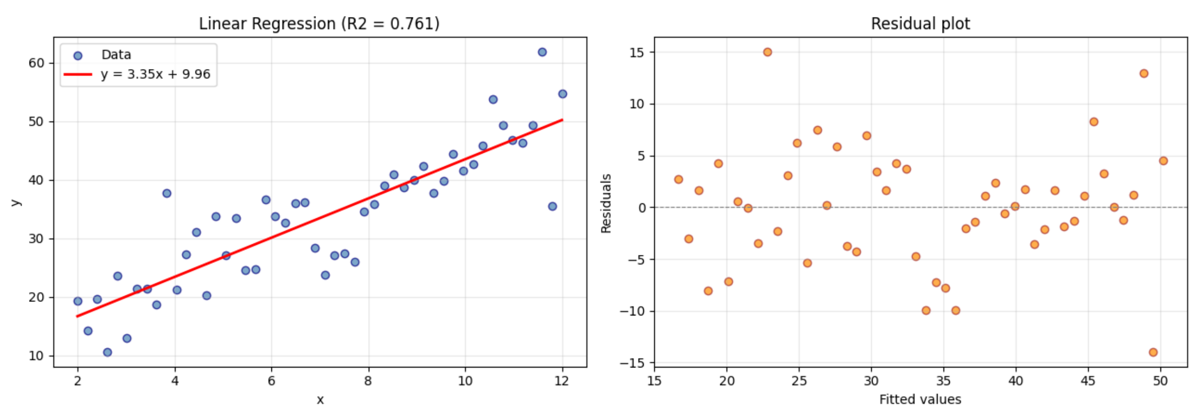
8.2.1 Scatter plot with regression line

```
fig, axes = plt.subplots(1, 2, figsize=(13, 4.5))

axes[0].scatter(x, y, alpha=0.7, color='steelblue', edgecolors='navy', label='Data')
axes[0].plot(x, y_fit, 'r-', linewidth=2, label=f'y = {coef[0]:.2f}x + {coef[1]:.2f}')
axes[0].set_xlabel('x')
axes[0].set_ylabel('y')
axes[0].set_title(f'Linear Regression (R2 = {R2:.3f})')
axes[0].legend()
axes[0].grid(True, alpha=0.3)

axes[1].scatter(y_fit, residuals, alpha=0.7, color='darkorange', edgecolors='brown')
axes[1].axhline(0, color='gray', linewidth=0.8, linestyle='--')
axes[1].set_xlabel('Fitted values')
axes[1].set_ylabel('Residuals')
axes[1].set_title('Residual plot')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



8.2.2 Conclusion

The residual plot allows us to verify regression assumptions: if residuals are randomly scattered around zero, the linear model is adequate. A pattern in the residuals would indicate a missing nonlinear term.

8.3 Estimating π with Monte Carlo

The Monte Carlo method estimates π by throwing random points in a unit square $[0, 1] \times [0, 1]$. The fraction that falls inside the quarter circle (radius 1) approximates $\pi/4$:

$$\pi \approx 4 \cdot \frac{N_{\text{inside}}}{N_{\text{total}}}$$

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)
N = 5000
x = np.random.uniform(0, 1, N)
y = np.random.uniform(0, 1, N)

inside = x**2 + y**2 <= 1
pi_est = 4 * np.sum(inside) / N

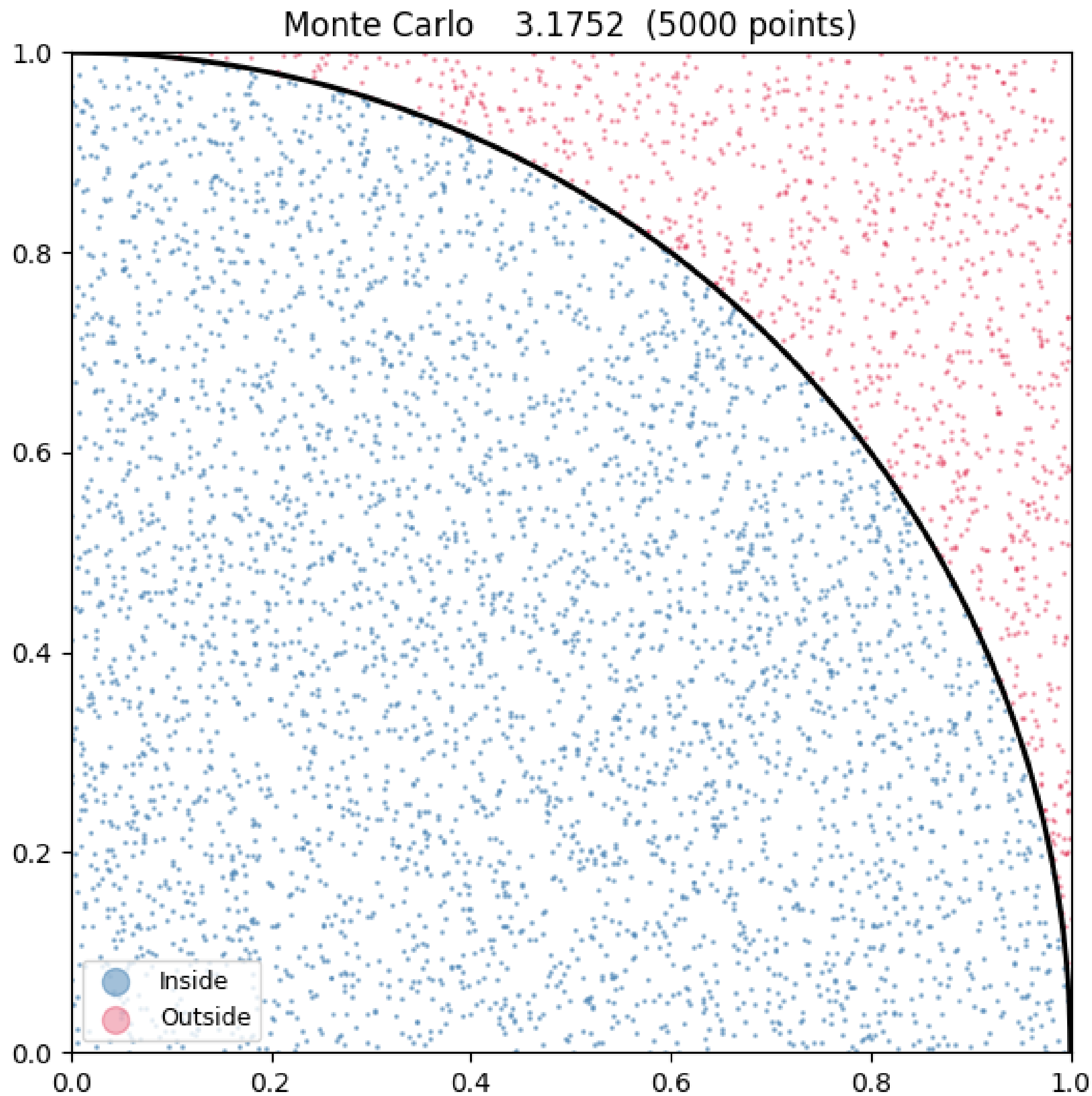
print(f'Total points:     {N}')
print(f'Inside circle:    {np.sum(inside)}')
print(f' estimate:       {pi_est:.6f}')
print(f'True  value:     {np.pi:.6f}')
print(f'Absolute error:   {abs(pi_est - np.pi):.6f}')
```

```
Total points:     5000
Inside circle:    3969
 estimate:       3.175200
True  value:     3.141593
Absolute error:   0.033607
```

8.3.1 Point Visualization

Blue points fall inside the quarter circle, red points outside. The black curve is the boundary $x^2 + y^2 = 1$.

```
plt.figure(figsize=(6, 6))
plt.scatter(x[inside], y[inside], s=1, color='steelblue', alpha=0.5, label='Inside')
plt.scatter(x[~inside], y[~inside], s=1, color='crimson', alpha=0.3, label='Outside')
theta = np.linspace(0, np.pi/2, 200)
plt.plot(np.cos(theta), np.sin(theta), 'k-', linewidth=2)
plt.xlim(0, 1)
plt.ylim(0, 1)
plt.gca().set_aspect('equal')
plt.title(f'Monte Carlo    {pi_est:.4f}  ({N} points)')
plt.legend(markerscale=10, fontsize=9)
plt.tight_layout()
plt.show()
```



8.3.2 Convergence as N Increases

The estimate improves as we increase the number of points. The error decreases as $O(1/\sqrt{N})$, characteristic of Monte Carlo methods.

8.4 Practice Questions: Statistics

Test your knowledge of fundamental statistics concepts. Expand each question to reveal the answer.

Question 1 — p-value

What exactly does a p-value of 0.03 mean in a hypothesis test?

A p-value of 0.03 means that, **if the null hypothesis H_0 were true**, the probability of observing a result as extreme or more extreme than the one obtained would be 3%.

Important: It is NOT the probability that H_0 is true. It is the probability of the data (or more extreme) given H_0.

Since 0.03 < 0.05 (the typical significance level), we would reject H_0.

Question 2 — Confidence interval

Interpret a 95% confidence interval for a mean: $[12.3, 15.7]$**.**

It does NOT mean the population mean has a 95% probability of being in that interval. The correct frequentist interpretation is:

> If we repeat the experiment many times, 95% of the intervals constructed with the same method will contain the true parameter value.

For this particular interval, the population mean is either inside or outside — there is no intermediate probability. What has 95% confidence is the **construction method**.

Question 3 — Normal distribution

What proportion of data falls within ±2 standard deviations of the mean in a normal distribution?

Using the empirical rule (68-95-99.7 rule):

The following table summarizes the main elements of this section.

Table. Practice Questions: Statistics.

Range	Proportion
$\mu \pm 1\sigma$	$\approx 68.27\%$
$\mu \pm 2\sigma$	$\approx 95.45\%$
$\mu \pm 3\sigma$	$\approx 99.73\%$

Therefore, approximately **95.45%** of data falls within $\mu \pm 2\sigma$.

Question 4 — Correlation vs. Causation

If two variables have a correlation of $r = 0.85$, can you conclude that one causes the other?

No. Correlation does not imply causation. A strong correlation only indicates linear association. Possible explanations include:

1. X **causes** Y — possible but not proven by r alone.
2. Y **causes** X — the direction could be reversed.
3. **A confounding variable** Z causes both.
4. **Coincidence** — especially with small datasets.

Establishing causation requires experimental studies (randomized controlled trials), not just observational data.

Question 5 — Type I and Type II Errors

Define Type I and Type II errors. Which one is directly controlled by the α level?

The following table summarizes the main elements of this section.

Table. Practice Questions: Statistics.

Type	Occurs when...	Called
Type I Error	We reject H_0 when it is true	False positive
Type II Error	We fail to reject H_0 when it is false	False negative

The **significance level** α directly controls the probability of committing a Type I error: $P(\text{Type I}) = \alpha$.

The probability of a Type II error is denoted β, and the power of a test is $1 - \beta$. Decreasing α generally increases β.

Question 6 — Central Limit Theorem

State the Central Limit Theorem and explain its practical importance.

The **Central Limit Theorem (CLT)** states that, for sufficiently large samples ($n \geq 30$, approximately), the distribution of the sample mean $\bar{X}$ approaches a normal distribution:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

regardless of the original population distribution.

Practical importance: It justifies using the normal distribution to construct confidence intervals and perform hypothesis tests about the mean, even when the population is not normal. It is the foundation of much of inferential statistics.

CHAPTER 9

DEGREE IN COMPUTER SCIENCE

In this section, you will find examples adapted to Computer Science Degree courses. The goal is to show how to integrate:

- Database diagrams (E-R model)
- UML class diagrams
- Algorithmic complexity analysis with executable code

Applicable subjects

This block can be adapted to programming, databases, and software engineering courses.

9.1 Entity-Relationship Diagram with Kroki (Mermaid)

E-R diagrams are fundamental in database design. With Kroki and `:type: mermaid`, you can include them directly in your book and make them work in both HTML and PDF.

9.1.1 Example: University database

Code:

````
```{kroki}
:type: mermaid
:align: center

erDiagram
    STUDENT {
        int id PK
        string name
        string email
    }
    COURSE {
        int id PK
        string name
        int credits
    }
    PROFESSOR {
        int id PK
        string name
````

(continues on next page)
````

(continued from previous page)

````
        string department
    }
    ENROLLMENT {
        int id PK
        float grade
        string term
    }

    STUDENT ||--o{ ENROLLMENT : "enrolls in"
    COURSE ||--o{ ENROLLMENT : "has"
    PROFESSOR ||--o{ COURSE : "teaches"
```
````

Result:

As shown in Fig. 9.1, the diagram is versioned as a static image.

- **Relationship types**: Use `||--o{` (one-to-many), `||--||` (one-to-one), `}o--o{` (many-to-many).
- **Primary keys**: Mark attributes with `PK`.
- **Foreign keys**: Mark attributes with `FK`.

9.1.2 Cardinality reference

The following table summarizes the main elements of this section.

Table. Cardinality reference.

Notation	Meaning
`	
`	
`}o--o{`	Many to many

9.2 UML Class Diagram with Kroki (Mermaid)

Class diagrams are essential for modeling object-oriented systems. With Kroki and `:type: mermaid`, you can represent classes, attributes, methods, and inheritance in both HTML and PDF.

9.2.1 Example: Geometric Shape Hierarchy

Code:

````
```{kroki}
:type: mermaid
:align: center

classDiagram
    class Shape {
````

(continues on next page)

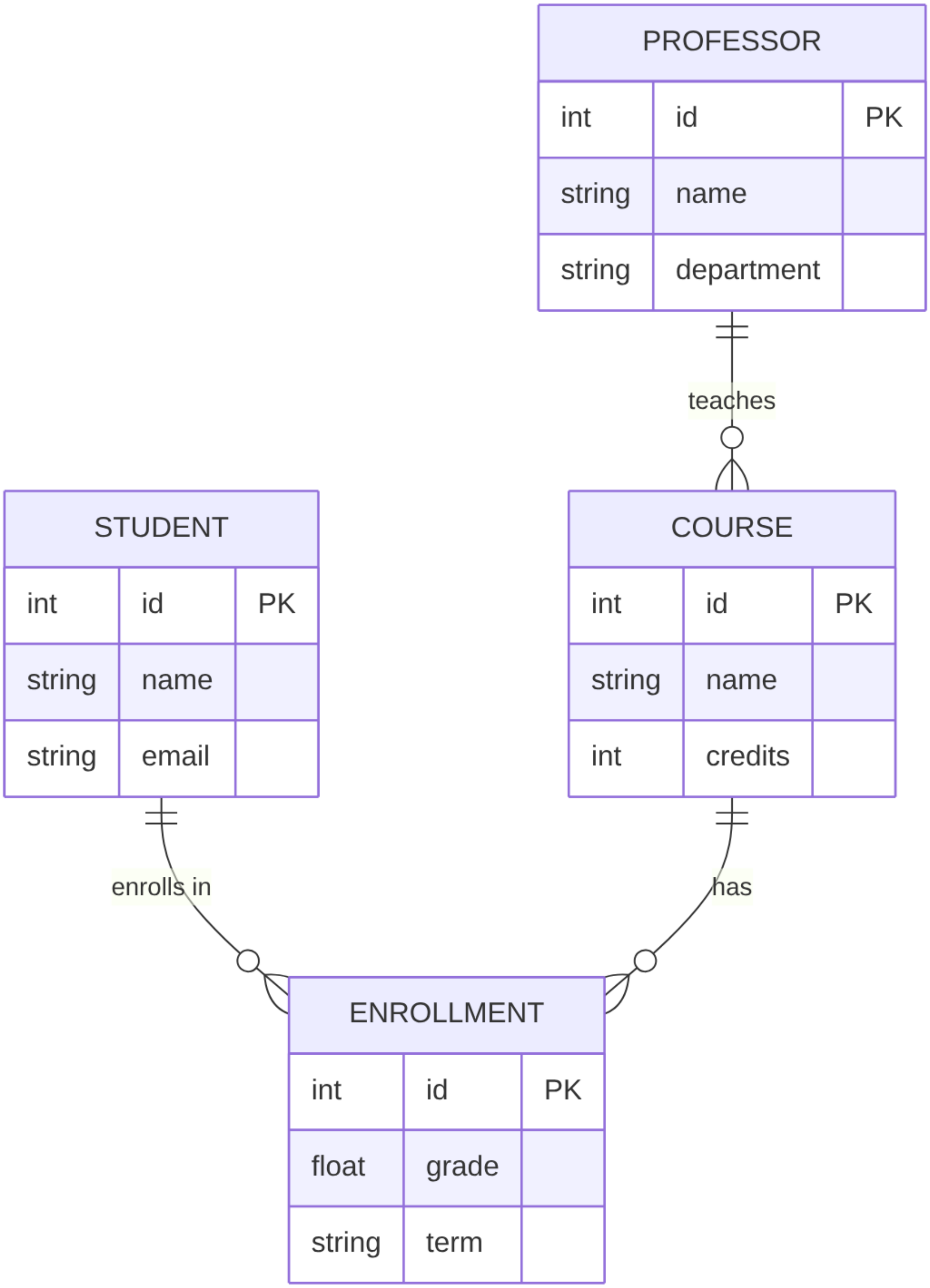


Figure9.1: Diagram: Example: University database.

(continued from previous page)

```
        <<abstract>>
        +String color
        +area() float
        +perimeter() float
    }

    class Circle {
        +float radius
        +area() float
        +perimeter() float
    }

    class Rectangle {
        +float width
        +float height
        +area() float
        +perimeter() float
    }

    class Triangle {
        +float base
        +float height
        +area() float
        +perimeter() float
    }

    Shape <|-- Circle
    Shape <|-- Rectangle
    Shape <|-- Triangle
```
```

**Result:**

As shown in Fig. 9.2, the diagram is versioned as a static image.

- **Methods**: Listed in the second section, with return type.
- **Abstract class**: Indicated with `<<abstract>>`.

### 9.2.2 Relationship types

The following table summarizes the main elements of this section.

**Table. Relationship types.**

| Notation | Meaning |
|---|---|
| `<\|--` | Inheritance (is a) |
| `*--` | Composition |
| `o--` | Aggregation |
| `-->` | Association |
| `..>` | Dependency |
```

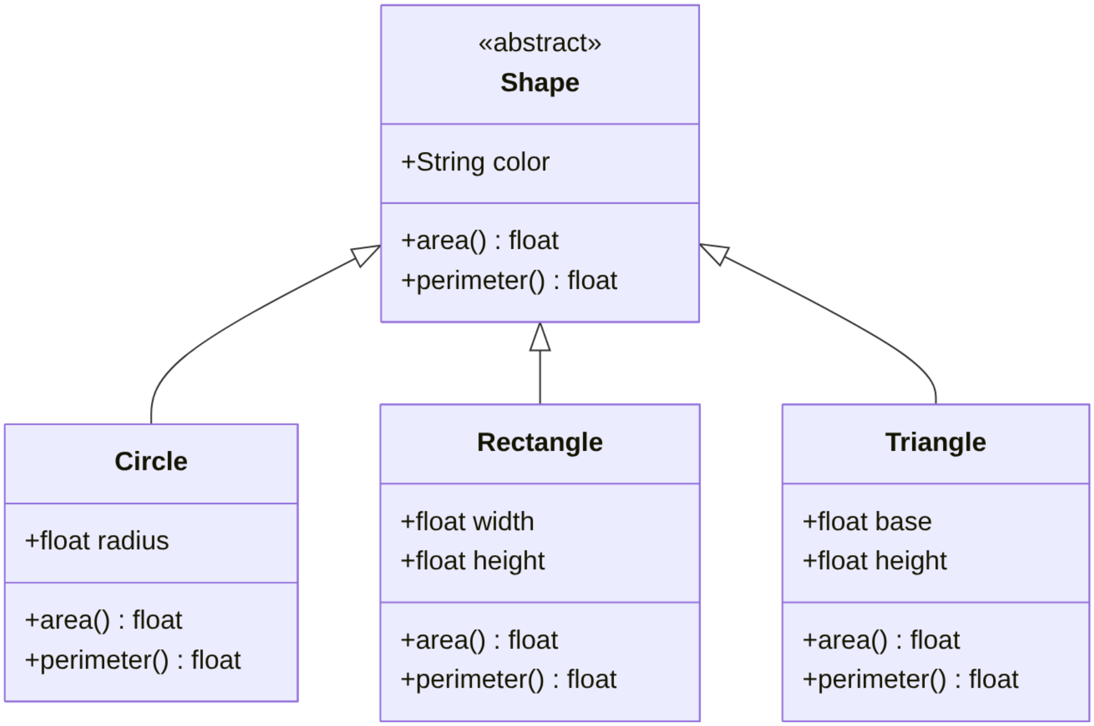


Figure9.2: Diagram: Example: Geometric Shape Hierarchy.

9.3 Diagrams with Kroki for Computer Science

Kroki lets you generate many useful diagram types in Computer Science without depending on browser-side JavaScript. All the following examples work in **HTML and PDF**.

9.3.1 1. Mermaid: layered architecture

As shown in Fig. 9.3, the diagram is versioned as a static image.

As shown in Fig. 9.4, the diagram is versioned as a static image.

As shown in Fig. 9.5, the diagram is versioned as a static image.

As shown in Fig. 9.6, the diagram is versioned as a static image.

```
## 5. Wavedrom: digital protocol timing


As shown in {numref}`fig-diagram-02-degrees-computer-science-degree-kroki-diagrams-
↪05`, the diagram is versioned as a static image.

```{figure} ../../../_static/generated/diagrams/en/02_degrees_computer_science_degree_
↪kroki_diagrams_05.svg
:name: fig-diagram-02-degrees-computer-science-degree-kroki-diagrams-05
:alt: Diagram: Wavedrom: digital protocol timing
:width: 60%
:align: center
```

(continues on next page)
```

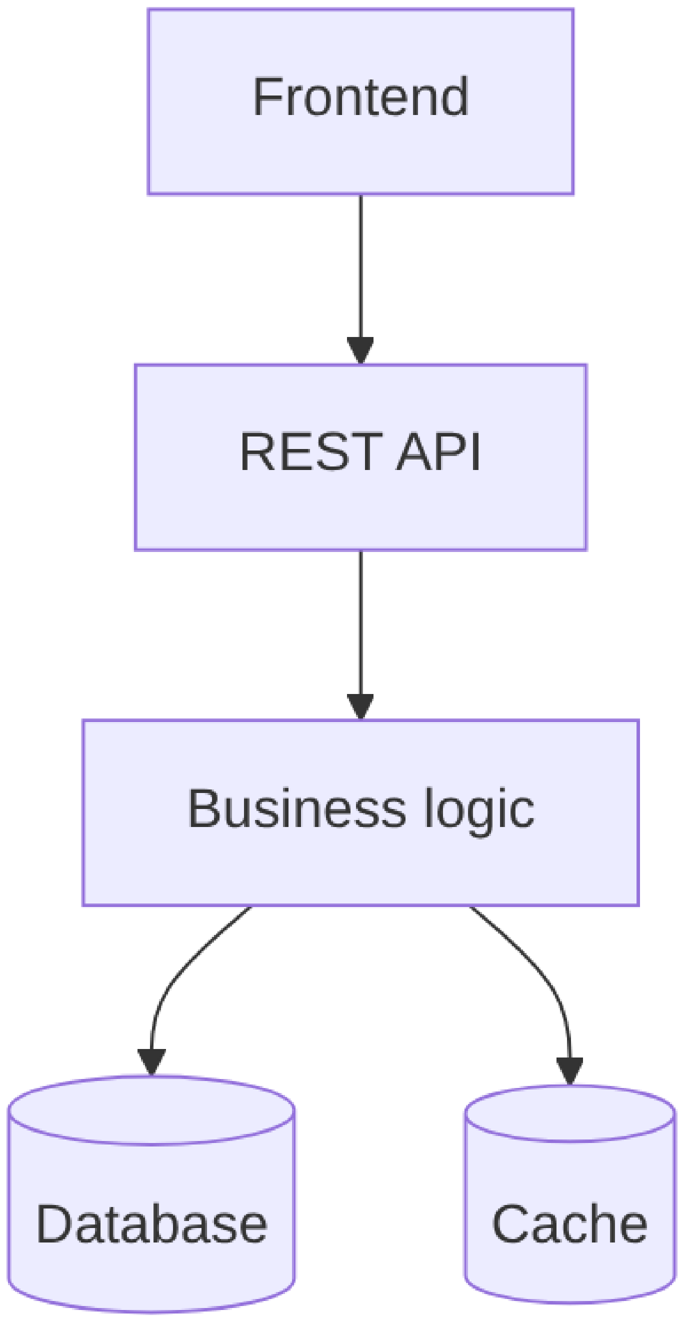


Figure9.3: Diagram: Mermaid: layered architecture.

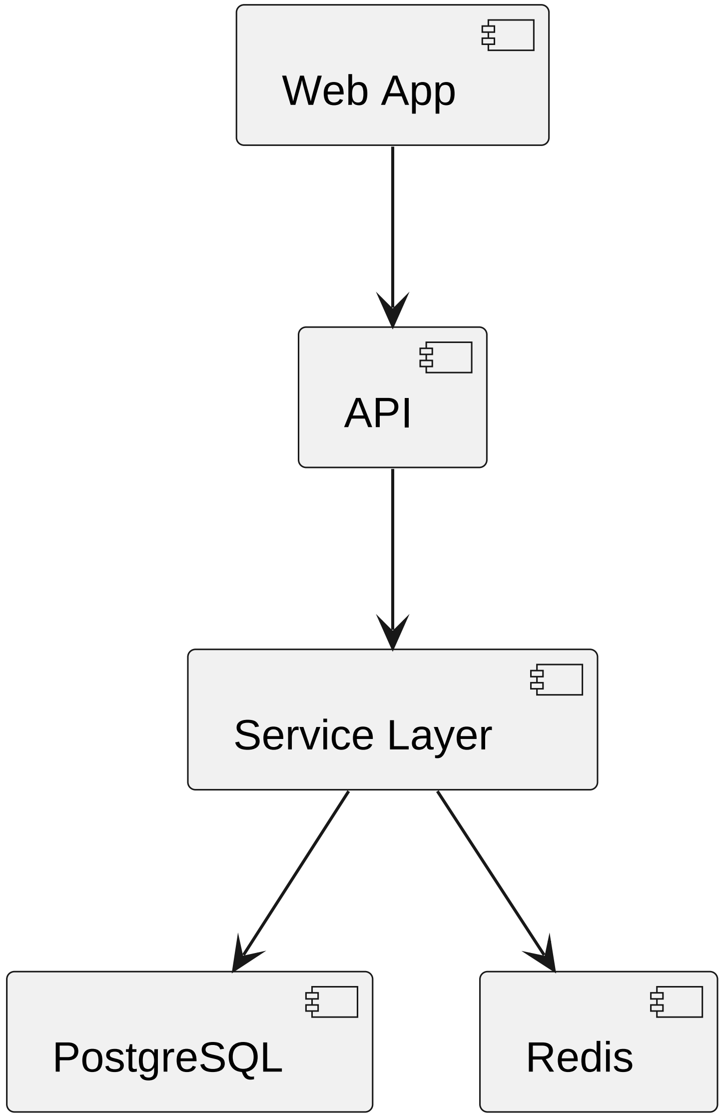


Figure9.4: Diagram: PlantUML: system components.

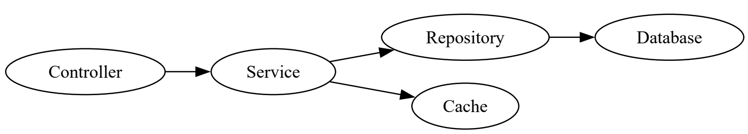


Figure9.5: Diagram: GraphViz: dependency graph.

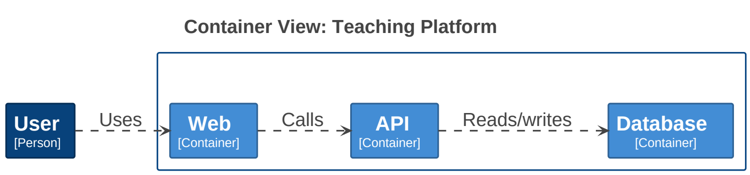


Figure9.6: Diagram: Structurizr: simplified C4 view.

(continued from previous page)

```
Diagram: Wavedrom: digital protocol timing.
```

- architecture
- UML
- dependency graphs
- protocols
- C4 diagrams

If the user asks for a fast and portable diagram, Kroki is a great option.

9.4 Algorithmic complexity: asymptotic growth and Big-O notation

Goal. Understand how algorithm cost is analysed as input size grows, and distinguish between time measured on a particular machine and asymptotic complexity.

Algorithmic complexity studies a cost function $T(n)$, where n denotes input size and $T(n)$ counts resources such as elementary operations, comparisons, memory accesses, or space usage. On a real computer, running time depends on the processor, compiler, programming language, memory hierarchy, and many implementation details. Academic analysis therefore abstracts away from those details and compares the **order of growth** of $T(n)$ as n tends to infinity [Ada09].

Formally, $f(n) \in O(g(n))$ if there exist constants $c > 0$ and $n_0 > 0$ such that

$$0 \leq f(n) \leq c\, g(n) \quad \text{for all } n \geq n_0.$$

Thus, $g(n)$ is an asymptotic upper bound for $f(n)$ up to multiplicative constants. The notation $\Omega(g(n))$ gives an asymptotic lower bound, and $\Theta(g(n))$ gives a tight bound: $f(n) \in \Theta(g(n))$ when both $f(n) \in O(g(n))$ and $f(n) \in \Omega(g(n))$.

Three points are especially important in teaching contexts:

1. Big-O does not measure seconds: it measures relative growth with respect to n.
2. Constants and lower-order terms are ignored only asymptotically. For small inputs they can matter a great deal.
3. The analysed case must be stated: worst case, best case, average case, or amortized cost. By default, many courses use Big-O for worst-case upper bounds.

9.4.1 Growth hierarchy

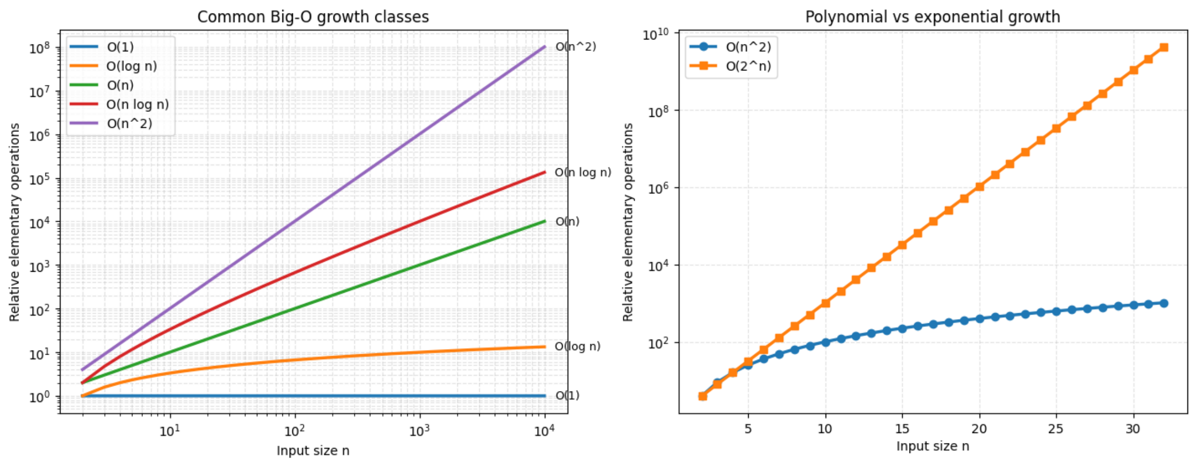
The following figure does not show real execution times. It shows theoretical growth functions in elementary operations. This avoids letting machine-specific noise hide the main idea: as n grows, differences between complexity classes become dominant.

The classes shown are common in algorithm analysis:

- $O(1)$: constant-cost access or decision.
- $O(\log n)$: halving the problem size, as in binary search.
- $O(n)$: one linear pass through the data.
- $O(n \log n)$: efficient comparison sorting, such as mergesort or heapsort.
- $O(n^2)$: algorithms with two nested loops over the input.

- $O(2^n)$: exponential exploration of a solution space.

```python
import numpy as np
import matplotlib.pyplot as plt

# Use theoretical growth functions, not wall-clock timings.
n = np.arange(2, 10001)
log_n = np.log2(n)
curves = {
    "O(1)": np.ones_like(n, dtype=float),
    "O(log n)": log_n,
    "O(n)": n,
    "O(n log n)": n * log_n,
    "O(n^2)": n**2,
}

n_exp = np.arange(2, 33)
quadratic_small = n_exp**2
exponential_small = 2.0**n_exp

fig, axes = plt.subplots(1, 2, figsize=(13, 5), constrained_layout=True)

ax = axes[0]
for label, values in curves.items():
    ax.loglog(n, values, linewidth=2.4, label=label)
    ax.annotate(
        label,
        xy=(n[-1], values[-1]),
        xytext=(8, 0),
        textcoords="offset points",
        va="center",
        fontsize=9,
    )
ax.set_title("Common Big-O growth classes")
ax.set_xlabel("Input size n")
ax.set_ylabel("Relative elementary operations")
ax.grid(True, which="both", linestyle="--", alpha=0.35)
ax.legend(loc="upper left")

ax = axes[1]
ax.semilogy(n_exp, quadratic_small, linewidth=2.4, marker="o", label="O(n^2)")
ax.semilogy(n_exp, exponential_small, linewidth=2.4, marker="s", label="O(2^n)")
ax.set_title("Polynomial vs exponential growth")
ax.set_xlabel("Input size n")
ax.set_ylabel("Relative elementary operations")
ax.grid(True, which="both", linestyle="--", alpha=0.35)
ax.legend(loc="upper left")

plt.show()

sample_n = np.array([10, 100, 1000, 10000])
print("n       log2(n)       n       n log2(n)       n^2")
for value in sample_n:
    print(
        f"{value:<6d} {np.log2(value):>8.2f} "
        f"{value:>8d} {value * np.log2(value):>13.2f} {value**2:>12d}"
    )
```




```
n        log2(n)       n      n log2(n)       n^2
10          3.32      10          33.22          100
100         6.64     100         664.39        10000
1000        9.97    1000        9965.78      1000000
10000      13.29   10000      132877.12    100000000
```

9.4.2 Interpretation

On logarithmic scales, straight lines make orders of magnitude easier to compare: $O(n)$ grows faster than $O(\log n)$, $O(n \log n)$ lies between linear and quadratic growth, and $O(n^2)$ separates quickly as input size increases. The second plot shows why exponential algorithms often become infeasible even for moderate values of n: 2^n very soon exceeds any fixed polynomial.

A practical consequence is that optimizing constants is not a substitute for choosing a better complexity class. Halving the running time of an $O(n^2)$ algorithm helps, but changing the approach to an $O(n \log n)$ algorithm can completely change the scale of problems that can be solved.

9.4.3 Exercises

1. For $n = 10^6$, compare n, $n \log_2 n$, and n^2 mentally. What practical difference is there between scanning a list, sorting it efficiently, and comparing all pairs?
2. Choose an algorithm from a course you know and identify which elementary operation dominates its cost. Then justify whether your analysis is worst case, average case, or amortized cost.

DEGREE IN GEOLOGY

In this section, you will find examples adapted to Geology Degree courses. The goal is to show how to integrate:

- Topographic profiles from elevation data
- Stratigraphic columns with executable code
- Rock cycle diagrams

Applicable subjects

This block can be adapted to cartography, stratigraphy, general geology, geomorphology, and petrology courses.

10.1 Topographic Profile

A topographic profile represents elevation variation along a section line. We will generate a synthetic profile and plot it with matplotlib.

```
import numpy as np
import matplotlib.pyplot as plt

distance = np.linspace(0, 10, 500)

elevation = (
    200
    + 80 * np.sin(0.8 * distance)
    + 40 * np.sin(2.5 * distance)
    + 20 * np.random.normal(0, 1, len(distance))
)

plt.figure(figsize=(10, 4))
plt.fill_between(distance, 0, elevation, color='#8B7355', alpha=0.4)
plt.plot(distance, elevation, color='#5C4033', linewidth=1.5)
plt.xlabel('Distance (km)')
plt.ylabel('Elevation (m)')
plt.title('Synthetic topographic profile')
plt.axhline(y=0, color='skyblue', linewidth=0.5)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

10.1.1 Expected result

A profile simulating terrain with elevation variations is produced. The brown fill represents the ground mass and the blue line marks sea level.

10.1.2 Conclusion

- Topographic profiles can be created from real data (CSV, GeoTIFF).
- This example uses synthetic data to avoid external file dependencies.
- For real data, simply replace the `distance` and `elevation` arrays.

10.2 Stratigraphic Column

A stratigraphic column represents the vertical succession of geological layers. Each layer is drawn as a colored rectangle according to its lithology.

```
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

layers = [
    {"name": "Limestone",     "thickness": 15, "color": "#4A90D9", "age": "Jurassic"},
    {"name": "Sandstone",     "thickness": 20, "color": "#E8C84A", "age": "Triassic"},
    {"name": "Clay",          "thickness": 10, "color": "#C97B3D", "age": "Triassic"},
    {"name": "Conglomerate",  "thickness": 12, "color": "#9B7653", "age": "Permian"},
    {"name": "Shale",         "thickness": 18, "color": "#6B6B6B", "age":
↪"Carboniferous"},
    {"name": "Granite",       "thickness": 25, "color": "#D44A4A", "age": "Precambrian
↪"},
]

fig, ax = plt.subplots(figsize=(5, 8))

y_base = 0
for layer in layers:
    ax.add_patch(plt.Rectangle((0, y_base), 1, layer["thickness"],
                               facecolor=layer["color"], edgecolor="black",␣
↪linewidth=0.8))
    ax.text(0.5, y_base + layer["thickness"] / 2,
            f"{layer['name']}\n({layer['age']}, {layer['thickness']} m)",
            ha='center', va='center', fontsize=8, fontweight='bold')
    y_base += layer["thickness"]

ax.set_xlim(-0.2, 1.4)
ax.set_ylim(-2, y_base + 2)
ax.set_ylabel('Cumulative thickness (m)')
ax.set_title('Synthetic stratigraphic column')
ax.set_xticks([])
ax.invert_yaxis()
plt.tight_layout()
plt.show()
```

10.2.1 Expected result

The plot shows stacked layers from bottom to top (oldest to youngest). Each color represents a different lithology: blue for limestone, yellow for sandstone, etc.

10.2.2 Conclusion

- You can add more layers by appending entries to the `layers` list.
- Colors are customizable by changing the hex values.
- For real data, replace the list with data read from a CSV file.

10.3 The Rock Cycle with Kroki (Mermaid)

The rock cycle describes the transformations between the three main rock types. With Kroki and `:type: mermaid`, we can represent this cycle as a flowchart that works in both HTML and PDF.

10.3.1 The cycle

Code:

````
```{kroki}
:type: mermaid
:align: center

flowchart LR
    Magma["Magma"]
    Igneous["Igneous Rock"]
    Sedimentary["Sedimentary Rock"]
    Metamorphic["Metamorphic Rock"]

    Magma -->|"Cooling and solidification"| Igneous
    Igneous -->|"Weathering and erosion"| Sedimentary
    Sedimentary -->|"Heat and pressure"| Metamorphic
    Metamorphic -->|"Melting"| Magma
    Igneous -->|"Heat and pressure"| Metamorphic
    Metamorphic -->|"Weathering and erosion"| Sedimentary
    Sedimentary -->|"Melting"| Magma
```
````

Result:

As shown in Fig. 10.1, the diagram is versioned as a static image.

The following table summarizes the main elements of this section.

Table. Transition explanation.

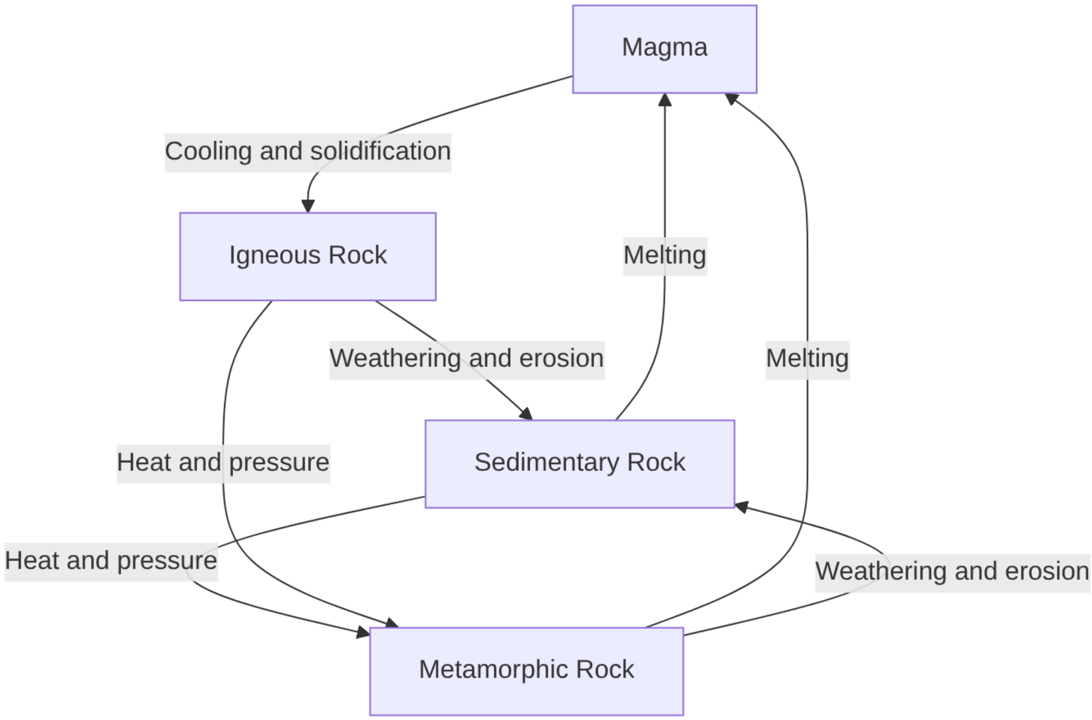


Figure10.1: Diagram: The cycle.

Transition	Geological process
Magma → Igneous	Cooling and solidification at the surface or underground
Igneous → Sedimentary	Weathering, erosion, transport, and sedimentation
Sedimentary → Metamorphic	Heat and pressure (metamorphism) without melting
Metamorphic → Magma	Complete melting at extreme temperatures
Igneous → Metamorphic	Direct metamorphism through heat and pressure
Metamorphic → Sedimentary	Uplift, weathering, and erosion

10.3.2 Key point

Any rock type can transform into any other. It is not a linear cycle, but a **transformation network** operating on geological timescales (millions of years).

CHAPTER 11

CHEMICAL ENGINEERING DEGREE

This section contains examples for Chemical Engineering courses. The cases combine material balances, separation operations, mass transfer, reaction engineering, and technical chart reading.

The pages connect balance formulation, process design, and computational calculation. They cite classic sources on balances, distillation, pipe friction, and diffusion-reaction in catalysts so each example has a verifiable foundation [FRB16, MT25, Moo44, Thi39, TS13].

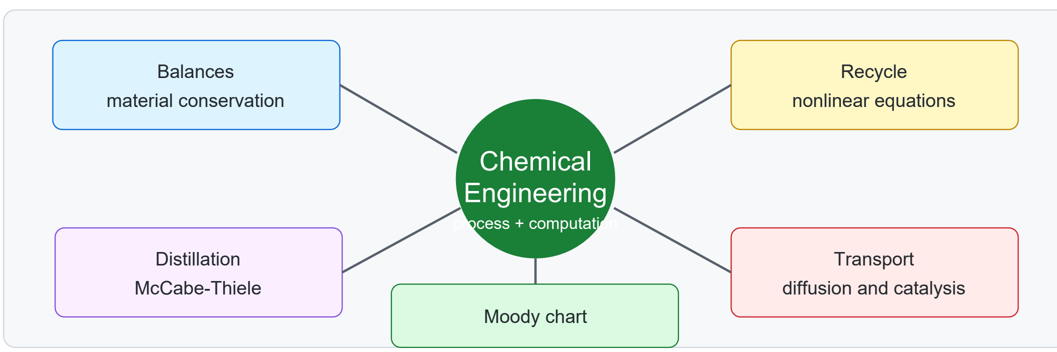


Figure11.1: Teaching resource map for the Chemical Engineering section.

The teaching goal is for students to:

- Turn a written process description into a clear flow diagram.
- Explore how separation designs change when parameters vary.
- Solve recycle streams with simple numerical methods.
- Connect transport, kinetics, and concentration profiles.
- Read professional engineering charts, such as the Moody chart, with computational support.

How to use the template in this section

Each example turns a plant situation into a model: first identify variables, then write equations, and finally interpret a code-generated figure or a static diagram.

11.1 Process Flow Diagram

A flow diagram helps turn a verbal process description into a structure that can be reviewed, discussed, and improved. For teaching notes, a simplified first draft is often enough: feeds, main unit operations, products, and possible recycle streams.

In process design, the flow diagram is the bridge between the chemical description and the quantitative balance. Industrial PFDs include streams, equipment, and operating data, but a teaching diagram can start with the same basic logic [FRB16, TS13].

For each unit, the core idea is:

$$\text{input} - \text{output} + \text{generation} - \text{consumption} = \text{accumulation}$$

As shown in Fig. 11.2, the process can be represented with a feed, a mixer, a reactor, a separator, and a recycle stream. The figure is not meant to replace an industrial PFD; it is a quick teaching draft.

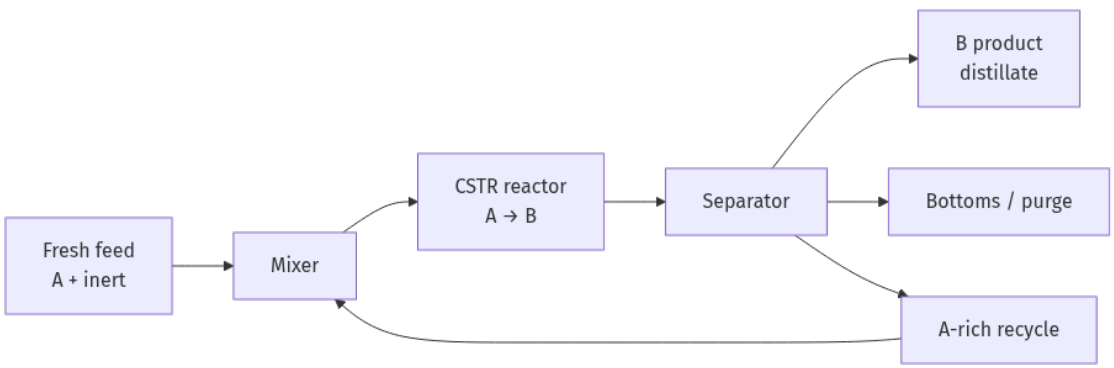


Figure11.2: Simplified process flow diagram with recycle.

11.1.1 Classroom Use

Students can be asked to identify:

- The inlet and outlet streams.
- The unit where the reaction occurs.
- The stream that is recycled and why it may improve reactant use.
- Which variables would be needed for a full material balance.

Short activity

Redraw the diagram to include a purge stream. Explain which operating problem that purge could solve if an inert component accumulates.

Checklist

Before writing equations, mark on the diagram: system boundary, known streams, unknown streams, reactions, and possible recycle loops. If one of these pieces is missing, the balance will be ambiguous.

11.2 McCabe-Thiele Diagram

The McCabe-Thiele diagram is a classic graphical tool for estimating theoretical stages in binary distillation. The method was introduced as a graphical design procedure for fractionating columns and remains useful for teaching equilibrium balances and operating lines [MT25].

This example uses constant relative volatility:

$$y^*(x) = \frac{\alpha x}{1 + (\alpha - 1)x}$$

The rectifying line is:

$$y = \frac{R}{R+1}x + \frac{x_D}{R+1}$$

What to modify

Change `R`, `alpha`, `xD`, `xF`, or `xB`. A higher reflux ratio moves the rectifying line closer to the diagonal and usually reduces the number of stages, but it increases energy cost.

```
import numpy as np
import matplotlib.pyplot as plt

# Editable parameters
alpha = 2.45    # relative volatility
xF = 0.45       # feed composition
xD = 0.92       # distillate composition
xB = 0.08       # bottoms composition
R = 2.2         # reflux ratio
max_stages = 60


def y_equilibrium(x, alpha):
    return alpha * x / (1 + (alpha - 1) * x)


def x_equilibrium_from_y(y, alpha):
    return y / (alpha - (alpha - 1) * y)


def y_rectifying(x, R, xD):
    return (R / (R + 1)) * x + xD / (R + 1)

# Saturated-liquid feed: q-line is vertical at xF. The stripping line is drawn
# through the feed intersection and the bottoms point on the diagonal.
y_feed = y_rectifying(xF, R, xD)
m_strip = (y_feed - xB) / (xF - xB)
b_strip = xB - m_strip * xB

def y_operating(x):
    return y_rectifying(x, R, xD) if x >= xF else m_strip * x + b_strip

# Staircase construction.
stair_x = [xD]
stair_y = [xD]
```

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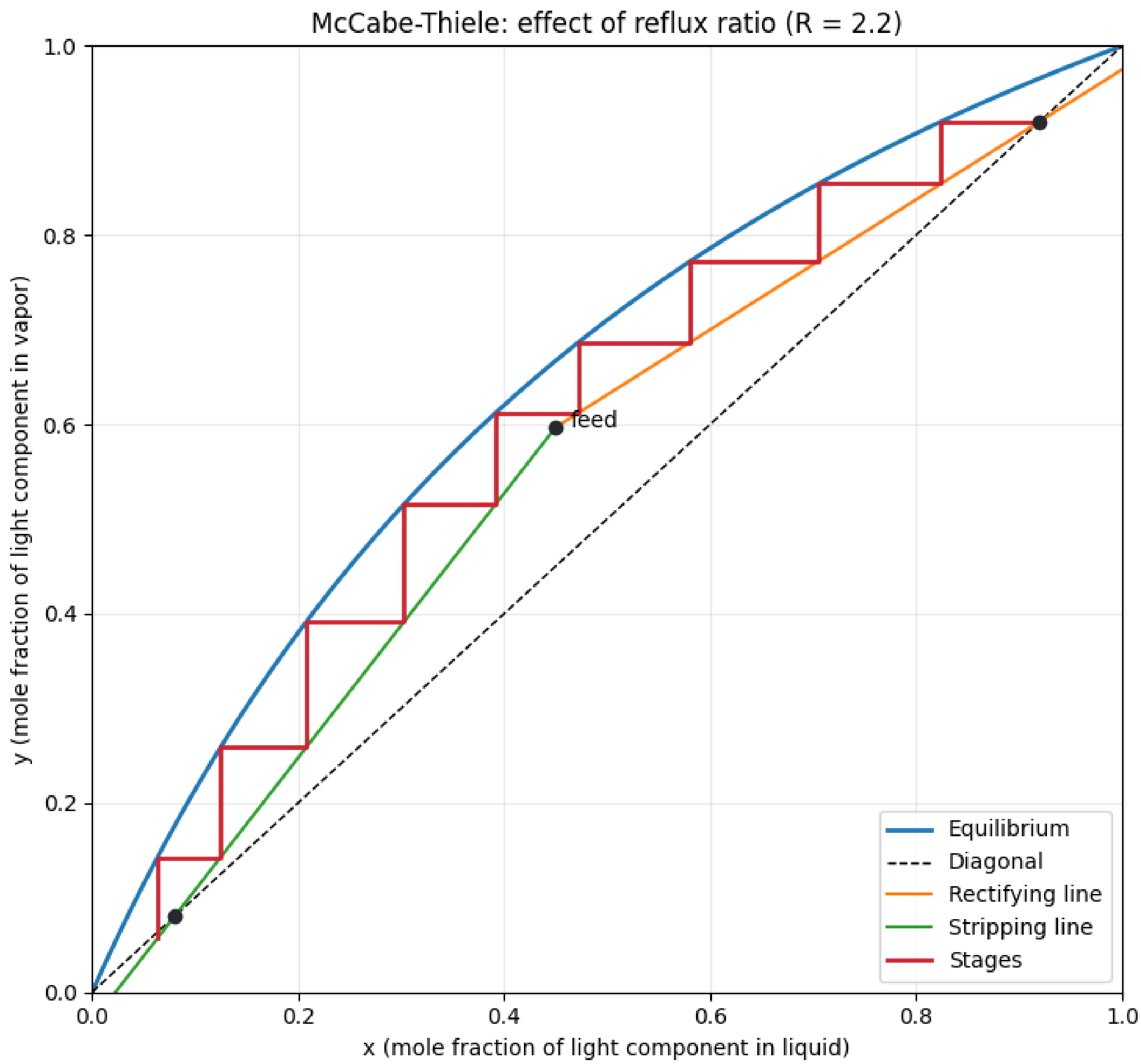
```
stages = 0
x_current = xD
y_current = xD
while x_current > xB and stages < max_stages:
    x_eq = x_equilibrium_from_y(y_current, alpha)
    stair_x.extend([x_eq, x_eq])
    stair_y.extend([y_current, y_operating(x_eq)])
    x_current = x_eq
    y_current = y_operating(x_eq)
    stages += 1

x = np.linspace(0, 1, 500)
plt.figure(figsize=(7.5, 7))
plt.plot(x, y_equilibrium(x, alpha), label="Equilibrium", linewidth=2)
plt.plot(x, x, "k--", label="Diagonal", linewidth=1)
plt.plot(x[x >= xF], y_rectifying(x[x >= xF], R, xD), label="Rectifying line")
plt.plot(x[x <= xF], m_strip * x[x <= xF] + b_strip, label="Stripping line")
plt.plot(stair_x, stair_y, color="#cf222e", linewidth=2, label="Stages")
plt.scatter([xF, xD, xB], [y_feed, xD, xB], color="#24292f", zorder=5)
plt.text(xF + 0.015, y_feed, "feed")
plt.xlim(0, 1)
plt.ylim(0, 1)
plt.xlabel("x (mole fraction of light component in liquid)")
plt.ylabel("y (mole fraction of light component in vapor)")
plt.title("McCabe-Thiele: effect of reflux ratio (R = " + str(R) + ")")
plt.grid(alpha=0.25)
plt.legend(loc="lower right")
plt.tight_layout()
plt.show()

print(f"Approximate theoretical stages: {stages}")
print(f"xD={xD:.2f}, xF={xF:.2f}, xB={xB:.2f}, alpha={alpha:.2f}, R={R:.2f}")
```




```
Approximate theoretical stages: 9
xD=0.92, xF=0.45, xB=0.08, alpha=2.45, R=2.20
```

11.2.1 Reading the Result

As the reflux ratio increases, the rectifying line moves closer to the diagonal and fewer stages are required. The energy cost increases because more liquid is recycled inside the column.

Guided reading

Count the red staircase steps between x_D and x_B. Each step represents one ideal stage: horizontal to equilibrium, then vertical to the operating line.

11.3 Recycle Solved with `fsolve`

Many chemical processes include recycle streams. This means that a variable depends on itself: the stream leaving the separator returns to the mixer and changes the reactor inlet. Material balances are the starting point for this kind of problem [FRB16].

The system is solved with `fsolve`, a SciPy function based on robust numerical algorithms for nonlinear equations [VGO+20].

The simplified balance combines reactor conversion and separation:

$$F_{A,\text{out}} = (F_{A0} + F_{A,\text{rec}})(1 - X), \qquad F_{A,\text{rec}} = rF_{A,\text{out}}$$

> **What to modify**
>
> Change `recycle_fraction`, `V`, or `k`. Recycle improves global conversion of fresh feed, but it also increases the internal flow that equipment must handle.

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import fsolve

# Editable parameters
F_A0 = 100.0        # fresh molar flow of A (mol/min)
Q0 = 120.0          # fresh volumetric flow (L/min)
C_ref = 1.0         # reference concentration in recycle stream (mol/L)
V = 450.0           # reactor volume (L)
k = 0.32            # first-order kinetic constant (1/min)
recycle_fraction = 0.65


def solve_recycle(recycle_fraction):
    def equations(unknowns):
        F_A_out, F_A_rec = unknowns
        Q_rec = F_A_rec / C_ref
        tau = V / (Q0 + Q_rec)
        X = k * tau / (1 + k * tau)
        eq_reactor = F_A_out - (F_A0 + F_A_rec) * (1 - X)
        eq_separator = F_A_rec - recycle_fraction * F_A_out
        return [eq_reactor, eq_separator]
    F_A_out, F_A_rec = fsolve(equations, [35.0, 20.0])
    F_A_product = F_A_out - F_A_rec
    global_conversion = 1 - F_A_product / F_A0
    return F_A_out, F_A_rec, F_A_product, global_conversion

base = solve_recycle(recycle_fraction)
print("Base-case solution")
print(f"F_A,out reactor = {base[0]:.2f} mol/min")
print(f"F_A,recycle     = {base[1]:.2f} mol/min")
print(f"F_A to product  = {base[2]:.2f} mol/min")
print(f"Global conversion of fresh A = {100*base[3]:.1f} %")

fractions = np.linspace(0, 0.88, 40)
results = np.array([solve_recycle(fr) for fr in fractions])
```

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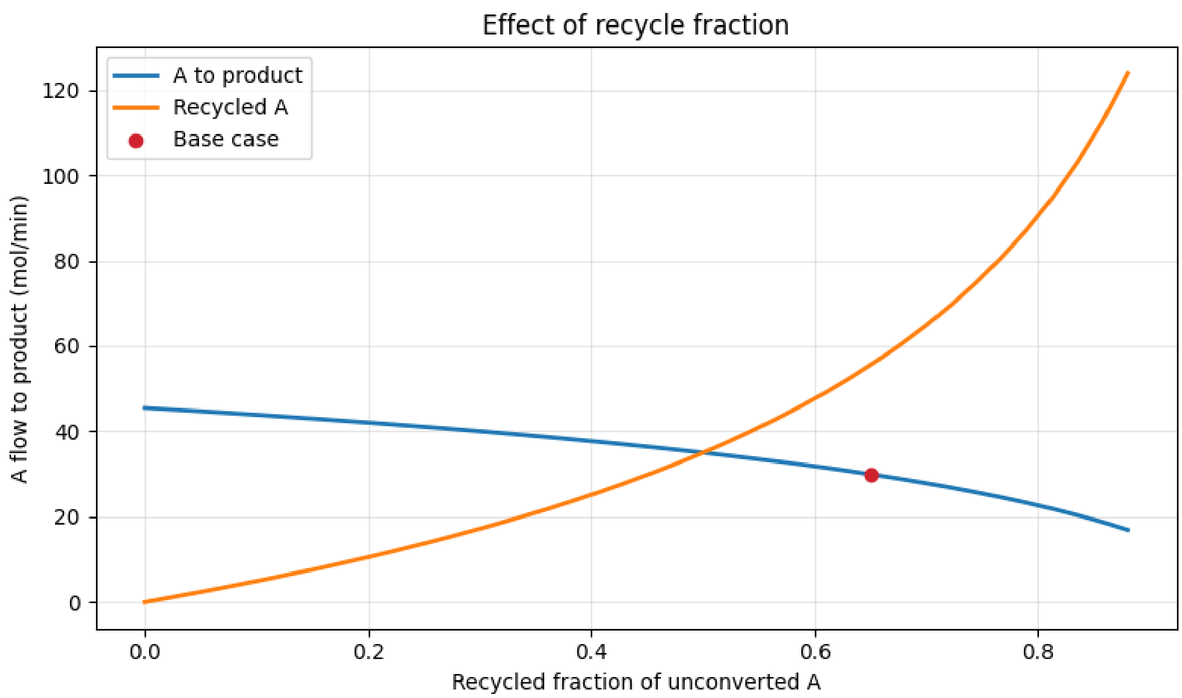
```
plt.figure(figsize=(8, 4.8))
plt.plot(fractions, results[:, 2], label="A to product", linewidth=2)
plt.plot(fractions, results[:, 1], label="Recycled A", linewidth=2)
plt.scatter([recycle_fraction], [base[2]], color="#cf222e", zorder=5, label="Base case
↪")
plt.xlabel("Recycled fraction of unconverted A")
plt.ylabel("A flow to product (mol/min)")
plt.title("Effect of recycle fraction")
plt.grid(alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()
```

```
Base-case solution
F_A,out reactor = 85.45 mol/min
F_A,recycle     = 55.54 mol/min
F_A to product  = 29.91 mol/min
Global conversion of fresh A = 70.1 %
```



11.3.1 Interpretation

Recycle increases overall reactant use, but it also increases the internal flow rate. In a real design, conversion, equipment size, energy cost, and inert accumulation must be balanced.

Guided reading

Notice that product flow does not increase indefinitely with recycle fraction. The system gains conversion, but the reactor handles more internal flow and a lower effective space time.

11.4 Concentration Gradient in a Catalyst Particle

When a reaction occurs inside a porous particle, the reactant concentration may be lower in the interior than at the surface. This idea is formalized with the Thiele modulus, introduced to relate catalytic activity and particle size, and with criteria such as Weisz-Prater for detecting diffusional limitations [Thi39, WP54].

For a first-order reaction in a spherical particle, a useful scale is:

$$\phi = R_p \sqrt{\frac{k}{D_e}}$$

The figure compares external transfer, internal diffusion, and kinetics. The vertical axis now adjusts automatically to the useful range of the profiles, and center and surface are marked so the radial progression is easier to read.

> **What to modify**
>
> Change `kf`, `De`, or `k` in the `scenarios` list. If D_e decreases or k increases, ϕ increases and the internal profile falls more strongly.

```
import numpy as np
import matplotlib.pyplot as plt

# Editable parameters
R_particle = 1.5e-3    # m
C_bulk = 1.0           # mol/L, used as reference
scenarios = [('Fast transfer', 0.03, 3.5e-06, 0.25), ('Low internal diffusion', 0.03,
↪7e-07, 0.25), ('Slow external film', 0.006, 3.5e-06, 0.25), ('Fast kinetics', 0.03,
↪3.5e-06, 0.9)]


def effectiveness_factor(phi):
    if phi < 1e-6:
        return 1.0
    return 3 / phi**2 * (phi / np.tanh(phi) - 1)


def profile(r_over_R, kf, De, k):
    phi = R_particle * np.sqrt(k / De)
    eta = effectiveness_factor(phi)
    C_surface = C_bulk / (1 + eta * k * R_particle / (3 * kf))
    r_safe = np.maximum(r_over_R, 1e-6)
    internal = np.sinh(phi * r_safe) / (r_safe * np.sinh(phi)) if phi > 1e-6 else np.
↪ones_like(r_safe)
    internal[0] = phi / np.sinh(phi) if phi > 1e-6 else 1.0
    return C_surface * internal / C_bulk, phi, eta, C_surface / C_bulk


r = np.linspace(0, 1, 250)
profiles = []
print("Scenario summary")
for name, kf, De, k in scenarios:
    C, phi, eta, Cs = profile(r, kf, De, k)
    profiles.append((name, C, phi, eta, Cs))
    print(f"{name}: phi={phi:.2f}, eta={eta:.2f}, C_surface/C_bulk={Cs:.2f}")
```

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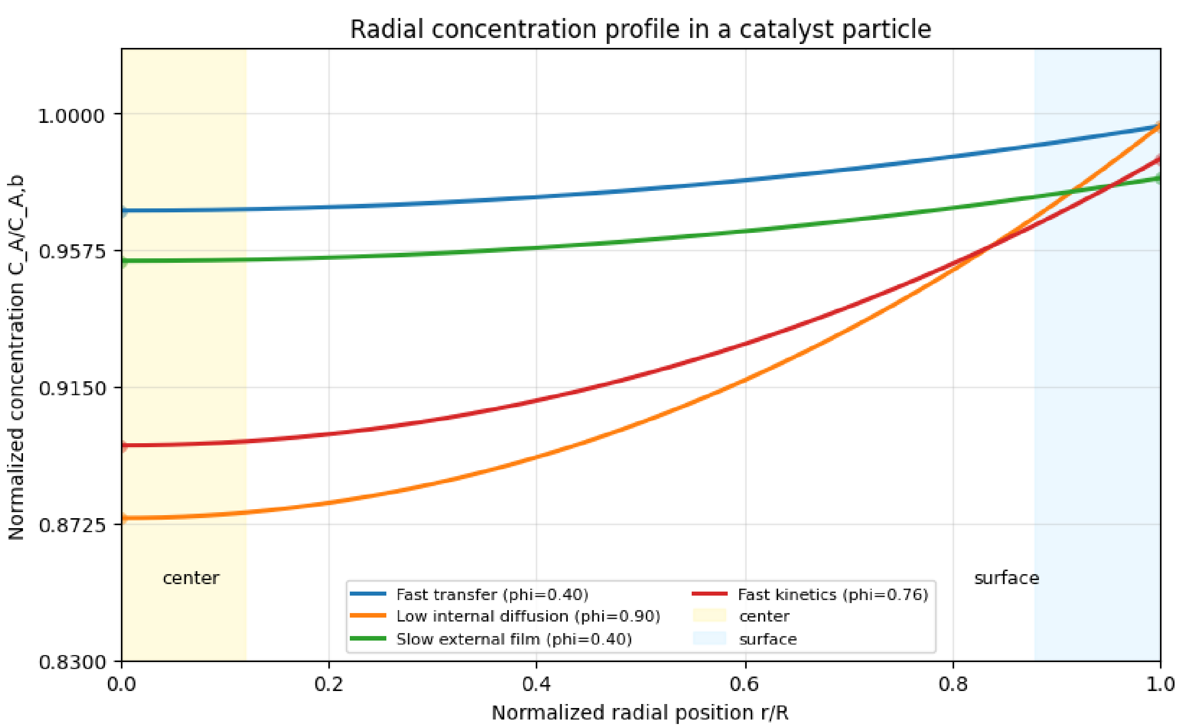
```
y_min = max(0.0, min(C.min() for _, C, _, _, _ in profiles) - 0.04)
y_max = 1.02

fig, ax = plt.subplots(figsize=(8.4, 5.2))
for name, C, phi, eta, Cs in profiles:
    ax.plot(r, C, linewidth=2.2, label=f"{name} (phi={phi:.2f})")
    ax.scatter([0, 1], [C[0], C[-1]], s=28)

ax.axvspan(0, 0.12, color="#fff8c5", alpha=0.55, label="center")
ax.axvspan(0.88, 1, color="#ddf4ff", alpha=0.55, label="surface")
ax.set_xlim(0, 1)
ax.set_ylim(y_min, y_max)
ax.set_xticks(np.linspace(0, 1, 6))
ax.set_yticks(np.linspace(round(y_min, 2), 1.0, 5))
ax.set_xlabel("Normalized radial position r/R")
ax.set_ylabel("Normalized concentration C_A/C_A,b")
ax.set_title("Radial concentration profile in a catalyst particle")
ax.grid(alpha=0.3)
ax.annotate("center", xy=(0.04, y_min + 0.02), fontsize=9)
ax.annotate("surface", xy=(0.82, y_min + 0.02), fontsize=9)
ax.legend(fontsize=8, ncol=2)
plt.tight_layout()
plt.show()
```

```
Scenario summary
Fast transfer: phi=0.40, eta=0.99, C_surface/C_bulk=1.00
Low internal diffusion: phi=0.90, eta=0.95, C_surface/C_bulk=1.00
Slow external film: phi=0.40, eta=0.99, C_surface/C_bulk=0.98
Fast kinetics: phi=0.76, eta=0.96, C_surface/C_bulk=0.99
```



11.4.1 Interpretation

If internal diffusion is slow or kinetics are fast, a strong gradient appears inside the particle. If external transfer is slow, even the surface concentration drops below the bulk-fluid concentration.

Guided reading

Compare the marked points at $r/R = 0$ and $r/R = 1$. The separation between them summarizes internal loss; the surface drop below 1 summarizes external resistance.

11.5 Fast Reading of the Moody Chart

The Moody chart links Reynolds number, relative roughness, and the Darcy friction factor. Moody popularized it as a pipe-flow calculation tool, and this example uses Haaland's explicit correlation to automate the reading [Haa83, Moo44].

The laminar region uses:

$$f = \frac{64}{Re}$$

The turbulent region is approximated with Haaland's explicit equation:

$$\frac{1}{\sqrt{f}} \approx -1.8 \log_{10}\left[\left(\frac{\varepsilon/D}{3.7}\right)^{1.11} + \frac{6.9}{Re}\right]$$

What to modify

Change `roughness_values` or add new design points in `points`. The logarithmic scale makes laminar flow, transition, and turbulence visible at the same time.

```
import numpy as np
import matplotlib.pyplot as plt


def friction_factor(Re, eps_rel):
    Re = np.asarray(Re, dtype=float)
    f = np.empty_like(Re)
    laminar = Re < 2300
    f[laminar] = 64 / Re[laminar]
    turbulent = ~laminar
    # Haaland equation for turbulent flow.
    f[turbulent] = (-1.8 * np.log10((eps_rel / 3.7)**1.11 + 6.9 / Re[turbulent]))**-2
    return f

Re = np.logspace(3, 8, 500)
roughness_values = [0.0, 1e-5, 1e-4, 1e-3, 5e-3]
points = [
    (4.0e3, 1e-4, "A"),
    (2.0e4, 1e-4, "B"),
    (1.5e5, 1e-3, "C"),
    (1.0e6, 5e-3, "D"),
]
```

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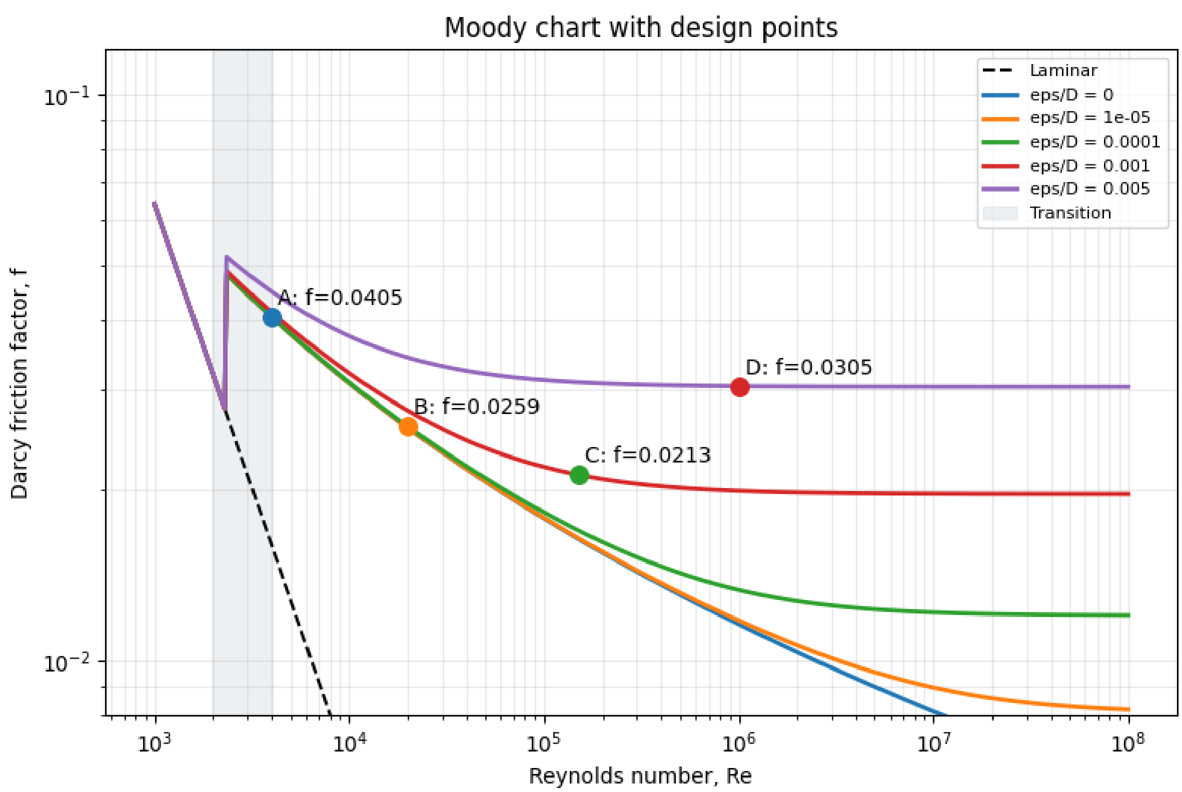
```
plt.figure(figsize=(8, 5.4))
plt.loglog(Re, 64/Re, "k--", linewidth=1.5, label="Laminar")
for eps in roughness_values:
    plt.loglog(Re, friction_factor(Re, eps), linewidth=2, label=f"eps/D = {eps:g}")

for Re_p, eps_p, name in points:
    f_p = friction_factor(np.array([Re_p]), eps_p)[0]
    plt.scatter(Re_p, f_p, s=70, zorder=5)
    plt.text(Re_p * 1.08, f_p * 1.05, f"{name}: f={f_p:.4f}")

plt.axvspan(2000, 4000, color="#d8dee4", alpha=0.45, label="Transition")
plt.xlabel("Reynolds number, Re")
plt.ylabel("Darcy friction factor, f")
plt.title("Moody chart with design points")
plt.grid(True, which="both", alpha=0.25)
plt.ylim(0.008, 0.12)
plt.legend(fontsize=8)
plt.tight_layout()
plt.show()
```



11.5.1 Interpretation

In laminar flow the friction factor depends only on Reynolds number. In turbulent flow, relative roughness shifts the curves and can dominate the behavior at high Reynolds numbers.

Guided reading

Compare points B and C. Both are turbulent, but C has a higher relative roughness and therefore lies on a higher friction curve.

CHEMISTRY DEGREE

This section contains examples for Chemistry courses. The cases combine acid-base equilibria, enzyme kinetics, periodic properties, infrared spectroscopy, and electrochemistry.

The activities are anchored in stable reference sources: IUPAC for electrochemical terminology, Henderson-Hasselbalch and the periodic table, NIST for spectroscopic data, and classic enzyme-kinetics papers [MM13, InternationalUoPaAChemistry14, InternationalUoPaAChemistry22, InternationalUoPaAChemistry25, NationalIoSaTechnology26].

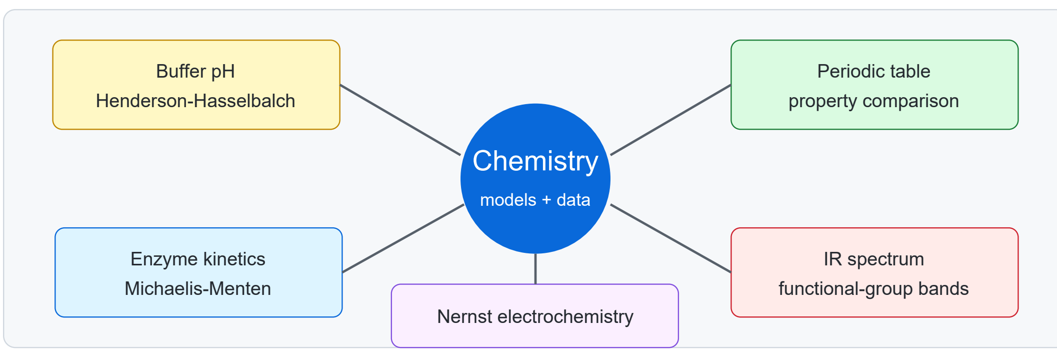


Figure12.1: Teaching resource map for the Chemistry section.

The teaching goal is for students to:

- Interpret pH curves in buffer systems.
- Compare kinetic models and the effect of experimental parameters.
- Consult chemical properties visually.
- Recognize characteristic bands in IR spectra.
- Apply the Nernst equation to redox pairs under non-standard conditions.

How to use the template in this section

Each page combines text, equations, citations, and a figure or interactive tool. The aim is for students not only to see the result, but also to change parameters, compare scenarios, and justify their reading with a bibliographic source.

12.1 pH Curve in a Buffer Solution

A buffer solution resists pH changes because it contains a weak acid and its conjugate base. The Henderson-Hasselbalch relationship estimates pH when the ratio between both forms is known [InternationalUoPaAChemistry14].

This example uses the acetic acid/acetate pair and calculates pH after adding strong acid or strong base:

$$pH = pK_a + \log_{10}\left(\frac{n_{\mathrm{A}^-}}{n_{\mathrm{HA}}}\right)$$

What to modify

Change `n_HA0`, `n_A0`, or the `additions` range. The $pK_a \pm 1$ band is usually the region where the buffer responds most gradually.

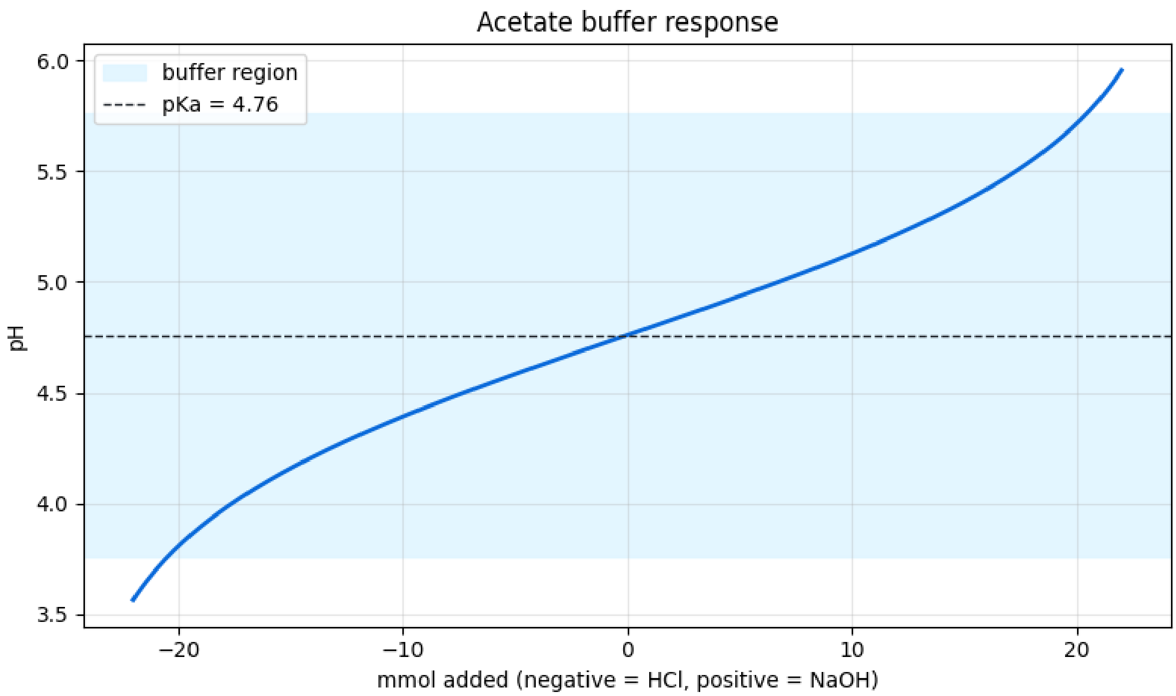
```
import numpy as np
import matplotlib.pyplot as plt

# Editable parameters
pKa = 4.76
n_HA0 = 25.0   # mmol acetic acid
n_A0 = 25.0    # mmol acetate
additions = np.linspace(-22, 22, 220)  # negative: strong acid; positive: strong base

pH = []
for added in additions:
    if added >= 0:
        n_HA = n_HA0 - added
        n_A = n_A0 + added
    else:
        n_HA = n_HA0 - added
        n_A = n_A0 + added
    n_HA = max(n_HA, 1e-6)
    n_A = max(n_A, 1e-6)
    pH.append(pKa + np.log10(n_A / n_HA))
pH = np.array(pH)

plt.figure(figsize=(8, 4.8))
plt.plot(additions, pH, linewidth=2, color="#0969da")
plt.axhspan(pKa - 1, pKa + 1, color="#ddf4ff", alpha=0.8, label="buffer region")
plt.axhline(pKa, color="#24292f", linestyle="--", linewidth=1, label=f"pKa = {pKa}")
plt.xlabel("mmol added (negative = HCl, positive = NaOH)")
plt.ylabel("pH")
plt.title("Acetate buffer response")
plt.grid(alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()

print(f"Initial pH = {pKa + np.log10(n_A0/n_HA0):.2f}")
print(f"pH after adding 10 mmol NaOH = {pH[np.argmin(np.abs(additions - 10))]:.2f}")
print(f"pH after adding 10 mmol HCl  = {pH[np.argmin(np.abs(additions + 10))]:.2f}")
```




```
Initial pH = 4.76
pH after adding 10 mmol NaOH = 5.13
pH after adding 10 mmol HCl  = 4.39
```

12.1.1 Interpretation

The most stable region appears when acid and conjugate base amounts are comparable. Near the ends, one of the two forms is depleted and pH changes much more sharply.

Guided reading

Find the blue band. Inside that band, the same amount of added HCl or NaOH causes a smaller pH change than outside it. That is the practical idea behind buffer capacity.

12.2 Michaelis-Menten Kinetics

Michaelis-Menten kinetics relates the initial rate of an enzyme reaction to substrate concentration. The classic model comes from Michaelis and Menten's work, which has been revisited and translated in a useful historical teaching source [JG11, MM13].

The base model is:

$$v = \frac{V_{\max}[S]}{K_M + [S]}$$

This example compares changes in $V_{\max}$, K_M, and the presence of a competitive inhibitor.

What to modify

Change `Vmax`, `Km`, `I`, or `Ki`. A competitive inhibitor raises the apparent K_M, so more substrate is needed to reach the same fraction of V_{max}.

```
import numpy as np
import matplotlib.pyplot as plt

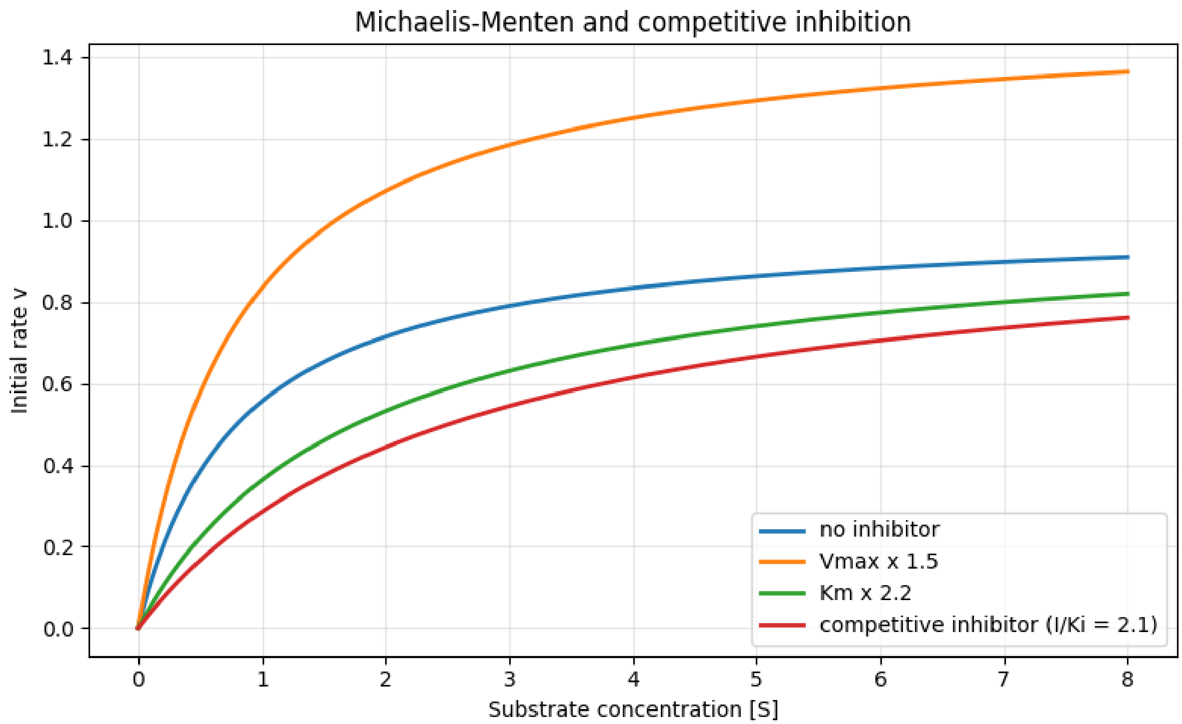
# Editable parameters
Vmax = 1.0
Km = 0.8
I = 1.5
Ki = 0.7
S = np.linspace(0, 8, 300)


def michaelis(S, Vmax, Km):
    return Vmax * S / (Km + S)

v_base = michaelis(S, Vmax, Km)
v_high_vmax = michaelis(S, 1.5 * Vmax, Km)
v_high_km = michaelis(S, Vmax, 2.2 * Km)
Km_app = Km * (1 + I / Ki)
v_inhibited = michaelis(S, Vmax, Km_app)

plt.figure(figsize=(8, 5))
plt.plot(S, v_base, linewidth=2, label="no inhibitor")
plt.plot(S, v_high_vmax, linewidth=2, label="Vmax x 1.5")
plt.plot(S, v_high_km, linewidth=2, label="Km x 2.2")
plt.plot(S, v_inhibited, linewidth=2, label="competitive inhibitor (I/Ki = " + f"{I/
↪Ki:.1f})")
plt.xlabel("Substrate concentration [S]")
plt.ylabel("Initial rate v")
plt.title("Michaelis-Menten and competitive inhibition")
plt.grid(alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()

print(f"Base Vmax = {Vmax:.2f}, Km = {Km:.2f}")
print(f"Competitive inhibitor: apparent Km = {Km_app:.2f}")
```




```
Base Vmax = 1.00, Km = 0.80
Competitive inhibitor: apparent Km = 2.51
```

12.2.1 Interpretation

Increasing $V_{\max}$ raises the maximum rate. Increasing K_M shifts the curve to the right. Under competitive inhibition, the apparent effect is an increase in K_M without changing $V_{\max}$.

Check question

Find in the figure where each curve reaches about half of its maximum rate. That visual reading helps interpret K_M as a concentration scale, not just as an algebraic parameter.

12.3 Interactive Periodic Table

The interactive table below summarizes a teaching selection of representative elements. It is not intended to be a complete database; it is a fast way to compare properties and chemical families during an explanation.

The values and categories in a teaching periodic table should be treated as a gateway to evaluated data. IUPAC keeps its periodic table updated with abridged standard atomic weights recommended by CIAAW [InternationalUoPaAChemistry22].

A useful qualitative reading is that many trends increase with effective nuclear charge:

$$Z_{\text{eff}} \uparrow \implies E_{\text{ion}} \uparrow \text{ and } r_{\text{at}} \downarrow$$

Interactive table. Basic properties of selected elements.

Interactive periodic table: see the HTML version to filter elements and compare properties. This page uses a teaching selection of representative elements.

The interactive table makes trends quick to compare: alkali metals have low ionization energies, halogens have high electronegativities, and noble gases usually have no defined electronegativity on this scale.

12.3.1 Activity

Select electronegativity and compare sodium, chlorine, and argon. Then explain why NaCl is interpreted very differently from a covalent molecule such as HCl.

Guided reading

Compare sodium, magnesium, and chlorine using ionization energy and electronegativity. The goal is not to memorize values, but to recognize patterns across groups and periods.

12.4 Interactive IR Spectrum

An infrared spectrum links absorption bands with functional groups. The tool below is a teaching simulation: when a functional group is enabled, approximate bands are shown in the regions where they commonly appear.

Real spectra should be checked against evaluated databases. The NIST Chemistry WebBook includes IR spectra for thousands of compounds as well as thermochemical and spectroscopic data [NationalIoSaTechnology26].

The horizontal axis uses wavenumber:

$$\tilde{\nu} = \frac{1}{\lambda}$$

Interactive figure. Approximate IR bands by functional group.

Interactive IR spectrum: see the HTML version to enable functional groups. Typical bands: O--H 3200--3600 cm^{-1}, C--H 2850--3000 cm^{-1}, C=O 1650--1750 cm^{-1}, C≡N 2210--2260 cm^{-1}.

The interactive figure helps separate two ideas: band position gives information about the bond, but width and intensity also depend on chemical environment and sample conditions.

Guided activity

Enable O-H and C=O at the same time. Then disable O-H and observe how the broad high-frequency band disappears. That width is often as informative as the peak position.

12.5 Nernst Equation

The Nernst equation links an equilibrium electrode potential to the composition of the contacting phases. IUPAC writes it in terms of activities and notes that analytical chemistry often uses concentrations as a practical approximation [InternationalUoPaAChemistry25].

This example uses the Daniell cell:

$$\mathrm{Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)}$$

For this reaction, $Q = [\mathrm{Zn}^{2+}]/[\mathrm{Cu}^{2+}]$ and $n = 2$. The model explored here is:

$$E = E^{\circ} - \frac{RT}{nF} \ln Q$$

What to modify

Change `T_values`, `[Cu2+]`, or `[Zn2+]` in the code cell. Notice that $Q = 1$ gives $E = E^{\circ}$, and that the logarithmic correction becomes steeper as temperature increases.

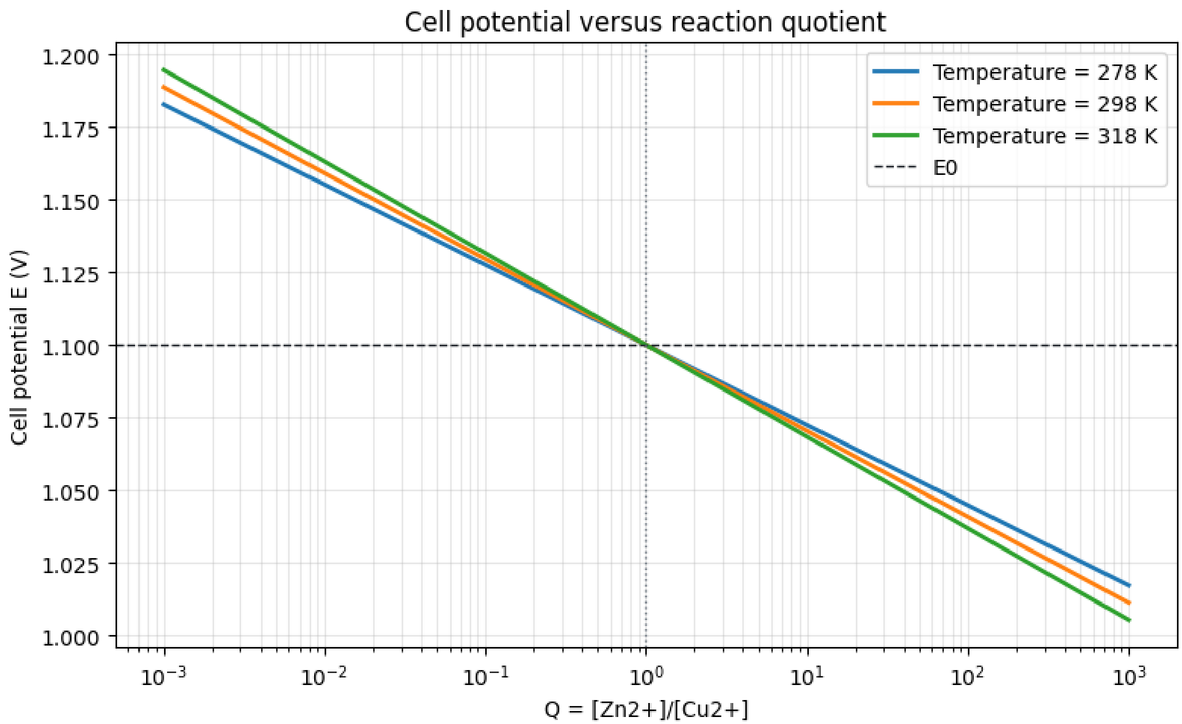
```
import numpy as np
import matplotlib.pyplot as plt

# Editable parameters
E0 = 1.10       # V, standard Daniell cell potential
n = 2
R = 8.314462618
F = 96485.33212
T_values = [278.15, 298.15, 318.15]
Q = np.logspace(-3, 3, 400)


def nernst(E0, T, n, Q):
    return E0 - (R * T) / (n * F) * np.log(Q)

plt.figure(figsize=(8, 5))
for T in T_values:
    plt.semilogx(Q, nernst(E0, T, n, Q), linewidth=2, label=f"Temperature = {T:.0f} K
↪")
plt.axhline(E0, color="#24292f", linestyle="--", linewidth=1, label="E0")
plt.axvline(1, color="#6e7781", linestyle=":", linewidth=1)
plt.xlabel("Q = [Zn2+]/[Cu2+]")
plt.ylabel("Cell potential E (V)")
plt.title("Cell potential versus reaction quotient")
plt.grid(alpha=0.3, which="both")
plt.legend()
plt.tight_layout()
plt.show()

Cu2 = 0.010
Zn2 = 1.000
T = 298.15
Q_case = Zn2 / Cu2
E_case = nernst(E0, T, n, Q_case)
print(f"[Cu2+] = {Cu2:.3f} M, [Zn2+] = {Zn2:.3f} M, T = {T:.2f} K")
print(f"Q = {Q_case:.1f}")
print(f"E = {E_case:.3f} V")
```




```
[Cu2+] = 0.010 M, [Zn2+] = 1.000 M, T = 298.15 K
Q = 100.0
E = 1.041 V
```

12.5.1 Interpretation

The computational figure shows that increasing Q shifts the reaction further toward products and lowers the cell potential. The vertical line at $Q = 1$ marks the simplified standard case; on both sides, the $\ln Q$ term corrects the value of E°.

Check question

If $[Cu^{2+}]$ decreases and $[Zn^{2+}]$ increases, Q increases. The equation then predicts a lower potential because the cell has less thermodynamic driving force left in the written direction.

Part III

Cross-disciplinary Examples

CROSS-DISCIPLINARY EXAMPLES

What are cross-disciplinary examples?

They are reusable teaching patterns for many degrees, not examples locked to a single discipline. The idea is to learn a technique —modeling a process, analyzing a survey, building a rubric, summarizing texts, simulating data, or solving a balance— and then adapt it to sciences, humanities, social sciences, health, engineering, or education.

13.1 Example families

The following table summarizes the main elements of this section.

Table. Example families.

Example	Cross-disciplinary pattern	Can be adapted to…
Growth and saturation models	Systems that grow toward a limit	Populations, cultural diffusion, technology adoption, learning, epidemics
Questionnaire analysis	Surveys and opinion scales	Psychology, education, sociology, teaching satisfaction, market studies
Text corpus frequencies	Counting, classifying, and summarizing terms	Languages, history, law, communication, document analysis, NLP
Rubrics and learning sequences	Transparent assessment and planning	Any course with essays, labs, projects, or presentations
Simulated data and variable relationships	Data exploration and correlations	Health, economics, education, environment, social sciences, laboratory work
Balances and conservation	Inputs, outputs, accumulation, and transformation	Chemistry, ecology, economics, logistics, energy, budgets, demography

How to use this section

Do not copy the discipline: copy the **pattern**. Change the vocabulary, the data, and the teaching question so it fits your course. That is where the cross-disciplinary value lives.

13.2 The common idea

All these examples follow the same philosophy:

1. describe a problem with a clear structure;
2. represent data, rules, or criteria as code or tables;
3. generate a reproducible teaching artifact;
4. allow AI to help adapt the example to another context.

This turns each page into a template: it is not "a Biology example" or "a Psychology example", but a recipe that can travel across fields.

13.3 Growth and Saturation Models

The logistic model describes any process that grows quickly at first and slows down as it approaches a limit: a population, technology adoption, diffusion of an idea, accumulated learning, or the early spread of a disease.

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$$

- N = accumulated quantity of the phenomenon
- r = intrinsic growth rate
- K = limit or maximum capacity of the system

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import odeint

def logistic(N, t, r, K):
    return r * N * (1 - N / K)

t = np.linspace(0, 50, 500)
N0 = 10

params = [(0.1, 500), (0.3, 500), (0.3, 1000)]
labels = ['r=0.1, K=500', 'r=0.3, K=500', 'r=0.3, K=1000']

fig, ax = plt.subplots(figsize=(8, 5))
for (r, K), label in zip(params, labels):
    sol = odeint(logistic, N0, t, args=(r, K))
    ax.plot(t, sol, label=label)

ax.set_xlabel('Time')
ax.set_ylabel('Population (N)')
ax.set_title('Logistic growth curves')
ax.legend()
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

13.3.1 Applications

- **Biology**: population dynamics, bacterial growth
- **Medicine**: epidemic spread (simplified SIR model)
- **Economics**: adoption of new technologies

> **Try it yourself**
>
> Change the values of `r` and `K` in the `params` list and observe how the curve changes. What happens when `r` is very high?

13.4 Questionnaire Analysis

We generate synthetic data from a Likert-scale survey (1-5) and analyze the results. The same pattern works for teaching satisfaction, classroom climate, public opinion, market studies, or self-assessment.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(42)

n_participants = 120
questions = [
    'Clarity of explanations',
    'Teaching materials',
    'Practical usefulness',
    'Pace of the class',
    'Teacher availability',
    'Fair assessment',
    'Class atmosphere',
    'Workload',
    'Online resources',
    'Would recommend the course'
]

data = {}
for q in questions:
    center = np.random.choice([2.5, 3.0, 3.5, 4.0])
    data[q] = np.clip(np.round(np.random.normal(center, 0.8, n_participants)), 1, 5).
↪astype(int)

df = pd.DataFrame(data)
df.head(10)
```

```
summary = df.agg(['mean', 'std']).T.round(2)
summary.columns = ['Mean', 'Std Deviation']
summary
```

```
fig, ax = plt.subplots(figsize=(10, 6))

frequencies = df.apply(lambda col: col.value_counts().reindex(range(1, 6), fill_
```

(continues on next page)

(continued from previous page)

```
↪value=0))
colors = ['#d73027', '#fc8d59', '#fee08b', '#91bfdb', '#4575b4']

x = np.arange(len(questions))
width = 0.15

for i, (val, color) in enumerate(zip(range(1, 6), colors)):
    ax.bar(x + i * width, frequencies.loc[val], width, label=str(val), color=color)

ax.set_xticks(x + 2 * width)
ax.set_xticklabels([q[:25] + '...' if len(q) > 25 else q for q in questions],
↪rotation=45, ha='right')
ax.set_ylabel('Number of responses')
ax.set_title('Response distribution by question')
ax.legend(title='Value', bbox_to_anchor=(1.05, 1))
plt.tight_layout()
plt.show()
```

13.4.1 Applications

- **Psychology**: attitude and perception analysis
- **Education**: student satisfaction evaluation
- **Medicine**: quality of life surveys

Adapt this example

Replace the `questions` list with your own survey items and adjust `n_participants`. The code generates data and plots automatically.

13.5 Text Corpus Frequencies

In any discipline that works with documents, **term frequency** indicates how many times each word or concept appears in a corpus. It helps detect dominant topics, key vocabulary, and shifts in focus across texts.

Why does frequency matter?

The most frequent words or concepts often reveal the structure of a discourse. The same technique can be applied to novels, legal decisions, interviews, scientific articles, historical newspapers, or open-ended survey responses.

13.5.1 Frequency table: sample corpus

The following table summarizes the main elements of this section.

Table. Frequency table: sample corpus.

Term	Grammatical category	Frequency	Cumulative percentage
the	Definite article	6 892	12.3 %
of	Preposition	5 417	22.0 %
and	Conjunction	4 756	30.5 %
to	Preposition	4 102	37.8 %
a	Indefinite article	3 945	44.8 %
in	Preposition	3 401	50.9 %
is	Verb	2 980	56.2 %
it	Pronoun	2 654	60.9 %
for	Preposition	2 432	65.3 %
that	Conjunction / Pronoun	2 180	69.2 %

13.5.2 Teaching applications

- **Humanities**: topic analysis in literary, historical, or philosophical texts
- **Social sciences**: study of interviews, open-ended surveys, or public discourse
- **Health sciences**: review of frequent terms in reports or simulated anamnesis
- **Engineering and computer science**: foundations of text mining and NLP
- **Education**: design of materials graded by lexical or conceptual difficulty

How to adapt this table

Replace the frequencies with those from your own corpus. Tools such as AntConc, Voyant Tools, or Python compute frequencies automatically from almost any collection of texts.

13.6 Rubrics and Learning Sequences

13.6.1 Rubric: Research project

The following table summarizes the main elements of this section.

Table. Rubric: Research project.

Criterion	Excellent (4)	Good (3)	Sufficient (2)	Insufficient (1)
Problem statement	Defines an original, relevant problem with clear context	Defines an adequate problem with some context	Defines a vague or poorly contextualized problem	No identifiable problem defined
Methodology	Rigorous, well-justified, replicable method	Adequate method with partial justification	Simple method without clear justification	No defined method
Data analysis	In-depth analysis with correct interpretation	Correct analysis with basic interpretation	Superficial analysis with minor errors	No analysis or major errors
Conclusions	Well-founded conclusions answering the problem	Coherent conclusions aligned with data	Partial or poorly connected conclusions	No conclusions or unrelated to the work
Presentation	Impeccable structure, clear writing, no errors	Good structure, minimal errors	Acceptable structure, some errors	Disorganized, frequent errors

Scoring

Minimum passing score: **10 / 20** (50 %). The total score is obtained by adding the values of each criterion.

13.6.2 Learning objective sequence

The following table summarizes the main elements of this section.

Table. Learning objective sequence.

Phase	Objective	Expected outcome
1. Exploration	Identify a topic of interest	List of 3 candidate topics
2. Delimitation	Formulate a research question	Concrete, viable question
3. Design	Select an appropriate methodology	Research protocol
4. Collection	Gather relevant data	Organized dataset
5. Analysis	Interpret the results	Findings report
6. Communication	Present the final work	Written document + oral defense

13.6.3 Cross-disciplinary applications

- **Humanities**: assessment of essays, text commentaries, and presentations
- **Social sciences**: assessment of case studies, reports, and debates
- **Experimental sciences**: assessment of lab work and field notebooks
- **Engineering**: assessment of projects, prototypes, and technical documentation
- **Health**: assessment of simulated clinical cases and diagnostic reasoning

Adapt the rubric

Replace the criteria in the first column with those from your course and adjust the descriptors. The structure works for almost any assessable product: essay, lab activity, report, project, poster, presentation, or portfolio.

13.7 Simulated Data and Variable Relationships

We generate a simulated dataset and explore correlations between variables. Although the example uses clinical variables, the pattern works for economics, education, environment, laboratory work, or social sciences.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(42)
n = 200

age = np.random.randint(30, 80, n)
cholesterol = 150 + 1.5 * age + np.random.normal(0, 25, n)
systolic_bp = 110 + 0.5 * age + np.random.normal(0, 10, n)
risk = (0.3 * (age - 30) + 0.1 * (cholesterol - 150) + np.random.normal(0, 5, n)).
↪clip(0, 100)

df = pd.DataFrame({
    'Age': age,
    'Cholesterol (mg/dL)': cholesterol.round(0),
    'Systolic BP (mmHg)': systolic_bp.round(0),
    'Cardiovascular risk (%)': risk.round(1)
})

df.describe().round(1)
```

```
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

axes[0].scatter(df['Age'], df['Cholesterol (mg/dL)'], alpha=0.5, c='steelblue')
axes[0].set_xlabel('Age (years)')
axes[0].set_ylabel('Cholesterol (mg/dL)')
axes[0].set_title('Age vs Cholesterol')

scatter = axes[1].scatter(
    df['Cholesterol (mg/dL)'], df['Cardiovascular risk (%)'],
    alpha=0.5, c=df['Age'], cmap='RdYlBu_r'
)
axes[1].set_xlabel('Cholesterol (mg/dL)')
axes[1].set_ylabel('Cardiovascular risk (%)')
axes[1].set_title('Cholesterol vs Risk (color = Age)')
plt.colorbar(scatter, ax=axes[1], label='Age')

plt.tight_layout()
plt.show()
```

```
correlations = df.corr().round(2)
correlations
```

13.7.1 Applications

- **Medicine**: exploratory analysis of clinical data
- **Psychology**: correlation between well-being variables
- **Biology**: relationship between physiological variables

> **Important note**
>
> These data are **simulated** and do not represent real patients. In actual clinical research, always verify the origin and quality of the data.

13.8 Balances and Conservation

13.8.1 Problem statement

A solution of reactant A with concentration $C_{A0} = 2.0$ mol/L is fed into a continuous stirred-tank reactor (CSTR) at a volumetric flow rate $Q = 10$ L/min. The reaction is:

$$A \longrightarrow B \quad \text{with} \quad r_A = -k\,C_A$$

where $k = 0.5$ min^{-1}. The reactor volume is $V = 100$ L.

Determine the outlet concentration C_A.

This problem is the minimal example of a very general strategy: define a system boundary, list inputs and outputs, and apply material conservation. It is the first step in many Chemical Engineering and compartment-model problems [FRB16].

> **Mental picture of the reactor**
>
> Think of the CSTR as a perfectly mixed box: everything leaving has the same concentration as the reactor contents. That is why the reaction term uses reactor C_A, not inlet C_{A0}.

13.8.2 Mass balance equation

The steady-state mass balance for component A is:

$$0 = Q\,C_{A0} - Q\,C_A + r_A\,V$$

Substituting the reaction rate expression:

$$0 = Q\,C_{A0} - Q\,C_A - k\,C_A\,V$$

13.8.3 Step-by-step solution

Step 1: Group terms with C_A.

$$Q\,C_{A0} = C_A\,(Q + kV)$$

Step 2: Solve for C_A.

$$C_A = \frac{Q\,C_{A0}}{Q + kV}$$

Step 3: Substitute numerical values.

$$C_A = \frac{10 \times 2.0}{10 + 0.5 \times 100} = \frac{20}{60} = 0.333 \text{ mol/L}$$

13.8.4 Conversion

The reactor conversion is defined as:

$$X = 1 - \frac{C_A}{C_{A0}} = 1 - \frac{0.333}{2.0} = 0.833\ (83.3\%)$$

Quick verification

Residence time: $\tau = V/Q = 100/10 = 10$ min. For a first-order CSTR: $X = k\tau/(1 + k\tau) = 5/6 = 0.833$. The results match.

Try another scenario

If the volume is doubled to $V = 200$ L, residence time increases and the predicted conversion becomes $X = 10/11 = 0.909$. The balance immediately shows the link between equipment size and conversion.

13.8.5 Cross-disciplinary applications

- **Chemical Engineering**: reactor design, stoichiometric calculations
- **Biology and environment**: bioreactor, ecosystem, and wastewater treatment models
- **Medicine**: pharmacokinetics and compartment models
- **Economics**: income, expenses, debt, and accumulation flows
- **Logistics**: inputs, outputs, and material storage
- **Energy**: power balances, consumption, and losses

Part IV

Examples from Other Books

CHAPTER 14

EXAMPLES FROM OTHER BOOKS

This section collects external books, Jupyter Books, TeachBooks, repositories, and teaching resources that can inspire the design of a new book.

The goal is not to provide a formal literature review, but to show real examples of structure, navigation, notebook use, exercises, AI policies, and web publication that can be adapted to courses in science and chemical science degrees.

Table14.1: Catalogue of external resources.

Area or degree	Resource	Type	What a teacher can learn	Link
All areas	Gallery of Jupyter Books	Jupyter Books directory	Community catalogue of books built with Jupyter Book, useful for exploring structures, navigation styles, repositories, and teaching examples across many disciplines.	Open
Biology and Botany	Plants & Python	Jupyter Book	Bilingual course with lessons in coding, plant biology, computation, and bioinformatics; a useful example of integrating biological content, notebooks, practice activities, and access through Binder or Colab.	Open
Medicine and bioimage analysis	Bio-image Analysis Notebooks	Jupyter Book	Practical book on biomedical image analysis with Python: fundamentals, segmentation, machine learning, 3D analysis, feature extraction, algorithm validation, and reproducible workflows.	Open
Chemical Sciences	eChem: Computational Chemistry from Laptop to HPC	Jupyter Book	How to organize an advanced computational chemistry book with notebooks, theory, molecular visualization, and complete workflows.	Open
Chemical Sciences	PQ-Jupyter	Notebook repository	Chemical process exercises with notebooks on balances, reactors, distillation, absorption, and extraction.	Open
Statistics Degree	Learning from Data	Jupyter Book	How to organize notes with examples on Bayesian statistics, regression, MCMC, Gaussian processes, and mini-projects.	Open
Biostatistics	A Little Bit of Everything in Biostatistics for Health Science Students	Jupyter Book	Introductory biostatistics manual for health science students, useful as an example of organizing concepts, data, descriptive analysis, inference, and applied activities.	Open
Physics Degree	ShowingPhysics	Jupyter Book	A broad collection of physics demonstrations with clear navigation through pedagogy, inquiry, and conceptual-development blocks.	Open
Physics Degree	Quantum Mechanics I	Jupyter Book	Quantum mechanics notes with notebooks, external material, references, and a brief guide to using the book.	Open
Geology and Geological Engineering Degrees	Geo-Python	Teaching website	A lesson-based course with exercises, Python programming, and an explicit policy for the use of AI tools.	Open
Geology and Geological Engineering Degrees	Earth Engine and Geemap	Jupyter Book	Geospatial data science book with Python, Google Earth Engine, interactive maps, local data, visualization, analysis, timelapse animations, and web apps.	Open
Computer Science Degree	Python for Engineers	TeachBook/Jupyter Book	Programming course with theory, exercises, part-based navigation, clear credits, and a teaching website maintained by TU Delft.	Open
Mathematics Degree	An Introduction to Python Jupyter Notebooks for College Math	Jupyter Book	A mathematics book organized by areas, with notebooks, exercise solutions, and reusable material for	Open

When reviewing these examples, it is useful to focus on three aspects: how the table of contents is structured, what type of activities are proposed, and how text, code, visualizations, and exercises are combined.

Part V

Basic Content

BASIC CONTENT

This section is your **quick reference** for the most common TeachBook elements. Each page shows the MyST code you need to write and explains what it produces when compiled.

Note

All examples in this section work in both **HTML** and **PDF**, unless explicitly stated otherwise.

15.1 What you will find here

The following table summarizes the main elements of this section.

Table. What you will find here.

Page	What you will learn
MyST Markdown	Basic syntax for writing book pages
Figures and images	Inserting and referencing images
Equations	Inline and display math
Citations and bibliography	BibTeX bibliography
Cross-references	Linking figures, tables, sections…
Tables	Simple and formatted tables
Admonitions and dropdowns	Notes, warnings, hints, collapsible blocks
Tabs and cards	Tabbed content (sphinx-design)
Source code	Code blocks with syntax highlighting
Videos	YouTube and local video (HTML + PDF)
Mermaid diagrams	Flowcharts, sequences, class diagrams…
HTML/PDF compatibility	What works where and how to ensure compatibility

15.2 How to use this section

1. Find the page you need.
2. Copy the example that closest matches what you want.
3. Adapt the content to your case.
4. Build and verify the result.

> **Tip**
>
> If you use an AI assistant, you can ask it to "generate a figure like the one in the *Figures* section but with my image" and save time.

MYST MARKDOWN: THE BASICS

MyST (Mark up Your Scientific Text) is a Markdown extension designed for scientific and technical documents. It is the format used by Jupyter Book. If you know basic Markdown, you already know 80 % of MyST.

16.1 Essential Syntax

16.1.1 Headings

```
# Main title (H1)
## Section (H2)
### Subsection (H3)
```

16.1.2 Text Formatting

```
**Bold** for strong emphasis.
*Italic* for terms or titles of works.
`inline code` for file names or commands.
```

16.1.3 Lists

```
- Unordered list item
- Another item
  - Sub-item with indentation

1. First step
2. Second step
3. Third step
```

16.1.4 Links and Images

```
[Link text](https://example.com)
```

```
![Alt text](path/to/image.png)
```

For images that you will explain or cite in the text, use the `{figure}` directive from the *Figures and images* section. This lets you add a caption, alternative text, and references with `{numref}`.

16.2 Directives

MyST adds **directives** that do not exist in standard Markdown:

Note

This is a note. It appears highlighted in blue.

Warning

This is a warning. It appears highlighted in red/orange.

Tip

This is a tip. It appears highlighted in green.

You can create custom blocks with `admonition`:

````
```{admonition} My custom title
Block content goes here.
```
````

16.3 Math

Inline: the equation $E = mc^2$ is written as `$E = mc^2$`.

Display:

$$\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}$$

Written as `$$ ... $$` on its own line.

16.4 Tables

```
The following table summarizes the main elements of this section.

**Table. Tables.**

| Column A | Column B | Column C |
|----------|----------|----------|
| data 1   | data 2   | data 3   |
| data 4   | data 5   | data 6   |
```

Result:

The following table summarizes the main elements of this section.

Table. Tables.

Column A	Column B	Column C
data 1	data 2	data 3
data 4	data 5	data 6

16.5 Code Blocks

````
```python
def greet(name):
    return f"Hello, {name}!"
```
````

Tip

Do not try to memorize all the syntax. Use an AI assistant to generate the MyST you need and learn as you go.

FIGURES AND IMAGES

Figures are essential in any teaching material. MyST offers two main directives: `{image}` (simple image) and `{figure}` (image with number and caption).

17.1 Centered image with alt text

The `{image}` directive inserts an image without numbering:

```
```{image} ../../_static/logo.png
:alt: Book logo
:width: 60%
:align: center
```
```

Result:

17.2 Numbered figure with caption

The `{figure}` directive adds automatic numbering (`Figure 1`, `Figure 2`...):

````
```{figure} ../../_static/logo.png
:width: 50%
:align: center
:name: fig-figures-1

Book logo.
```
````

Result:

The Fig. 17.1 visually summarizes this part of the explanation.

Figure17.1: Book logo.

17.3 Cross-reference to a figure

Use `{numref}` to reference figures by their number:

```
The {numref}`fig-logo-en` shows the book logo.
```

Result: The Fig. 17.1 shows the book logo.

17.4 Margin figure

Add `:figclass: margin` to place an image in the margin (HTML only):

```
```{figure} ../../_static/logo.png
:figclass: margin
:width: 100%
:name: fig-figures-3

```

(continues on next page)
```

(continued from previous page)

```
Logo in the margin.
```
```

Result:

The Fig. 17.2 visually summarizes this part of the explanation.


**Figure17.2:** Logo in the margin.

## 17.5 Useful parameters

The following table summarizes the main elements of this section.

**Table. Useful parameters.**

| Parameter | What it does | Example |
|---|---|---|
| `:width:` | Width (`50%`, `200px`) | `:width: 80%` |
| `:align:` | Alignment (`left`, `center`, `right`) | `:align: center` |
| `:alt:` | Alt text (accessibility) | `:alt: Diagram` |
| `:name:` | Label for `{numref}` | `:name: fig-map` |
| `:figclass:` | CSS class (`margin`) | `:figclass: margin` |
```

Tip

Always add `:alt:` for accessibility. It is the text that screen readers see and displays if the image fails to load.

EQUATIONS AND MATHEMATICS

MyST supports LaTeX math via the `dollarmath` extension, already enabled in this template. Works in **HTML and PDF**.

18.1 Inline math

Use `$...$` within a line of text:

```
Energy is calculated as $E = mc^2$, where $m$ is mass.
```

Result: Energy is calculated as $E = mc^2$, where m is mass.

18.2 Display equation (unnumbered)

Use `$$...$$` on its own line:

```
$$
\int_0^\infty e^{-x^2} \, dx = \frac{\sqrt{\pi}}{2}
$$
```

Result:

$$\int_0^\infty e^{-x^2} \, dx = \frac{\sqrt{\pi}}{2}$$

18.3 Numbered equation with label

Add a label `(label)=` before the equation to reference it:

```
$$
\frac{\partial u}{\partial t} = \alpha \nabla^2 u
$$ (eq-heat)
```

Result:

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u \tag{18.1}$$

18.4 Cross-reference to an equation

Use `{eq}` to cite an equation by its number:

```
Equation {eq}`eq-heat` describes heat diffusion.
```

Result: Equation (18.1) describes heat diffusion.

18.5 Common math symbols

The following table summarizes the main elements of this section.

Table. Common math symbols.

Symbol	LaTeX code	Symbol	LaTeX code
α	`\alpha`	β	`\beta`
γ	`\gamma`	δ	`\delta`
π	`\pi`	θ	`\theta`
λ	`\lambda`	σ	`\sigma`
$\sum$	`\sum`	$\prod$	`\prod`
$\int$	`\int`	$\oint$	`\oint`
∞	`\infty`	∇	`\nabla`
$\frac{a}{b}$	`\frac{a}{b}`	$\sqrt{x}$	`\sqrt{x}`
$\leq$	`\leq`	$\geq$	`\geq`
$\approx$	`\approx`	$\neq$	`\neq`
$\vec{v}$	`\vec{v}`	$\hat{x}$	`\hat{x}`

Tip

For complex math, ask your AI assistant to generate LaTeX code from a natural language description: “generate the Gauss integral formula in LaTeX.”

CITATIONS AND BIBLIOGRAPHY

This template includes `sphinxcontrib-bibtex`, which manages bibliographic references using BibTeX files.

19.1 Bibliography file

Bibliography entries are stored in `book/_static/references.bib`. Each entry follows this format:

```
@book{einstein1920,
  author    = {Albert Einstein},
  title     = {Relativity: The Special and General Theory},
  year      = {1920},
  publisher = {Henry Holt and Company}
}

@article{dirac1930,
  author  = {Paul A. M. Dirac},
  title   = {A Theory of Electrons and Protons},
  journal = {Proceedings of the Royal Society A},
  year    = {1930},
  volume  = {126},
  pages   = {360--365}
}
```

19.2 Textual citation: `{cite:t}`

Inserts the author name followed by the year:

```
{cite:t}`einstein1920` formulated the theory of relativity.
```

Result: Einstein [Ein20] formulated the theory of relativity.

19.3 Parenthetical citation: `{cite:p}`

Inserts author and year in parentheses:

```
Quantum mechanics revolutionized physics {cite:p}`dirac1930`.
```

Result: Quantum mechanics revolutionized physics [Dir30].

19.4 Multiple citations

Separate multiple keys with commas:

```
Several authors contributed {cite:p}`einstein1920,dirac1930`.
```

Result: Several authors contributed [Dir30, Ein20].

19.5 Printing the bibliography

The template uses two bibliography levels:

- **Global**: the final `Bibliography` page gathers the entries cited in the book, not every entry from `book/_static/references.bib`.
- **Local**: an individual page can show only its own references in an HTML-only block.

19.5.1 Global book bibliography

The global bibliography is already prepared on the final `Bibliography` page:

````
```{bibliography}
:cited:
```
````

Do not add this block to normal pages: it should exist only on the global bibliography page.

> **Note**
>
> If you add local bibliographies to web pages, some entries will also appear on the global `Bibliography` page. This is expected: the local bibliography helps with page-level reading, while the global page gathers the whole book bibliography. In PDF, only the final global bibliography is printed.

19.5.2 Local page bibliography

For a local bibliography on the web, cite with `{cite:t}` or `{cite:p}` in the normal page text and add a filtered `{bibliography}` directive inside `{only} html` at the end. This lets HTML show page-level references, while the PDF keeps only the final global bibliography.

~~~~~
Text with a local citation {cite:p}`citation_key`.

````{only} html
## Bibliography for this page

```{bibliography}
:filter: docname in docnames
```
````
~~~~~

You can also show only selected keys from the document in HTML:

~~~~~
````{only} html
```{bibliography}
:filter: docname in docnames and key in {"ziman2000", "hodson1996"}
```
````
~~~~~

In this local example, Ziman [Zim00] describes science as a social practice, not just as a collection of results. Laboratory work should be connected to explanation-building, not only to following instructions [Hod96].

19.6 Workflow

1. Add BibTeX entries to `_static/references.bib`.
2. Cite with `{cite:t}` or `{cite:p}` on any page.
3. If you want a local bibliography visible in HTML, add an `{only} html` block with a filtered `{bibliography}` directive at the end of the page.
4. In PDF, do not generate local bibliographies: the template creates a temporary `.bib` with the used citations and shows them only on the global `Bibliography` page.

Warning

Each citation key (e.g., `einstein1920`) must be unique across the entire `.bib` file. Duplicates will cause a build failure.

CHAPTER 20

CROSS-REFERENCES

Cross-references allow you to link figures, tables, equations, and sections within the book. This is **essential** for long documents: if you rearrange chapters, links update automatically.

20.1 Label a section

Write `(label)=` just before the heading:

```
(my-topic)=
## My Important Topic

Content here...
```

20.2 Reference a section

Use `{ref}` to create a link with the heading text:

```
See the {ref}`my-topic` section for details.
```

Result: See the *Figures and Images* section for details.

20.3 Label and reference figures

Use `:name:` inside the `{figure}` directive and `{numref}` to cite by number:

````
```{figure} ../../_static/logo.png
:width: 40%
:name:  to cite by number:



Project logo.
```

The {numref}`fig-logo-ref-en` shows the logo.
````

Figure20.1: Project logo.

Result:

The Fig. 20.1 visually summarizes this part of the explanation.

The Fig. 20.1 shows the logo.

20.4 Label and reference equations

Use `(eq-label)=` after the equation and `{eq}` to cite:

```
$$
F = ma
$$ (eq-newton-en)

Equation {eq}`eq-newton-en` is Newton's second law.
```

Result:

$$F = ma \tag{20.1}$$

Equation (20.1) is Newton's second law.

20.5 Label and reference tables

Use `:name:` inside `{table}` and `{numref}` to cite:

````
```{table} Physical constants
:name: tab-constants-en
:align: center

| Constant | Symbol | Value |
|----------|--------|-------|
| Speed of light | $c$ | $3 \times 10^8$ m/s |
| Planck   | $h$     | $6.63 \times 10^{-34}$ J·s |
```

The {numref}`tab-constants-en` lists fundamental constants.
````

Result:

Table20.1: Physical constants

Constant	Symbol	Value
Speed of light	c	3×10^8 m/s
Planck	h	6.63×10^{-34} J·s

The Table 20.1 lists fundamental constants.

20.6 Roles summary

The following table summarizes the main elements of this section.

Table. Roles summary.

Role	What it references	Example
`{ref}`	Sections (uses heading text)	``{ref}`my-topic` ``
`{numref}`	Figures and tables (uses number)	``{numref}`fig-logo-ref-en` ``
`{eq}`	Numbered equations	``{eq}`eq-newton-en` ``

Tip

Use descriptive label names: `fig-chromatogram`, `eq-bernoulli`, `tab-raw-data`. They are easier to remember than `fig1` or `eq2`.

TABLES

Tables are the most direct way to present structured data. MyST supports standard Markdown tables and the `{table}` directive for adding numbering.

21.1 Simple Markdown table

```
The following table summarizes the main elements of this section.

**Table. Simple Markdown table.**

| Element | Symbol | Atomic number |
|---------|--------|---------------|
| Hydrogen | H      | 1             |
| Helium   | He     | 2             |
| Lithium  | Li     | 3             |
```

Result:

The following table summarizes the main elements of this section.

Table. Simple Markdown table.

Element	Symbol	Atomic number
Hydrogen	H	1
Helium	He	2
Lithium	Li	3

21.2 Numbered table with caption

Use `{table}` for automatic numbering and cross-referencing:

```
```{table} Common laboratory solvents
:name: tab-solvents
:align: center

| Solvent | Formula | Boiling point (°C) |
|---------|---------|--------------------|
| Water   | H₂O     | 100                |
```

(continues on next page)
```

(continued from previous page)

```
| Ethanol | C₂H₅OH  | 78                 |
| Acetone | CH₃COCH₃| 56                 |
| Ether   | C₄H₁₀O  | 35                 |
```
```

Result:

**Table21.1:** Common laboratory solvents

| Solvent | Formula | Boiling point (°C) |
|---|---|---|
| Water | $H_2O$ | 100 |
| Ethanol | $C_2H_5OH$ | 78 |
| Acetone | $CH_3COCH_3$ | 56 |
| Ether | $C_4H_{10}O$ | 35 |

## 21.3 List-table directive

MyST does not support per-column alignment in Markdown tables. Use `{list-table}` for more control:

````
```{list-table} Experimental data
:header-rows: 1
:name: tab-list-en
:align: center

* - Sample
  - pH
  - Conductivity (mS/cm)
* - A
  - 7.2
  - 1.45
* - B
  - 6.8
  - 2.10
* - C
  - 8.1
  - 0.95
```
````

Result:

**Table21.2:** Experimental data

| Sample | pH | Conductivity (mS/cm) |
|---|---|---|
| A | 7.2 | 1.45 |
| B | 6.8 | 2.10 |
| C | 8.1 | 0.95 |
````

21.4 Tips for readable tables

- **Fewer than 6 columns**: wide tables are hard to read on screen.
- **Align numbers**: makes visual comparison easier.
- **Units in the header**: do not repeat units in each cell.
- **Long tables**: if you exceed 15 rows, consider using a chart instead.

Tip

The Table 21.1 shows an example with units in the header. Notice how numeric columns make comparison easier.

ADMONITIONS AND DROPDOWNS

Admonitions are **highlighted boxes** that draw the reader's attention. They are perfect for hints, warnings, key definitions, and exercise solutions.

22.1 Note (`{note}`)

````
```{note}
The exam will be held on June 15th in room 3.2.
```
````

Result:

Note

The exam will be held on June 15th in room 3.2.

22.2 Warning (`{warning}`)

````
```{warning}
Do not mix hydrochloric acid with bleach. It produces toxic gas.
```
````

Result:

Warning

Do not mix hydrochloric acid with bleach. It produces toxic gas.

22.3 Tip (`{tip}`)

````
```{tip}
To memorize the periodic table, use a color association technique.
```
````

Result:

> **Tip**
>
> To memorize the periodic table, use a color association technique.

22.4 Important (`{important}`)

````
```{important}
The submission deadline is Friday at 23:59. Late submissions are not accepted.
```
````

Result:

> **Important**
>
> The submission deadline is Friday at 23:59. Late submissions are not accepted.

22.5 Caution (`{caution}`)

````
```{caution}
This reagent is corrosive. Wear gloves and safety goggles.
```
````

Result:

> **Caution**
>
> This reagent is corrosive. Wear gloves and safety goggles.

22.6 Custom admonition

Use `{admonition}` with your own title:

````
```{admonition} Definition: covalent bond
A covalent bond is the union between two atoms that share one or more pairs of electrons.
```
````

Result:

Definition: covalent bond

A covalent bond is the union between two atoms that share one or more pairs of electrons.

22.7 Dropdown (collapsible content)

Add `:class: dropdown` to hide content until the reader clicks:

```
```{admonition} Solution to exercise 3
:class: dropdown

The molar mass of water (H₂O) is:

$$M = 2 \times 1.008 + 16.00 = 18.016 \text{ g/mol}$$
```
```

Result:

Solution to exercise 3

The molar mass of water (H_2O) is:

$$M = 2 \times 1.008 + 16.00 = 18.016 \text{ g/mol}$$

22.8 Teaching use cases

The following table summarizes the main elements of this section.

Table. Teaching use cases.

Type	Common use
`{note}`	Complementary information, reminders
`{warning}`	Hazards, common errors
`{tip}`	Study tips, tricks
`{important}`	Deadlines, mandatory requirements
`{caution}`	Lab safety precautions
`dropdown`	Exercise solutions, lengthy steps

Tip

Combine admonitions with math, code, or images inside the block. All MyST content works inside an admonition.

CHAPTER 23

TABS AND CARDS

The **sphinx-design** extension (already installed) lets you organize content into tabs and cards. Ideal for comparing options, showing code in multiple languages, or presenting alternative steps.

23.1 Simple tabs ({tab-set} / {tab-item})

````
::::{tab-set}
:::{tab-item} Python
```python
print("Hello, world")
```
:::

:::{tab-item} R
```r
print("Hello, world")
```
:::

:::{tab-item} MATLAB
```matlab
disp('Hello, world')
```
:::
::::
````

Result:

Python

```
print("Hello, world")
```

R

```
print("Hello, world")
```

MATLAB

```
disp('Hello, world')
```

23.2 Grouped tabs ({tab-set} / {tab-item})

Tab groups allow synchronization:

```
::::{tab-set}
:::{tab-item} Analytical method
Solving the differential equation directly.
:::

:::{tab-item} Numerical method
Approximating the solution step by step with a computer.
:::
::::
```

Result:

Analytical method

Solving the differential equation directly.

Numerical method

Approximating the solution step by step with a computer.

23.3 Tabs for procedure steps

```
::::{tab-set}
:::{tab-item} Step 1: Prepare the sample
Weigh 0.5 g of the solid compound with 0.001 g precision.
:::

:::{tab-item} Step 2: Dissolve
Add 50 mL of solvent and stir for 5 minutes.
:::

:::{tab-item} Step 3: Measure
Read absorbance at 280 nm using the spectrophotometer.
:::
::::
```

Result:

Step 1: Prepare the sample

Weigh 0.5 g of the solid compound with 0.001 g precision.

Step 2: Dissolve

Add 50 mL of solvent and stir for 5 minutes.

Step 3: Measure

Read absorbance at 280 nm using the spectrophotometer.

23.4 Cards (`{card}`)

Cards present information in visual blocks:

````
```{card} Lab 1: Spectroscopy
Determination of protein concentration using the Bradford method.
+++
Week 3
```
````

Result:

Lab 1: Spectroscopy   Determination of protein concentration using the Bradford method.

Week 3

Warning

Tabs and cards work in **HTML** but **not in PDF**. In PDF, the content of all tabs is shown sequentially.

CODE BLOCKS

Code blocks are fundamental in science: they show computational formulas, analysis scripts, and step-by-step procedures.

24.1 Syntax highlighting

Add the language after the backticks:

````
```python
import numpy as np

data = np.array([3.2, 4.1, 3.8, 4.5, 3.9])
mean = np.mean(data)
print(f"Mean: {mean:.2f}")
```
````

Result:

```python
import numpy as np

data = np.array([3.2, 4.1, 3.8, 4.5, 3.9])
mean = np.mean(data)
print(f"Mean: {mean:.2f}")
```

24.2 Line numbers

Use the `{code-block}` directive with `:linenos:` to show line numbers. This option does not work inside a normal block written as ` ```python`; in that case it is interpreted as one more line of code.

````
```{code-block} python
:linenos:

def calculate_std(data):
    n = len(data)
    mean = sum(data) / n
    variance = sum((x - mean) ** 2 for x in data) / n
    return variance ** 0.5
```
````

Result:

```python
def calculate_std(data):
    n = len(data)
    mean = sum(data) / n
    variance = sum((x - mean) ** 2 for x in data) / n
    return variance ** 0.5
```

24.3 Highlight specific lines

Use `:emphasize-lines:` inside `{code-block}` to highlight specific lines:

````
```{code-block} python
:emphasize-lines: 3

def read_data(filename):
    with open(filename, 'r') as f:
        lines = f.readlines()  # This line will be highlighted
    return [float(l.strip()) for l in lines]
```
````

Result:

```python
def read_data(filename):
    with open(filename, 'r') as f:
        lines = f.readlines()
    return [float(l.strip()) for l in lines]
```

24.4 Other languages

Highlighting works with many languages:

```bash
# Bash example
curl -O https://example.com/data.csv
head -5 data.csv
```

```sql
-- SQL example
SELECT name, grade
FROM students
WHERE grade >= 5.0
ORDER BY grade DESC;
```

24.5 Common languages in science

The following table summarizes the main elements of this section.

Table. Common languages in science.

Language	When to use it
`python`	Data analysis, numerical computing
`r`	Statistics, biostatistics
`matlab`	Engineering, signal processing
`sql`	Database queries
`bash`	Task automation
`json` / `yaml`	Configuration files

Tip

This template includes `sphinx-copybutton`: a **Copy** button appears in the corner of each code block. Readers can copy code with one click.

CHAPTER 25

VIDEOS

Videos greatly enrich teaching materials. However, videos **cannot be embedded in PDF**, so you must always provide a text alternative.

25.1 YouTube video (HTML + PDF)

ALWAYS use this dual pattern with `{raw} html` and `{raw} latex`:

````
```{raw} html
<iframe width="560" height="315" src="https://www.youtube.com/embed/dQw4w9WgXcQ" frameborder="0" allowfullscreen></iframe>
```

```{raw} latex
\begin{center}
\textbf{Video:} \url{https://www.youtube.com/watch?v=dQw4w9WgXcQ}
\end{center}
```
````

Result:

Video: https://www.youtube.com/watch?v=dQw4w9WgXcQ

25.2 How to get a YouTube video ID

1. Open the video in your browser.
2. The URL has this format: `https://www.youtube.com/watch?v=VIDEO_ID`
3. Use `VIDEO_ID` in the embed URL: `https://www.youtube.com/embed/VIDEO_ID`

For example, if the URL is `https://www.youtube.com/watch?v=abc123`, the iframe uses `https://www.youtube.com/embed/abc123`.

25.3 Local video (HTML5)

For videos hosted in `_static/`:

```
```{raw} html
<video width="560" controls>
  <source src="_static/my_video.mp4" type="video/mp4">
  Your browser does not support HTML5 video.
</video>
```

```{raw} latex
\begin{center}
\textbf{Local video:} see the digital version to play it.
\end{center}
```
```

Recommended path for local book videos:

- `book/_static/videos/my_video.mp4`

25.4 Externally generated videos

The base installation of this template avoids heavy video-generation dependencies. If you need an animation, a screen recording, or a recorded explanation, generate the `.mp4` with the external tool of your choice and save the result in `book/_static/videos/`.

Then insert it with the local video pattern above and always add the `{raw} latex` block so the PDF keeps a textual reference.

25.5 Video with description

It is good practice to add context before the video:

```
The following video shows the crystallization process of copper sulfate:

[insert video block here]

Duration: 4:32 | Topic: Crystallization
```

25.6 Recommended complete pattern

```
The following video shows the experimental setup:

```{raw} html
<iframe width="560" height="315" src="https://www.youtube.com/embed/VIDEO_ID"
↪frameborder="0" allowfullscreen></iframe>
```

```{raw} latex
```

(continues on next page)
```

(continued from previous page)

```
\begin{center}
\textbf{Video: Experimental setup}\\
\url{https://www.youtube.com/watch?v=VIDEO_ID}
\end{center}
```

Duration: 5:15 | **Topic:** Chemical kinetics
```

**Warning**

NEVER use only `{raw} html` without the `{raw} latex` block. The PDF will contain no reference to the video and the reader will lose information.
```

DIAGRAMS WITH KROKI (MERMAID)

Kroki converts text into SVG diagrams during the book build. If you use `:type: mermaid`, you can write Mermaid syntax and get a diagram that works in both **HTML** and **PDF**.

26.1 Flowchart (`flowchart`)

````
```{kroki}
:type: mermaid
:align: center

%%{init: {"themeVariables": {"fontSize": "12px", "fontFamily": "Arial"}}}%%
flowchart LR
    A[Sample] --> B[Preparation]
    B --> C[Measurement]
    C --> D[Analysis]
    D --> E[Results]
```
````

Result:

As shown in Fig. 26.1, the diagram is versioned as a static image.

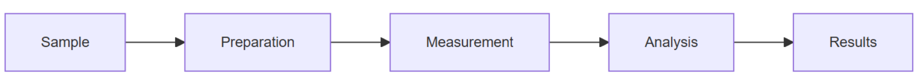


Figure26.1: Diagram: Flowchart (`flowchart`).

````
```{kroki}
:type: mermaid
:align: center

%%{init: {"themeVariables": {"fontSize": "12px", "fontFamily": "Arial"}}}%%
sequenceDiagram
    participant S as Student
    participant T as Teacher
    participant L as Laboratory
    S->>T: Submit report
    T->>S: Corrections
    S->>L: Book equipment
    L->>S: Confirmation
```
````

Result:

As shown in Fig. 26.2, the diagram is versioned as a static image.

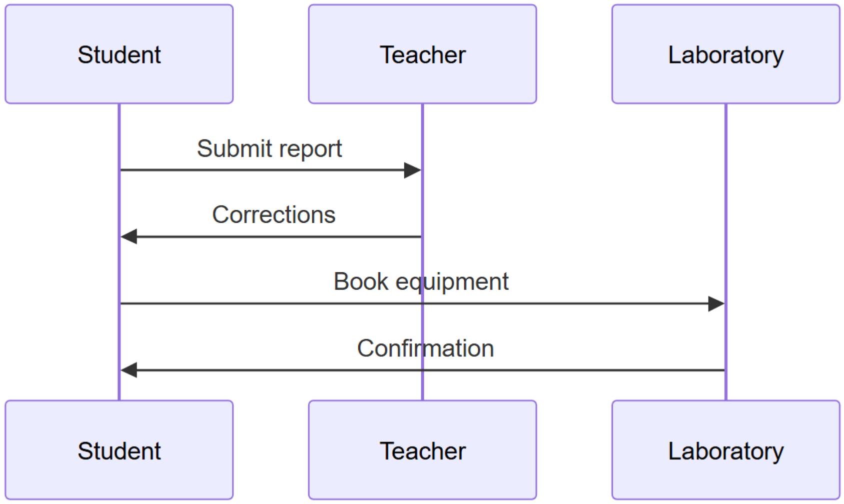


Figure26.2: Diagram: Sequence diagram (`sequenceDiagram`).

```
```{kroki}
:type: mermaid
:align: center

%%{init: {"themeVariables": {"fontSize": "11px", "fontFamily": "Arial"}}}%%
classDiagram
    direction LR
    class Experiment {
        +name
        +run()
    }
    class Sample {
        +type
        +prepare()
    }
    Experiment --> Sample : uses
```
```

Result:

As shown in Fig. 26.3, the diagram is versioned as a static image.

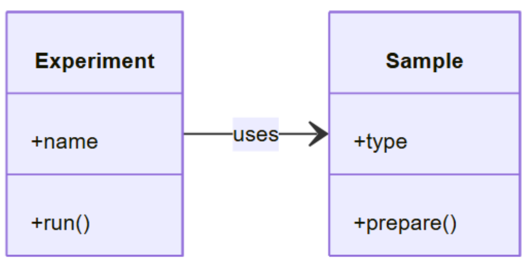


Figure26.3: Diagram: Class diagram (`classDiagram`).

````
```{kroki}
:type: mermaid
:align: center

%%{init: {"themeVariables": {"fontSize": "11px", "fontFamily": "Arial"}}}%%
erDiagram
    STUDENT ||--o{ ENROLLMENT : makes
```
````

Result:

As shown in Fig. 26.4, the diagram is versioned as a static image.

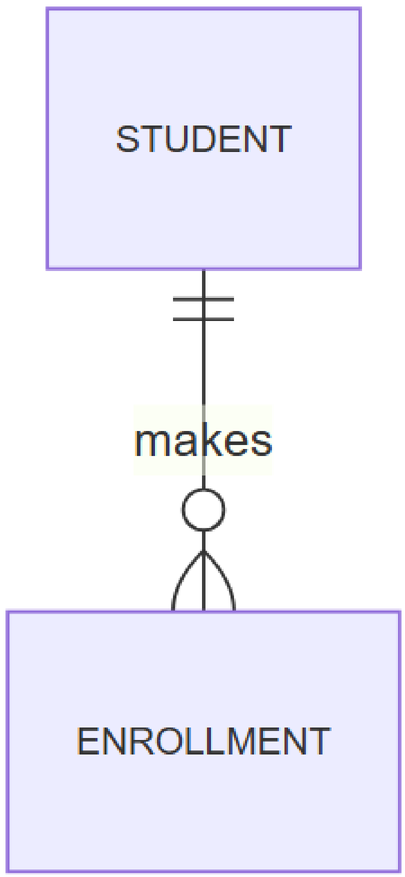


Figure26.4: Diagram: Entity-relationship diagram (`erDiagram`).

```
```{kroki}
:type: mermaid
:align: center

%%{init: {"themeVariables": {"fontSize": "11px", "fontFamily": "Arial"}}}%%
stateDiagram-v2
    direction LR
    [*] --> Pending
    Pending --> InProgress : start
```

(continues on next page)
```

(continued from previous page)

```
    InProgress --> Completed : finish
    InProgress --> Error : failure
    Error --> Pending : retry
    Completed --> [*]
```
```

Result:

As shown in Fig. 26.5, the diagram is versioned as a static image.

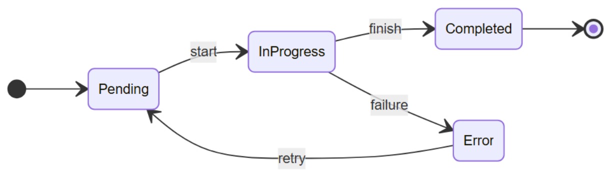



**Figure26.5:** Diagram: State diagram (`stateDiagram-v2`).

The following table summarizes the main elements of this section.

**Table. Flowchart directions.**

| Code | Direction |
|---|---|
| `flowchart LR` | Left to right |
| `flowchart RL` | Right to left |
| `flowchart TB` | Top to bottom |
| `flowchart BT` | Bottom to top |

> **Tip**
>
> Always use `{kroki}` with `:type: mermaid` instead of `{mermaid}`. This makes the diagram work in both HTML and PDF without manual fallbacks.
```

HTML AND PDF COMPATIBILITY

Your TeachBook is published in two formats: **HTML** (interactive web) and **PDF** (printable version). Not everything works the same in both. This page helps you write content that always works.

27.1 Compatibility table

The following table summarizes the main elements of this section.

Table. Compatibility table.

Element	HTML	PDF	Strategy
Text, lists, links	✅	✅	Use freely
Images (`{image}`, `{figure}`)	✅	✅	Use freely
LaTeX math (`$...$`, `$$...$$`)	✅	✅	Use freely
Tables	✅	✅	Use freely
Admonitions (`{note}`, `{tip}`...)	✅	✅	Use freely
Dropdowns (`:class: dropdown`)	✅	✅ expanded	In PDF it appears expanded
Syntax-highlighted code	✅	✅	Use freely
BibTeX citations	✅	✅	Use freely
Cross-references	✅	✅	Use freely
Kroki diagrams (Mermaid, PlantUML, etc.)	✅	✅	Use `{kroki}` with `:type: mermaid`
Videos (iframe/YouTube)	✅	❌	Use `{raw} latex` with URL
Tabs (`{tabbed}`, `{tab-set}`)	✅	❌	In PDF: sequential content
Interactive HTML (`<details>`)	✅	❌	Use `{raw} latex` alternative
Thebe (live code)	✅	❌	Code visible as text in PDF

27.2 Universal pattern: HTML + LaTeX fallback

For any content that only works in HTML, use this pattern:

```
```{raw} html
<!-- Interactive content here -->
<div>...</div>
```

```{raw} latex
\begin{center}
```

(continues on next page)
```

(continued from previous page)

```
Alternative text for the PDF.
\end{center}
```
```

## 27.3 Example: video with fallback

```
```{raw} html
<iframe width="560" height="315" src="https://www.youtube.com/embed/VIDEO_ID"
↪frameborder="0" allowfullscreen></iframe>
```

```{raw} latex
\begin{center}
\textbf{Video:} \url{https://www.youtube.com/watch?v=VIDEO_ID}
\end{center}
```
```

## 27.4 Example: diagram with Kroki

```
```{kroki}
:type: mermaid
:align: center

flowchart LR
    A --> B --> C
```
```

## 27.5 Practical rules

1. **If it is text, image, equation, or table**: do not worry, it works in both.
2. **If it is interactive (video, custom HTML)**: ALWAYS add a text alternative.
3. **If it is a diagram**: use Kroki. It works in both HTML and PDF.
4. **If it is a dropdown**: in PDF it appears expanded (no problem).
5. **If it uses tabs**: in PDF everything appears sequentially (verify it still makes sense).

**Tip**

Before publishing, always build both formats (`build_book.py` for HTML and `export_pdf.py` for PDF) and verify the content is coherent in both.
```

Part VI

Interactivity and Assessment

INTERACTIVITY AND ASSESSMENT

This section covers the different ways to add **interactive content** and **assessment** to your TeachBook.

28.1 What will you find here?

Interactivity in a TeachBook ranges from simple dropdowns to live executable code in the browser. This section covers:

The following table summarizes the main elements of this section.

Table. What will you find here?.

Resource	Type	Works in PDF?
Thebe + Binder	Live code with a remote kernel	No
Interactive notebooks	Code executed at build time	Partial
Widgets (ipywidgets)	Interactive controls	No
Quizzes (jupyterquiz)	Self-correcting questions	No
Parametric questions	Exercises with random values	Partial
Exercises with solutions	Dropdowns with answers	Yes
Interactive HTML	Pure JS elements (no frameworks)	No
H5P (optional)	External interactive content	No

Tip

Don't try to use everything at once. Start with **exercises with solutions** (simple and PDF-compatible) and add interactivity as you need it.

Warning

Features marked as experimental may change in future versions. Always verify that interactive content works correctly in your environment before publishing.

28.2 Thebe + Binder: Live Code

Thebe allows readers to run Python code from the web version of the book. In this project it connects to a **remote Binder kernel**.

> **Warning**
>
> Thebe is an **HTML-only** feature. It is not available in PDF export.

> **Warning**
>
> This feature is **optional** and requires additional configuration of a Jupyter server or Binder. You do not need it for the basic TeachBook setup.

28.2.1 How does it work?

1. Click the **"Live Code"** button below.
2. Binder starts a remote environment (this may take a few seconds the first time).
3. Code cells become editable and executable — try it!

28.2.2 Try it: executable code

Click the button to activate interactive mode. Then you can edit and run the cells.

```python
import numpy as np

# Create an array from 0 to 2*pi
x = np.linspace(0, 2 * np.pi, 100)

# Calculate sine and cosine
y_sin = np.sin(x)
y_cos = np.cos(x)

print(f"Mean of sine: {np.mean(y_sin):.4f}")
print(f"Mean of cosine: {np.mean(y_cos):.4f}")
print(f"Max value of sine: {np.max(y_sin):.4f}")
print("It works! Try changing np.sin to another function.")
```

```python
import matplotlib.pyplot as plt
import numpy as np

# Modifiable parameters
frequency = 2     # Hz
amplitude = 1.0   # Scale
phase = 0         # Radians

t = np.linspace(0, 2*np.pi, 200)
```

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```
signal = amplitude * np.sin(frequency * t + phase)

plt.figure(figsize=(8, 3))
plt.plot(t, signal, 'b-', linewidth=2, label=f'{amplitude}*sin({frequency}t + {phase})
↪')
plt.title('Interactive Sine Wave')
plt.xlabel('Time (rad)')
plt.ylabel('Amplitude')
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()
```

28.2.3 How to add this to your pages

1. Activation button

Add this directive where you want the “Live Code” button to appear:

````
```{thebe-button}
```
````

2. Executable cell

Use `code-block` with the `thebe` class:

````
```{code-block} python
:class: thebe

import numpy as np
print("Hello from the browser!")
```
````

You may also find examples written with the `{code-cell}` directive:

````
```{code-cell} python
import numpy as np
print(np.sqrt(16))
```
````

That syntax turns the block into an executable cell when Thebe is configured.

3. `_config.yml` configuration

Your `_config.yml` must include the `sphinx-thebe` extension:

```yaml
sphinx:
  extra_extensions:
    - sphinx_thebe
  config:
    thebe_config:
      always_load: false
      repository_url: https://github.com/binder-examples/jupyter-stacks-datascience
      repository_branch: master
```

This configuration uses **Thebe + Binder**, meaning the code runs in a remote Binder environment.

> **Simpler alternative**
>
> If you don't want to set up Thebe, you can use **Google Colab**. Upload your notebook to Colab and share the link in your TeachBook. Students can run the code in the cloud with no installation.

> **Summary**
>
> Thebe is powerful but optional. Start with static content and notebooks, and consider Thebe when your students need to experiment with code in real time.

28.3 Interactive Notebooks

A **Jupyter Notebook** is a document that mixes explanatory text and executable code in cells. It is ideal for teaching: you can explain a concept and show the calculation in the same document.

This notebook demonstrates how to create interactive content that students can modify. Change the variable values and re-run the cells to see the results.

28.3.1 Basic example: y = x^2

This first example computes and plots the function $y = x^2$. It shows the minimal notebook structure: one cell with data, one cell with a plot, and a short interpretation.

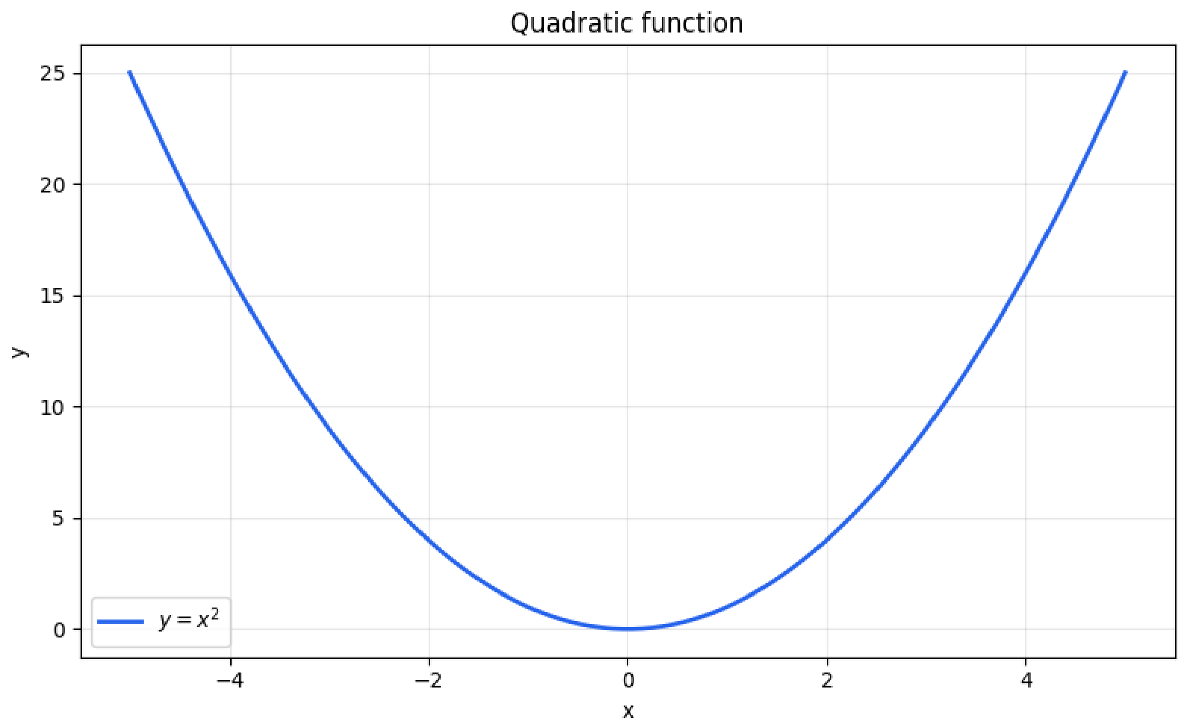
```python
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-5, 5, 100)
y = x ** 2

print(f"Minimum value of y: {y.min()}")
print(f"Value at x=3: {3**2}")
```

```
Minimum value of y: 0.002550760126517668
Value at x=3: 9
```

```
fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(x, y, label=r"$y = x^2$", color="#2563eb", linewidth=2)
ax.set_xlabel("x")
ax.set_ylabel("y")
ax.set_title("Quadratic function")
ax.legend()
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



28.3.2 Observations

- The function $y = x^2$ is a parabola with its vertex at the origin.
- Change the exponent in `y = x ** 2` to `x ** 3` and see how the plot changes.
- Notebooks let you experiment directly: modify the values and re-run the cell.

```
import numpy as np
import matplotlib.pyplot as plt

amplitude = 2.0
frequency = 1.5
phase = np.pi / 4

t = np.linspace(0, 4 * np.pi, 500)
y = amplitude * np.sin(frequency * t + phase)

plt.figure(figsize=(8, 4))
plt.plot(t, y, color='#2563eb', linewidth=2)
```

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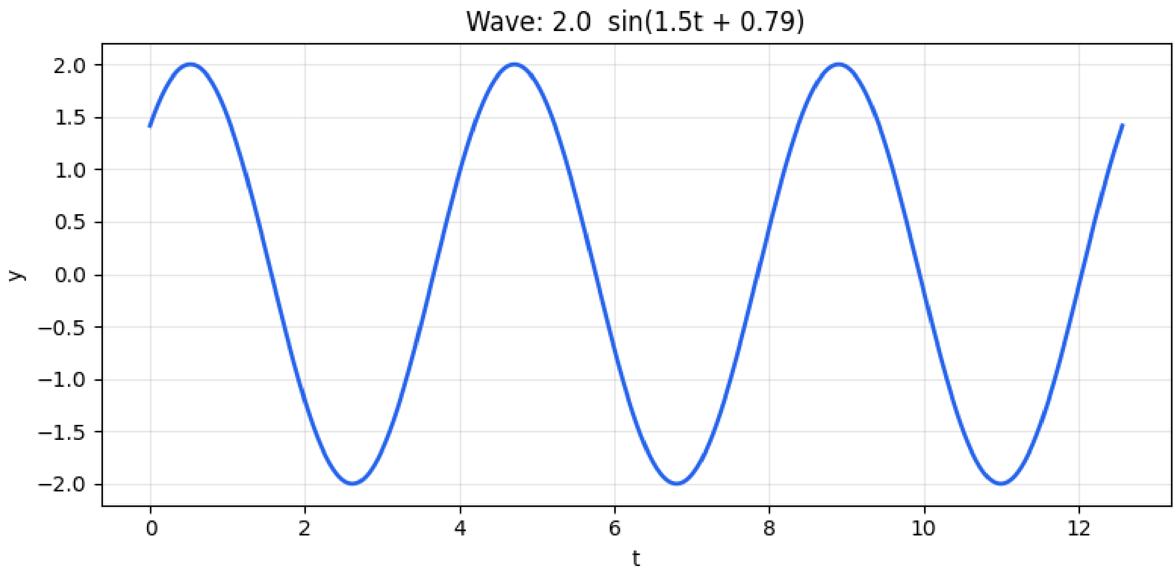
```
plt.title(f'Wave: {amplitude}  sin({frequency}t + {phase:.2f})')
plt.xlabel('t')
plt.ylabel('y')
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



28.3.3 Wave superposition

Modify the frequencies of the two waves to see phenomena like constructive and destructive interference.

```
freq1 = 3.0
freq2 = 3.2

t = np.linspace(0, 10, 1000)
wave1 = np.sin(2 * np.pi * freq1 * t)
wave2 = np.sin(2 * np.pi * freq2 * t)

fig, axes = plt.subplots(3, 1, figsize=(8, 8), sharex=True)

axes[0].plot(t, wave1, color='#2563eb')
axes[0].set_ylabel('Wave 1')
axes[0].set_title(f'f1 = {freq1} Hz')

axes[1].plot(t, wave2, color='#dc2626')
axes[1].set_ylabel('Wave 2')
axes[1].set_title(f'f2 = {freq2} Hz')

axes[2].plot(t, wave1 + wave2, color='#16a34a')
axes[2].set_ylabel('Sum')
axes[2].set_title('Wave 1 + Wave 2 (beats)')
axes[2].set_xlabel('Time (s)')

for ax in axes:
```

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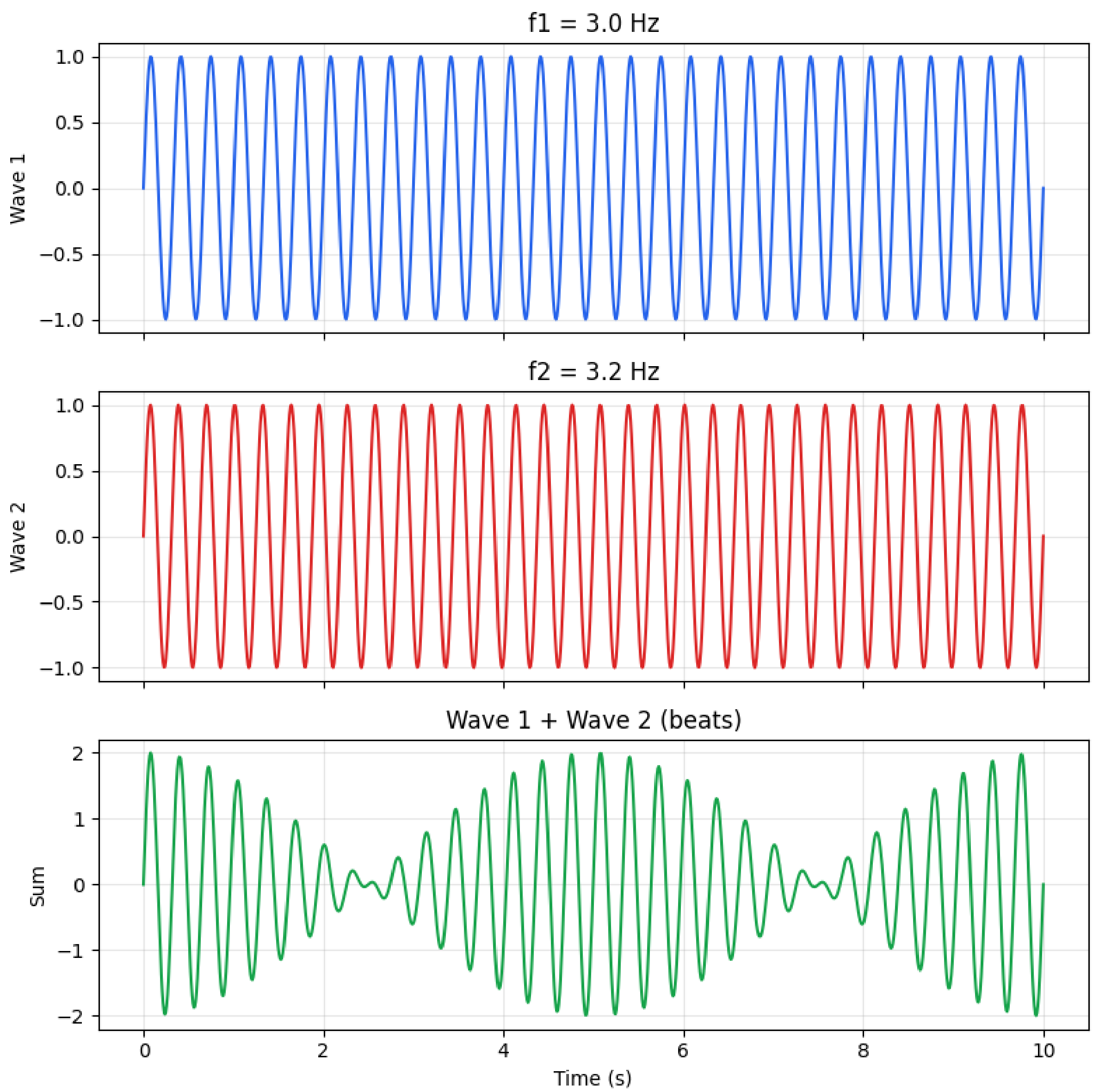
```
    ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



28.3.4 Exercise

1. Set `freq1 = freq2 = 5.0`. What do you observe?
2. Set `freq1 = 10.0` and `freq2 = 10.5`. What is the beat frequency?
3. Change the amplitude of the second wave and observe the result.

28.4 Interactive Widgets (ipywidgets)

Widgets let you add interactive controls such as sliders, dropdowns, and buttons. They are very useful for letting students explore parameters visually.

```
import numpy as np
import matplotlib.pyplot as plt
from ipywidgets import interact, FloatSlider, Dropdown

@interact(freq=FloatSlider(min=0.5, max=10.0, step=0.1, value=2.0, description=
↪'Frequency'),
          amp=FloatSlider(min=0.1, max=5.0, step=0.1, value=1.0, description=
↪'Amplitude'))
def plot_wave(freq, amp):
    t = np.linspace(0, 2 * np.pi, 500)
    y = amp * np.sin(freq * t)
    plt.figure(figsize=(8, 4))
    plt.plot(t, y, color='#2563eb', linewidth=2)
    plt.ylim(-5.5, 5.5)
    plt.title(f'{amp:.1f}  sin({freq:.1f}  t)')
    plt.xlabel('t')
    plt.grid(True, alpha=0.3)
    plt.tight_layout()
    plt.show()
```

```
interactive(children=(FloatSlider(value=2.0, description='Frequency', max=10.0,␣
↪min=0.5), FloatSlider(value=1.…
```

28.4.1 Function selector

Use the dropdown to choose which function to plot.

```
@interact(func=Dropdown(options=['sin', 'cos', 'tan', 'exp', 'log'], value='sin',␣
↪description='Function'))
def plot_func(func):
    t = np.linspace(-np.pi, np.pi, 500)
    funcs = {'sin': np.sin, 'cos': np.cos, 'tan': np.tan, 'exp': np.exp, 'log': np.
↪log}
    y = funcs[func](t)
    y_clipped = np.clip(y, -5, 5)
    plt.figure(figsize=(8, 4))
    plt.plot(t, y_clipped, color='#16a34a', linewidth=2)
    plt.title(f'f(t) = {func}(t)')
    plt.xlabel('t')
    plt.ylim(-5.5, 5.5)
    plt.grid(True, alpha=0.3)
```

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```
    plt.tight_layout()
    plt.show()
```

```
interactive(children=(Dropdown(description='Function', options=('sin', 'cos', 'tan
↪', 'exp', 'log'), value='sin…
```

28.5 Quizzes with jupyterquiz

jupyterquiz lets you embed multiple-choice and numeric questions directly in a notebook. Questions are automatically graded upon answering.

Requirement: `pip install jupyterquiz`

```
from jupyterquiz import display_quiz

physics_questions = [
    {
        "question": "What is the SI unit for force?",
        "type": "multiple_choice",
        "answers": [
            {"answer": "Joule (J)", "correct": False, "feedback": "The joule is a␣
↪unit of energy."},
            {"answer": "Newton (N)", "correct": True, "feedback": "Correct. 1 N = 1␣
↪kgm/s2."},
            {"answer": "Pascal (Pa)", "correct": False, "feedback": "The pascal is a␣
↪unit of pressure."},
            {"answer": "Watt (W)", "correct": False, "feedback": "The watt is a unit␣
↪of power."}
        ]
    },
    {
        "question": "What is the approximate acceleration due to gravity at Earth's␣
↪surface?",
        "type": "numeric",
        "answers": [
            {"type": "value", "value": 9.8, "correct": True, "feedback": "Correct. g ␣
↪9.8 m/s2."},
            {"type": "range", "range": [9.78, 9.83], "correct": True, "feedback":
↪"Correct, within the expected range."},
            {"type": "value", "value": 10, "correct": False, "feedback": "A common␣
↪approximation, but not the standard value."}
        ]
    },
    {
        "question": "Newton's Second Law is expressed as:",
        "type": "multiple_choice",
        "answers": [
            {"answer": "F = ma", "correct": True, "feedback": "Correct. Force equals␣
↪mass times acceleration."},
            {"answer": "F = mv", "correct": False, "feedback": "mv is momentum."},
            {"answer": "F = mg", "correct": False, "feedback": "This is the weight␣
↪formula, a specific case."},
            {"answer": "E = mc2", "correct": False, "feedback": "This is Einstein's␣
↪mass-energy equivalence."}
```

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```
        ]
    }
]

display_quiz(physics_questions)
```

```
<IPython.core.display.HTML object>
```

```
<IPython.core.display.Javascript object>
```

28.6 Parametric Questions

Parametric questions generate **random values** on each execution, so each student gets a different problem with the same structure. Re-run the cells to get new values.

```
import random
import math

a = random.randint(1, 10)
b = random.randint(1, 10)
c = random.randint(-5, 5)

print(f"Solve the quadratic equation: {a}x2 + {b}x + {c} = 0")
print(f"\nCalculate the discriminant  = b2 - 4ac and the solutions.")

discriminant = b**2 - 4*a*c
print(f"\n--- Solution (expand to reveal) ---")
print(f" = {b}2 - 4{a}({c}) = {discriminant}")
if discriminant > 0:
    x1 = (-b + math.sqrt(discriminant)) / (2*a)
    x2 = (-b - math.sqrt(discriminant)) / (2*a)
    print(f"x1 = {x1:.4f}")
    print(f"x2 = {x2:.4f}")
elif discriminant == 0:
    x = -b / (2*a)
    print(f"Double root: x = {x:.4f}")
else:
    real = -b / (2*a)
    imag = math.sqrt(-discriminant) / (2*a)
    print(f"x1 = {real:.4f} + {imag:.4f}i")
    print(f"x2 = {real:.4f} - {imag:.4f}i")
```

```
Solve the quadratic equation: 10x2 + 5x + -1 = 0

Calculate the discriminant  = b2 - 4ac and the solutions.

--- Solution (expand to reveal) ---
 = 52 - 410(-1) = 65
x1 = 0.1531
x2 = -0.6531
```

28.6.1 Parallel resistors

Calculate the equivalent resistance and total current.

```
R1 = random.choice([10, 22, 33, 47, 68, 100, 150, 220, 330, 470])
R2 = random.choice([10, 22, 33, 47, 68, 100, 150, 220, 330, 470])
R3 = random.choice([10, 22, 33, 47, 68, 100, 150, 220, 330, 470])
V = random.choice([5, 9, 12, 24])

print(f"Three resistors in parallel: R1 = {R1} , R2 = {R2} , R3 = {R3} ")
print(f"Applied voltage: V = {V} V")
print(f"\nCalculate the equivalent resistance Req and total current I.")

Req = 1 / (1/R1 + 1/R2 + 1/R3)
I_total = V / Req

print(f"\n--- Solution ---")
print(f"1/Req = 1/{R1} + 1/{R2} + 1/{R3}")
print(f"Req = {Req:.2f} ")
print(f"I = V/Req = {V}/{Req:.2f} = {I_total:.4f} A = {I_total*1000:.2f} mA")
```

```
Three resistors in parallel: R1 = 470 , R2 = 10 , R3 = 470
Applied voltage: V = 9 V

Calculate the equivalent resistance Req and total current I.

--- Solution ---
1/Req = 1/470 + 1/10 + 1/470
Req = 9.59
I = V/Req = 9/9.59 = 0.9383 A = 938.30 mA
```

28.7 Exercises with Solutions

Patterns for creating exercises with hidden solutions that students can reveal when ready. This approach works in both **HTML** and **PDF**.

28.7.1 Basic pattern

Use `{admonition}` for the exercise statement and `{admonition}` with `:class: dropdown` for the solution:

````
```{admonition} Exercise 1 — Basic derivative
:class: important

Calculate the derivative of $f(x) = 3x^4 - 2x^2 + 7x - 1$.
```

```{admonition} Solution
:class: dropdown

Apply differentiation rules term by term:

$$f'(x) = 12x^3 - 4x + 7$$
```
````

Exercise 1 — Basic derivative

Calculate the derivative of $f(x) = 3x^4 - 2x^2 + 7x - 1$.

Solution

Apply differentiation rules term by term:

$$f'(x) = 12x^3 - 4x + 7$$

28.7.2 Exercises with difficulty levels

Add difficulty hints to guide the student:

Exercise 2 — Integral (level: basic)

Calculate $\int_0^3 (2x + 1)\, dx$.

Solution

$$\int_0^3 (2x + 1)\, dx = \left[x^2 + x\right]_0^3 = (9 + 3) - 0 = 12$$

Exercise 3 — Integral (level: intermediate)

Calculate $\int_0^\pi x \sin(x)\, dx$. *Hint: integration by parts.*

Solution

Use integration by parts with $u = x$ and $dv = \sin(x)\, dx$:

$$\int_0^\pi x \sin(x)\, dx = \left[-x \cos(x)\right]_0^\pi + \int_0^\pi \cos(x)\, dx = \pi + 0 = \pi$$

Exercise 4 — Integral (level: advanced)

Calculate $\int_{-\infty}^\infty e^{-x^2}\, dx$. *Hint: use polar coordinates.*

Solution

Let $I = \int_{-\infty}^{\infty} e^{-x^2} dx$. Then:

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx\, dy$$

Converting to polar: $I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r\, dr\, d\theta = 2\pi \cdot \frac{1}{2} = \pi$

Therefore: $I = \sqrt{\pi}$

Tip

Dropdowns appear expanded in PDF export, which is desirable for the printed version. On the web, the student must click to reveal the solution.

28.8 Interactive HTML (no frameworks)

You can create interactive elements using **plain HTML and JavaScript**, without React, Vue, or any framework. All code is self-contained on the page itself.

Warning

Content in `{raw} html` blocks does **not appear in PDF**. Always add a `{raw} latex` block with alternative text for the printed version.

28.8.1 When to use interactive HTML

Use interactive HTML when you need simple behavior on the web: dropdowns, small controls, styled tables, or self-contained calculators. You do not need React, Vue, or any framework.

The basic directive is:

````
```{raw} html
<your HTML here>
```
````

Everything you write inside is inserted as-is into the web page. To keep the book working in PDF, pair that block with a `{raw} latex` alternative containing the essential information.

28.8.2 Asking a chat tool for a self-contained visualization

A practical way to create small interactive resources is to ask a tool such as Gemini, Claude Code, or ChatGPT to generate a **self-contained HTML** visualization. This means that the result must include everything in one block: HTML, CSS, and JavaScript, with no external files and no frameworks.

A useful prompt would be:

```
Generate a self-contained interactive HTML visualization to explain ordinary least squares (OLS).
It must work inside a Jupyter Book page using a raw html block.
Requirements:
- Do not use React, Vue, Angular, npm, or external libraries.
- Include all CSS and JavaScript in the same block.
- Use unique IDs with the prefix ols-demo-en- to avoid conflicts with other parts of the page.
- Do not include <!DOCTYPE html>, <html>, <head>, or <body>; only the fragment that will be inserted into the page.
- Add controls to change slope and intercept.
- Show the points, regression line, residuals, and sum of squared errors.
- The code must be clear and easy to paste into a ```{raw} html block.
```

Then paste the generated fragment inside a `{raw} html` block and add a `{raw} latex` version with the essential explanation for PDF. For example:

OLS visualization: In the HTML version, the reader can change the slope and intercept of a line, observe the residuals between the points and the prediction, and see how the sum of squared errors changes. The optimal fit button computes the ordinary least squares line.

28.8.3 Collapsible content with <details>

Hint: Remember that the derivative of e^x is e^x.

28.8.4 Slider with live output

Temperature converter: $T_{\circ\mathrm{F}} = T_{\circ\mathrm{C}} \times \frac{9}{5} + 32$

28.8.5 Simple calculator

Ohm's Law calculator: $V = I \times R$

Rules for interactive HTML

1. **No JS frameworks**: do not use React, Vue, Angular, or npm.
2. **No external files**: all JS goes inline inside the `{raw} html` block.
3. **Always include LaTeX fallback**: add a `{raw} latex` block with essential content for PDF.
4. **Inline styles**: use `style=""` on tags to avoid depending on external CSS.

28.9 H5P (Optional)

H5P is an interactive content platform that offers quizzes, drag-and-drop activities, video interactions, and much more.

> ⚠ **Warning**
>
> H5P is **optional** and **not a dependency** of this template. It requires an external server to host the content (H5P.com, Moodle, WordPress with H5P plugin, etc.). It is not self-contained.

28.9.1 What does H5P offer?

The following table summarizes the main elements of this section.

Table. What does H5P offer?.

Activity type	Description
Quizzes	Multiple choice, fill in the blanks
Drag & Drop	Drag elements to correct zones
Interactive video	Add pauses with questions on a video
Interactive presentation	Slides with integrated questions
Summary	Text with interspersed questions

28.9.2 How to embed H5P in your TeachBook

Create your content on the H5P platform and use an `iframe` to display it:

````
```{raw} html
<iframe src="https://H5P_URL_HERE" width="800" height="600" frameborder="0"␣
↪allowfullscreen="allowfullscreen" style="border: 1px solid #e5e7eb; border-radius:␣
↪8px;"></iframe>
```

```{raw} latex
\textbf{Interactive H5P activity:} Available in the web version of the book.
Link: \url{https://H5P_URL_HERE}
```
````

28.9.3 Compatible platforms

- **H5P.com**: Official service (paid for advanced use).
- **Moodle**: Free plugin, ideal if your university already uses Moodle.
- **WordPress**: Free H5P plugin.
- **Lumi**: Desktop application to create H5P without a server.

Recommended alternative

If you don't need the complexity of H5P, **exercises with solutions** (using `{admonition}` with `dropdown`) cover most assessment needs and work in both HTML and PDF, with no external server required.

Part VII

Web Publication

CHAPTER 29

WEB PUBLICATION

Web publication turns the generated book into a static site that can be opened from any browser. This template supports two publication routes for different needs:

- **GitHub Pages**: the recommended option for getting started. GitHub builds the book, generates the PDFs, and publishes the website automatically when changes are pushed to the main branch.
- **Own server**: the right option when the book must live under an institutional domain, for example `libro.usal.es`, or on university-managed hosting.

Both routes use the same technical output: the `book/_build/html` folder generated by `scripts/build_book.py`. The difference is where that folder is copied and which service serves the files publicly.

Important

Never write usernames, passwords, or access credentials inside the repository. Deployments to own servers must use GitHub Actions secrets.

The following sections summarize the two publication flows supported by the template.

29.1 GitHub Pages

GitHub Pages is the default deployment route for this template. It is the simplest flow because it does not require an own server: GitHub Actions builds the book and GitHub Pages serves the resulting website.

29.1.1 What the workflow does

The `.github/workflows/deploy.yml` file runs when changes are pushed to `main` and can also be started manually from the **Actions** tab. In each run it:

1. prepares Python 3.12;
2. installs the project environment with `scripts/setup_env.py`;
3. checks UTF-8 encoding and static assets;
4. installs the lightweight LaTeX toolchain;
5. generates the PDFs for all languages;
6. builds the multi-language HTML site;
7. publishes `book/_build/html` to GitHub Pages.

This order matters: PDFs are generated first and the website is built afterwards, so download links always point to current files.

29.1.2 Initial activation

In a new repository created from the template, the recommended setup is:

1. open **Settings** in GitHub;
2. go to **Pages**;
3. choose **Source: GitHub Actions**;
4. save the configuration;
5. push the first change to `main`.

After the first `push`, GitHub runs the `deploy-book` workflow. If it finishes successfully, the public URL appears on the GitHub Pages settings page and in the action summary.

29.1.3 Public and private repositories

The workflow is configured to read repository contents with minimum permissions. In private repositories, publication depends on the GitHub Pages features available for the account or organization. If GitHub Pages is not available for that repository, the book can still be built, but it will not be published through this route.

29.1.4 When to use GitHub Pages

Use GitHub Pages when:

- you want to publish quickly without requesting a server;
- the book can live under a GitHub URL;
- each change in `main` should update the website automatically;
- you are using the template for the first time and need the lowest-configuration path.

29.2 Own Server

Publishing to an own server lets you deploy the same book under an institutional domain, for example `libro.usal.es`, using SFTP. This option is useful when the website must integrate with university infrastructure or when you need control over the final domain.

29.2.1 What the server needs

The server must allow SFTP access with username and password, and it must provide a public directory where the generated website can be copied. In many hosting setups this directory is called `public_html`, although the exact name depends on the institutional configuration.

The SFTP deployment does not configure the domain by itself. There are two separate pieces:

- **Server**: receives the HTML, CSS, JavaScript, image, and PDF files.
- **Domain**: points an address such as `libro.usal.es` to the correct server.

The template builds and uploads the files. DNS and hosting configuration depend on the institution's web service.

29.2.2 Required secrets

The `.github/workflows/sftp-deploy.yml` workflow uses GitHub Actions secrets. They must be configured in **Settings → Secrets and variables → Actions**:

Secret	Purpose
`SFTP_SERVER`	SFTP server name or connection domain.
`SFTP_USERNAME`	SFTP username.
`SFTP_PASSWORD`	SFTP password.
`SFTP_PORT`	SFTP port. Optional; if missing, `22` is used.

> **Warning**
>
> Do not write these values in `.yml` files, Markdown, notebooks, or scripts. They must exist only as repository secrets.

29.2.3 Running the deployment

The own-server deployment is manual. To run it:

1. open the **Actions** tab in GitHub;
2. select **sftp-deploy-book**;
3. click **Run workflow**;
4. enter the remote directory, `public_html` by default;
5. confirm the run.

The workflow builds the complete book, generates the PDFs, and uploads `book/_build/html` to the selected remote directory.

29.2.4 Remote directory cleanup

The SFTP deployment synchronizes the server with cleanup: remote files that no longer exist in the local build are deleted. This prevents old pages from remaining published by accident.

Before deployment, the workflow validates that the remote directory is not a dangerous path such as `/`, `.`, `..`, or an absolute path. Even so, check the `remote_dir` value carefully, because that directory will be made to match `book/_build/html`.

29.2.5 When to use an own server

Use an own server when:

- you need to publish under an institutional domain;
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Part VIII

Information

CHAPTER 30

ABOUT

You can cite this book as:

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30.1 How this book is made

This website is written in Markdown files and Jupyter notebooks, and converted to HTML using tools from TeachBooks and Jupyter Book.

The source files are stored in a public GitHub repository:

- https://github.com/elloza/teachbook_usal_template

The published version of the book can be viewed at:

- https://elloza.com/teachbook_usal_template/

30.2 About the authors

30.2.1 Authors

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30.2.2 Institution

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CREDITS AND LICENSES

31.1 Project credits

This book and the course template are built with TeachBooks, on top of the Jupyter Book and MyST Markdown ecosystem.

TeachBooks provides documentation, components and working patterns that make it easier to create open, interactive and reproducible teaching books.

31.2 Acknowledgements

We thank the TeachBooks project and Delft University of Technology (TU Delft) for their work on open tools for educational books. The credits page in the TeachBooks Manual is used as a reference for explicitly acknowledging the tools, communities and institutional support on which this material builds.

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32.3 DOI

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